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MIL-STD-2222(AS)
12 AUGUST 1992

MILITARY STANDARD

INTERFACE CONTROL DOCUMENTATION FOR AVIONIC SYSTEMS

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FOREWORD

1. This military standard is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department, Code SR3, Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. This standard provides information for understanding the Interface Control Document (ICD) system, the ICD requirements and participant obligations for the support of the ICD process. This standard serves as a useful tool for government personnel, contractors and airframers by specifying hardware and software detailed interfaces. This standard can be used to document the interfaces between various avionic systems.

4. The Naval Air Systems Command has adopted a unique format for the ICD. This format includes "A" size sheets measuring 8.5 inches by 11 inches and "B" size sheets measuring 11 inches by 17 inches. The "A" size sheets contain the text data information and the "B" size sheets contain the interface schematic diagrams for the ICD. Samples of "A" and "B" sized ICDs are provided in appendix B.

5. Statement of work information is given in 4.10. Use of this information will enable equipment manufacturers (henceforth refer d to as contractors) and airframers to readily complete a contractual bid .o support the ICD work. Typical contractor and airframer ICD tasks are as follows:

   a. Contractors and airframers shall provide data for the ICD comment issues and approve the finalized ICDs.

   b. Contractors and airframers shall furnish interface design data at an agreed future date (both contractors and airframers must precisely agree as to what missing data is required, when it will be furnished, and which contractor or airframer is responsible).

6. Benefits of the ICDs are as follows:

   a. The documents provide a mutual interface agreement which is approved by the two affected contractors and airframers at a very early stage of the program.

   b. The "living" documents exist for the duration of the program so that changes can be accommodated.

   c. The format optimizes the information on sheets that are readily reproducible.

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d. The documents are easily understood and eliminate unnecessary verbiage.

e. The ICDs provide a means to control cost and schedule.

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1. SCOPE

1.1 Scope. This document provides guidelines for developing and using standardized ICDs. It informs the users of these documents how to initiate an ICD, of the coordination process, and of the methods implemented to change these contractual instruments. It also provides information to contractors and airframers to allow a thorough understanding of the type of data required for ICDs on new systems.

1.2 Applicability. This standard is intended for use by any government agency, contractor or airframer who desires to use an ICD or generate data to be incorporated into an ICD. The document is not intended to be a cookbook approach on how to draft an ICD. It is, instead, intended to assist government, contractor and airframer personnel with the use of existing ICDs or for estimating the scope of information for a prospective ICD.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-1553	Aircraft Internal Time Division Command/response Multiplex Data Base

(Unless otherwise indicated, copies of the military standard is available from the DODSSP, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS AND ABBREVIATIONS

3.1 Definitions.

3.1.1 Change notice. A change request which has been approved and signed by two contractors or contractor and airframer, and the government.

3.1.2 Change request. Document submitted to the government by a user requesting a change in an ICD.

3.1.3 Design specification differences table. The table in the ICD which highlights all the design specification differences which may cause incompatibilities between contractor(s) and airframer.

3.1.4 Government field activity. The activity responsible for the review and coordination of the interface design documentation.

3.1.5 Interface control document. The document that defines interface requirements and responsibilities between two contractors or a contractor and airframer for specific systems or platforms.

3.1.6 Interface Coordination Working Group. Group composed of the government field activity, contractors and airframers which reviews and coordinates an ICD prior to approval by the government field activity.

3.1.7 Interface design data requirement. The portion of the ICD that identifies missing data and potential conflicts.

3.1.8 Statement of work. Document submitted by the government field activity to the affected contractors and airframers with the data requirements to be included in the ICD.

3.1.9 Unreleased documents. Initial drafts of ICDs which are circulated by the government field activity to the appropriate airframers and contractors for comment.

3.2 Abbreviations. The abbreviations used in this document are defined as follows:

"A"	-	Size 8.5" x 11" sheet
ACA	-	Associate Contractor Agreement
AFC	-	Airframe Change
AMSDL	-	Acquisition Management Systems and Data Requirements Control List
ASPJ	-	Airborne Self-Protection Jammer
ASR	-	Advanced Special Receiver
ATE	-	Automated Test Equipment
AV	-	Avionics
"B"	-	Size 11" x 17" sheet

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BOL	-	Bofors Launcher
CLC	-	Command Launch Computer
CMDS	-	Countermeasures Dispensing System
CN	-	Change Notice
CR	-	Change Request
DoD	-	Department of Defense
DODISS	-	Department of Defense Index of Specifications and Standards
DWG	-	Drawing
ECP	-	Engineering Change Proposal
EMC	-	Electromagnetic Compatibility
EMI	-	Electromagnetic Interference
FAR	-	Federal Acquisition Regulation
GEN-X	-	Generic Expendable
HARM	-	High Speed Anti-Radiation Missile
ICD	-	Interface Control Document
ICWG	-	Interface Coordination Working Group
IDDR	-	Interface Design Data Requirement
ISM	-	Industrial Security Manual
MLV	-	Memory Loader Verifier
MUX	-	Multiplexed
MWS	-	Missile Warning System
NAVAIR	-	Naval Air Systems Command
PMA	-	Program Management Activity
RF	-	Radio Frequency
RWR	-	Radar Warning Receiver
RWS	-	Radar Warning System
SE	-	Support Equipment
STD	-	Standard
TBS	-	To Be Supplied
WRA	-	Weapon Replaceable Assembly

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4. GENERAL REQUIREMENTS

4.1 ICD purpose. The ICD clearly establishes the interface requirements and the interface responsibilities between two contractors or contractor and airframer. This unique format optimizes information on readily reproducible sheets.

The Government is the central authority for control of any changes; yet the two affected contractors or contractor and airframer signify compliance by signing the ICD. The format of the ICD provides a disciplined method for identifying, assigning and scheduling responsibilities for undefined interfaces (i.e., Interface Design Data Requirement (IDDR) as described in 4.6).

The ICDs are interface management tools. They provide the means for all parties (Government, contractor and airframer personnel) to maintain the latest information on the interfaces. Changes to the interface are not allowed unless first coordinated and agreed to by all parties of the affected ICD(s). The coordination process is described in 4.5.

The ICD is an effective tool used to minimize system integration problems by ensuring that all interfaces are clearly understood by all contractors and airframers early in the design stage. Cost and schedule savings have been experienced by this disciplined approach from design start to final systems integration.

4.2 ICD types. All ICDs follow the same basic format. There are instances where data in one ICD directly affects another ICD. This data is accompanied with "flags" to indicate the affected ICDs.

4.2.1 Systems on platforms. System interfaces on the aircraft contain all types of data. A sample of the AN/ALE-47 to F/A-18C/D ICD is provided in appendix B. Common data (e.g., mechanical installation data) are marked with flags to indicate other affected platform ICDs. This flag information is essential to the system supplier since the equipment is installed on many aircraft. A partial list of ICDs for systems installed on various platforms is provided in appendix B.

4.2.2 System to system. There are several systems, supplied by different contractors, which interface directly with each other. Some of these may communicate via aircraft wiring, but the airframer has no other interest in the interface. These interfaces are between the two system suppliers. The AN/ALE-47 to AN/ALR-67 ICD is provided as a sample of direct system-to-system ICDs in appendix B. For two systems that interface indirectly through a third system (see 4.2.3), reference shall be made only in the appropriate ICD where information is contained for the direct system interface.

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4.2.3 Data exchanges. These ICDs are similar to the system to system ICDs. The major difference is that data actually passes through a third system. This third system may modify the data (e.g., alter MIL-STD-1553 header data). Although the actual interfaces of these systems are to a third unit and not to each other, the data is often most critical to the system suppliers. A sample of an indirect system-to-system ICD defining the AN/ALE-47 to AN/ALQ-126B interface is provided in appendix B. A partial list of ICDs for system-to-system ICDs is provided in appendix B.

These intermediate ICDs are used as a vehicle for an agreement and as the baseline for the data in all subsequent ICDs (i.e., each of the associated platform ICDs will use the intermediate ICD as the source of data and formats).

4.3 Requirements for ICDs. An ICD is required whenever a system interface exists between two contractors or contractor and airframer. Keeping an interface between only two contractors or contractor and airframer has the advantage of simplifying the interface, accelerating the time it requires to obtain agreements, and increasing the likelihood that the ICD will be current.

4.4 Initiation of ICDs. The government will be informed whenever a contractor or airframer develops a new system or identifies an undefined interface. The government will assess the case and determine if an ICD should be written. The ICD will be written to the level necessary, describing all interface design data, to ensure that the equipment being designed will integrate properly.

4.5 Coordination process.

4.5.1 ICD process. The ICD development and management process is depicted on figure 1. This figure is referred to throughout this standard; the seven blocks, numbered in the upper right-hand corner, are referred to as "block n of figure 1."

4.5.2 Government field activity. Interfaces between two contractors or contractor and airframer are documented and coordinated by the government field activity. The government field activity provides coordination to obtain a well-defined interface that advocates the essential requirements of the involved contractors and airframers. The first steps are for the government field activity to review the interface design specification documentation supplied by the contractors and airframers and meet with them. The first ICD is then drafted.

4.5.3 Unreleased documents. Unreleased ICDs are documents distributed to all parties for comments prior to signing (refer to block 1 of figure 1).

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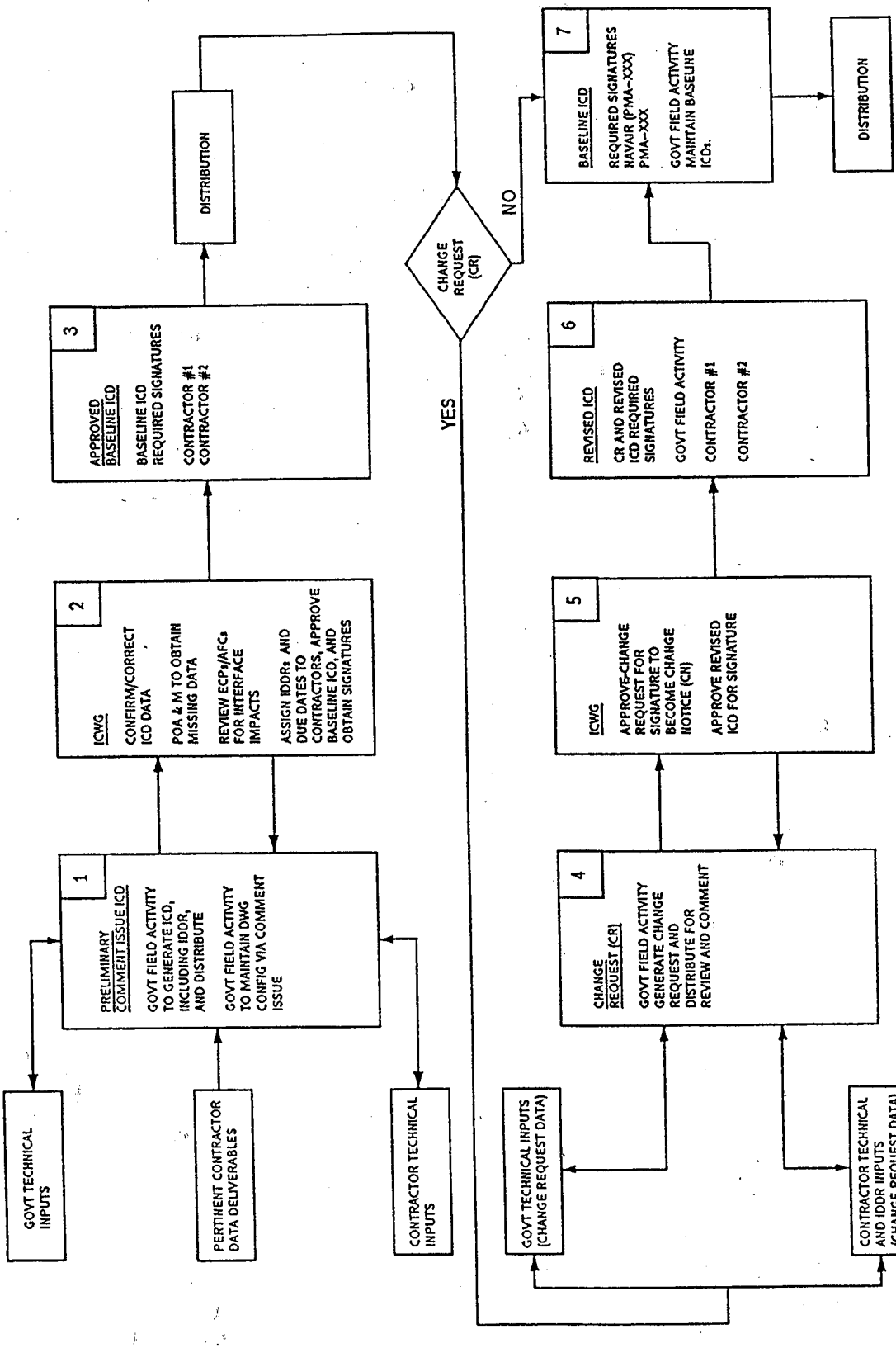


FIGURE 1. ICD development and management process.

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4.5 4 Preliminary. Documents stamped "PRELIMINARY" are first drafts. These documents are generally not for external distribution but may, at times, be informally presented as working papers to quickly show the data required. Preliminary documents are the very first versions of the documents being prepared. Upon reaching a subjective degree of maturity, the documents become comment issues.

4.5.5 Comment issues. Documents stamped "COMMENT ISSUE #1" are the first issues of the document provided for comments. They are as yet not agreed to by the two contractors or contractor and airframer, and are actually working documents. Comment issues are numbered for control. A comment issue may be updated at the government field activity's discretion to have the most up-to-date document available.

Comment issues are presented for comment and approval by the contractors and airframers.

4.5.6 Required signatures for released documents. The ICD is signed by accountable representatives from each of the two contractors or contractor and airframer. It is in the program's best interest to have early agreement for the baseline ICD with unknowns assigned due dates for completion. If changes to the ICD are required, the procedures in blocks 4, 5, and 6 shall be followed. Otherwise, proceed to the NAVAIR PMA for signatures. (Two PMA signatures may be required if the contractors and airframers have different contract managers. Refer to block 7 of figure 1.)

After the ICD is signed by all parties, the designated government field activity prepares it for distribution. At this time the document is released, under strict configuration control, and can be changed only by using the procedures of 4.9.

4.5.7 Interface Coordination Working Group meetings. The Interface Coordination Working Group (ICWG) meetings are convened primarily to obtain all parties' concurrence of the previously coordinated ICD and obtain the required signatures (refer to block 2 of figure 1). The ICWG procedures are outlined in appendix A.

4.6 Interface Design Data Requirement (IDDR). IDDRs are used in ICDs to identify missing data and potential interface conflicts. The detailed requirements for IDDRs are specified in 5.2.

4.7 ICD document structure. ICDs are to be structured to be independent of each other. The format of these documents should be kept as simple as possible while outlining the ICD requirements.

4.8 ICD data content. The ICD should contain all data that will impact the interface being described by the ICD. Possible areas of concern include, but are not limited to, hydraulics, thermodynamics, antenna patterns, RF interfaces, safety hazards, etc. Examples of RF interfaces are provided in appendix B.

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4.9 Change procedures. The change procedures are applicable to released ICDs. Blocks 4, 5, and 6 of figure 1 show a flow diagram of the change procedure.

4.9.1 Change request. Users submit a draft change request (CR) with justification, to the government field activity. The CR is the first step of stringent configuration management control. The government field activity prepares, coordinates and distributes the CR for review and comment. Within two weeks, each contractor or airframer identifies any cost and schedule impacts, along with any comments.

If necessary, the government field activity incorporates comments into the CR and redistributes it to the two contractors or contractor and airframer. If any additional changes are identified, the CR is updated, and this cycle is repeated. The CR is only a request to change the ICD; this request does not imply concurrence even if accepted by the two contractors or contractor and airframer until it becomes a change notice (CN).

4.9.2 Standard CR form. Blank pages of the CR are contained on figures 2, 3 and 4 (change request cover sheet, the first sheet of the change request and follow-on sheet). The circled numbers indicate the information that will be in each area. These numbers are defined in accordance with the following and may appear on all three pages:

1. Short title of the change.
2. The number of the change (of which the first characters are the document number).
3. & 4. The two contractors or contractor and airframer.
5. The PMA sponsor(s).
6. The ICD preparer who made the request.
7. The title of the document proposed to be modified.
8. The number of the document proposed to be modified.
9. The revision number of the document to which the change pertains and the revision number in which it will be incorporated.
10. Identifying federal code of contractor or airframer originating the ICD.

A sample of the CR and follow-on pages are contained on figures 5, 6, and 7.

4.9.3 Change notice. A CR becomes a CN after it is signed by the two contractors (or contractor and airframer) and the government. The CN is the official notification that a change has been approved and becomes effective upon receipt from the government.

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The ICD is changed to the next revision in accordance with the CN, on a judgment basis depending on the complexity of the change and the length of time since the last update. A particular ICD revision may contain the accumulation of several CNs. The revised ICD is distributed to all parties. Each ICD contains the CN incorporation history in the upper right-hand corner on the first sheet.

The approved CN looks the same as the CR except the words "CHANGE REQUEST" are replaced with the words "CHANGE NOTICE."

4.10 Statement of work. The statement of work defines the efforts required by a contractor or airframer to support the development and maintenance of an ICD. (Specific requirements for information to be included in the statement of work are given in 5.5.)

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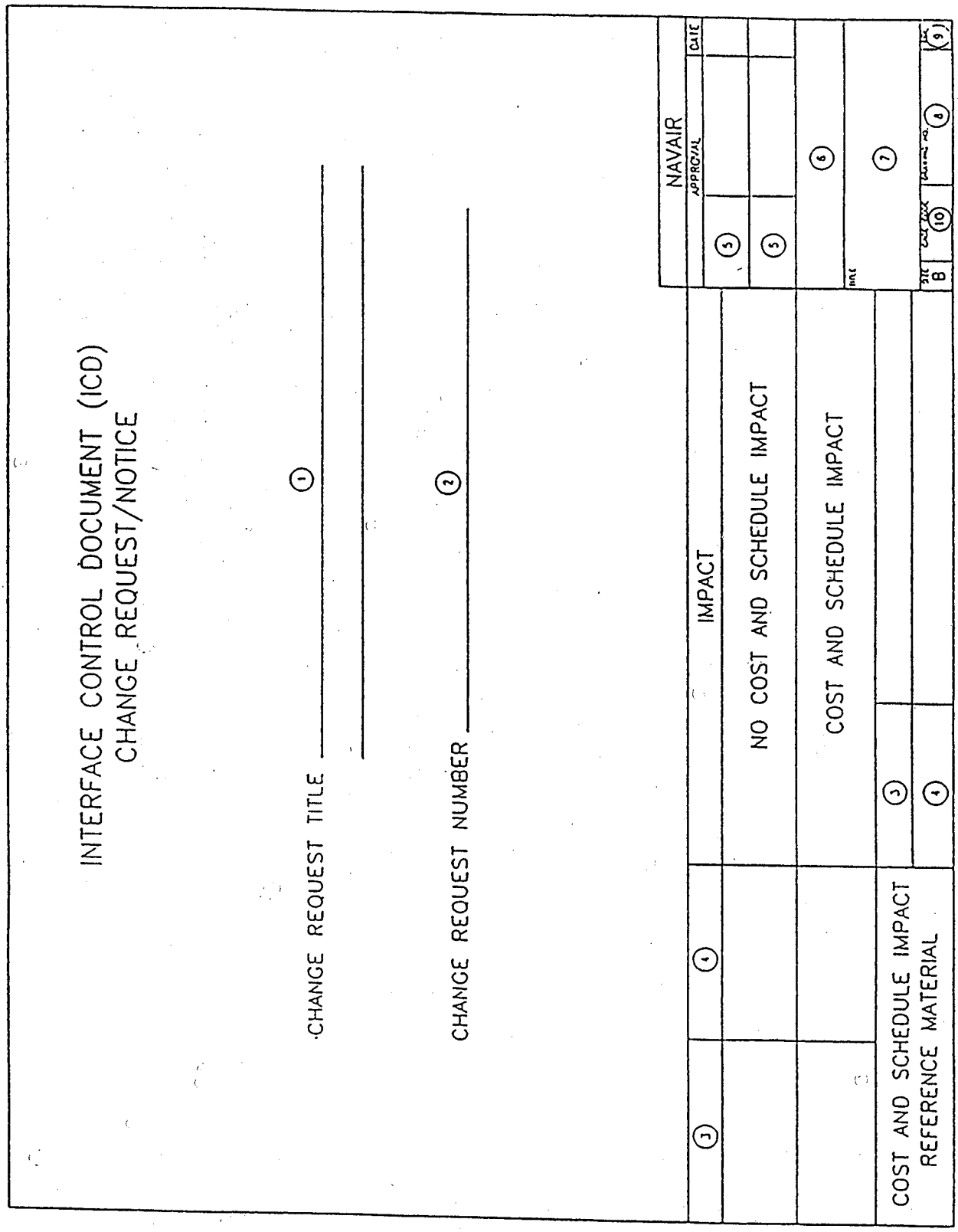


FIGURE 2. Change request cover sheet.

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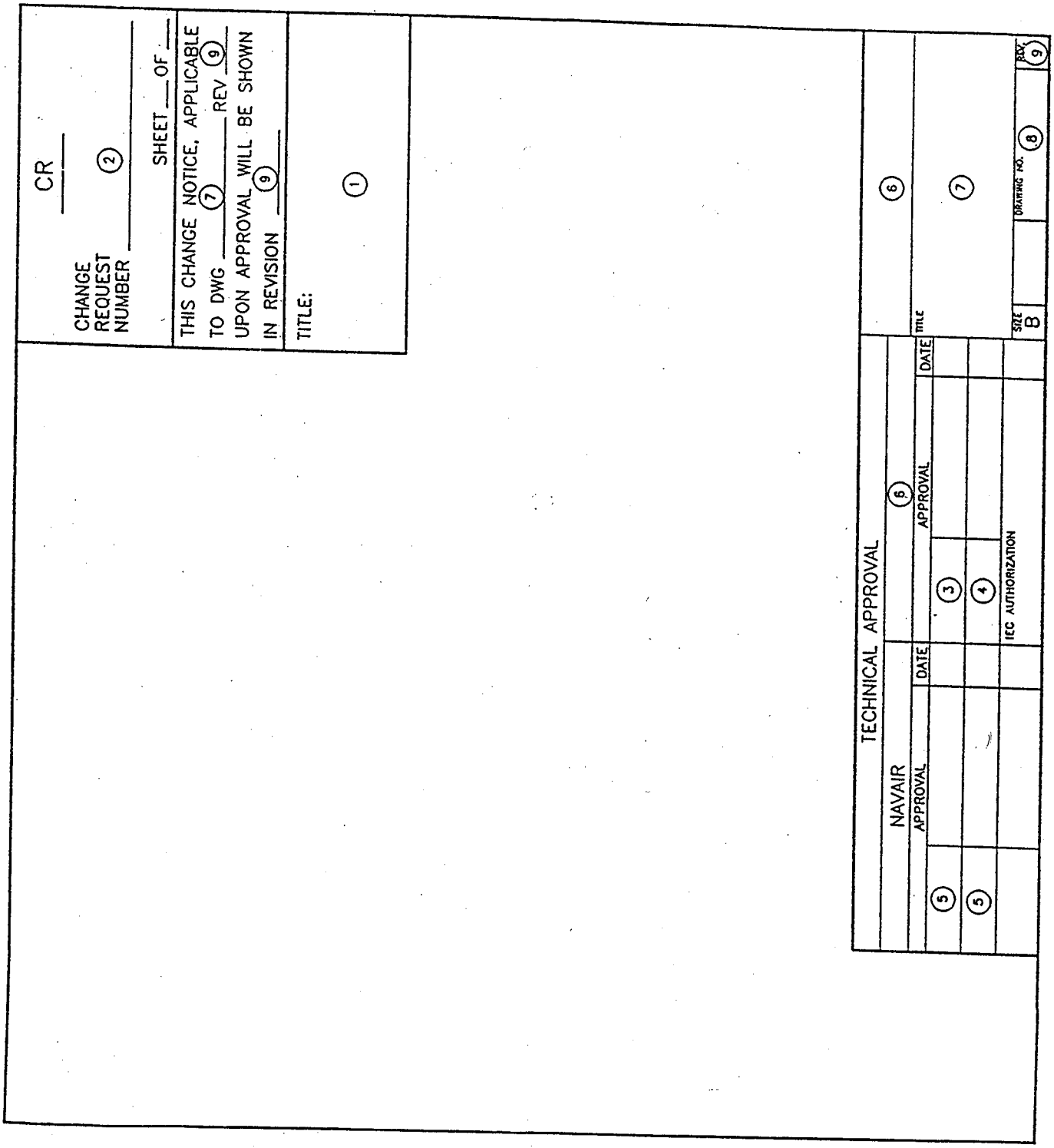


FIGURE 3. First sheet change request sample.

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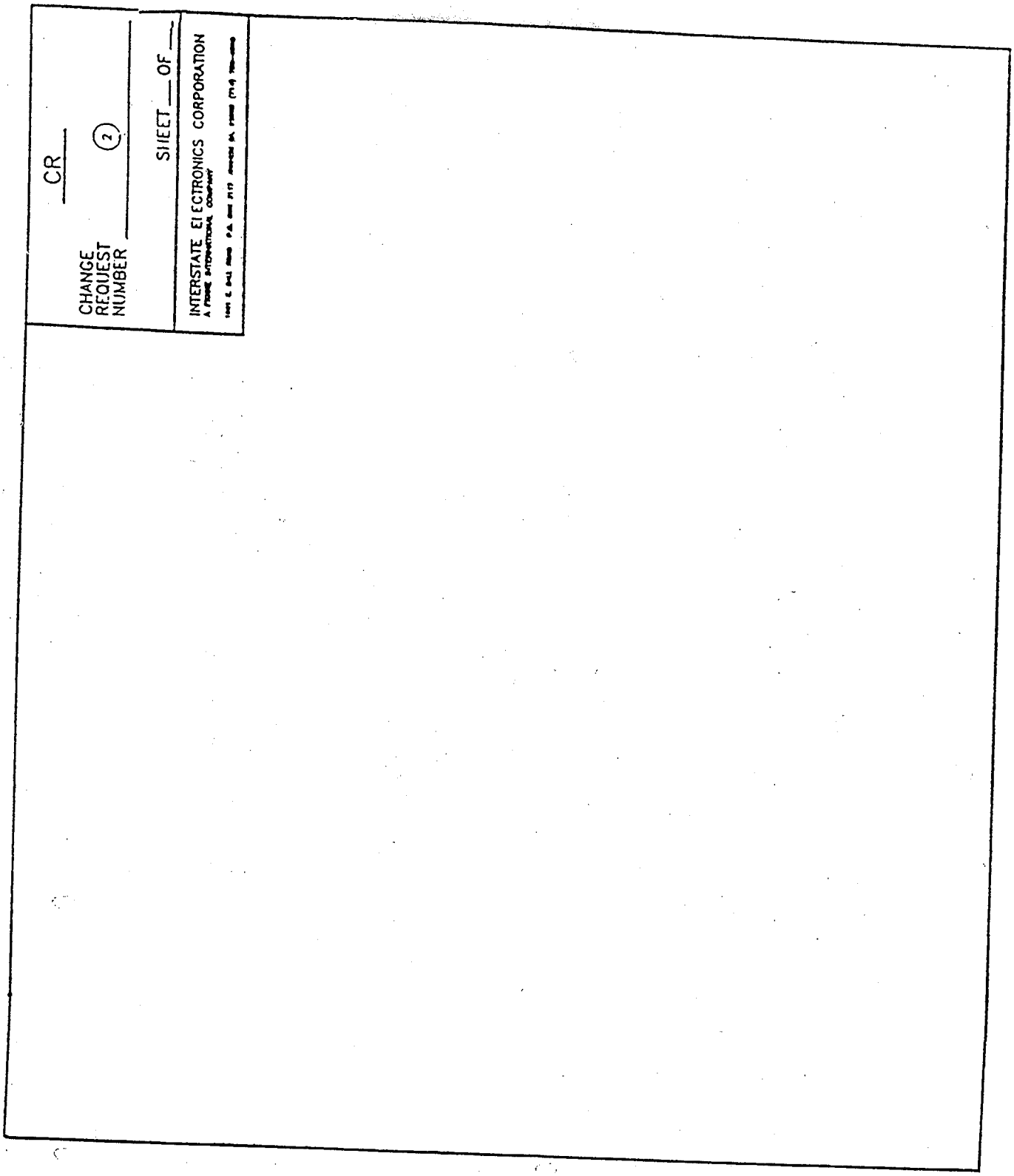


FIGURE 4. Follow-on sheet of change request.

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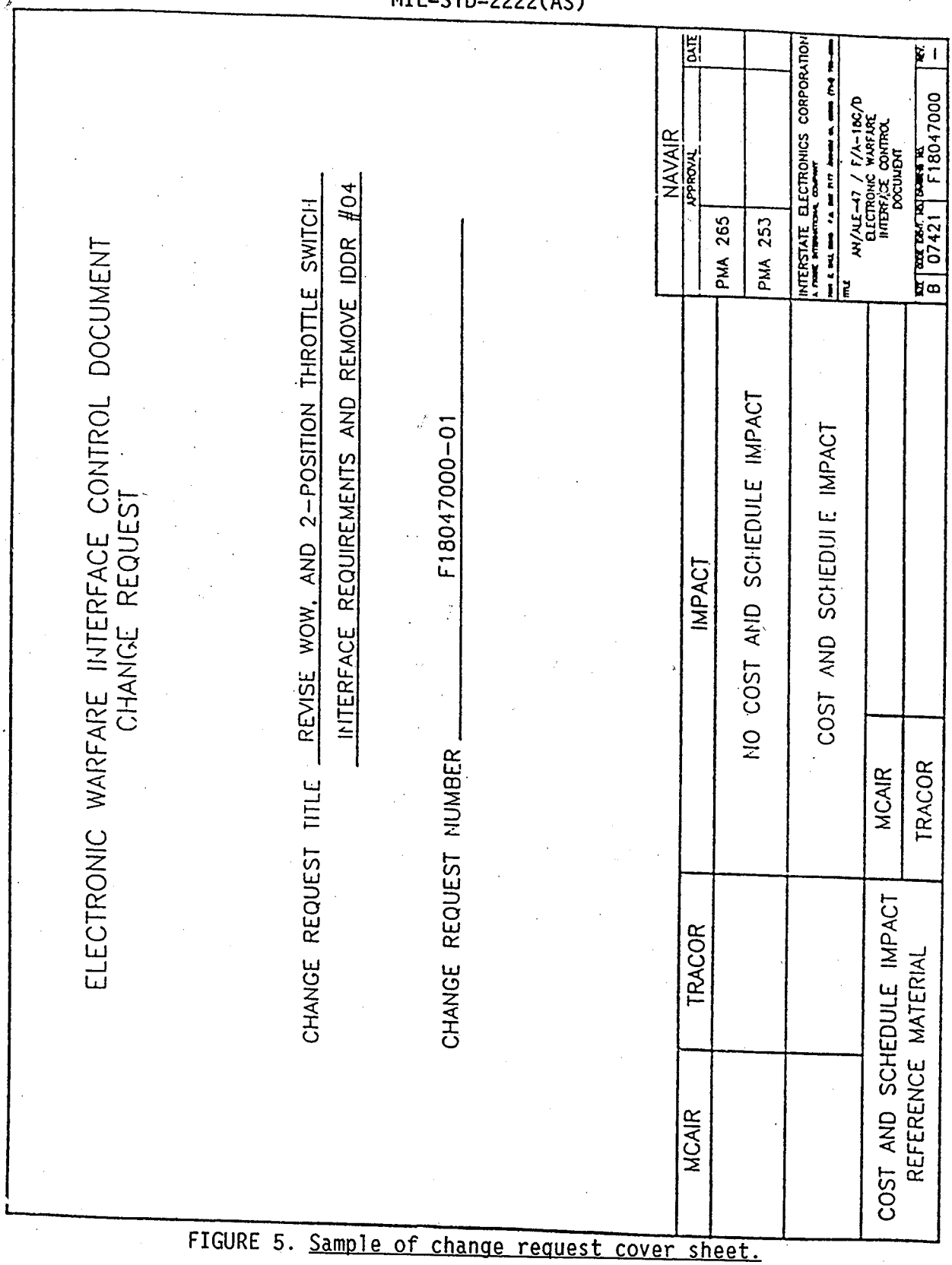


FIGURE 5. Sample of change request cover sheet.

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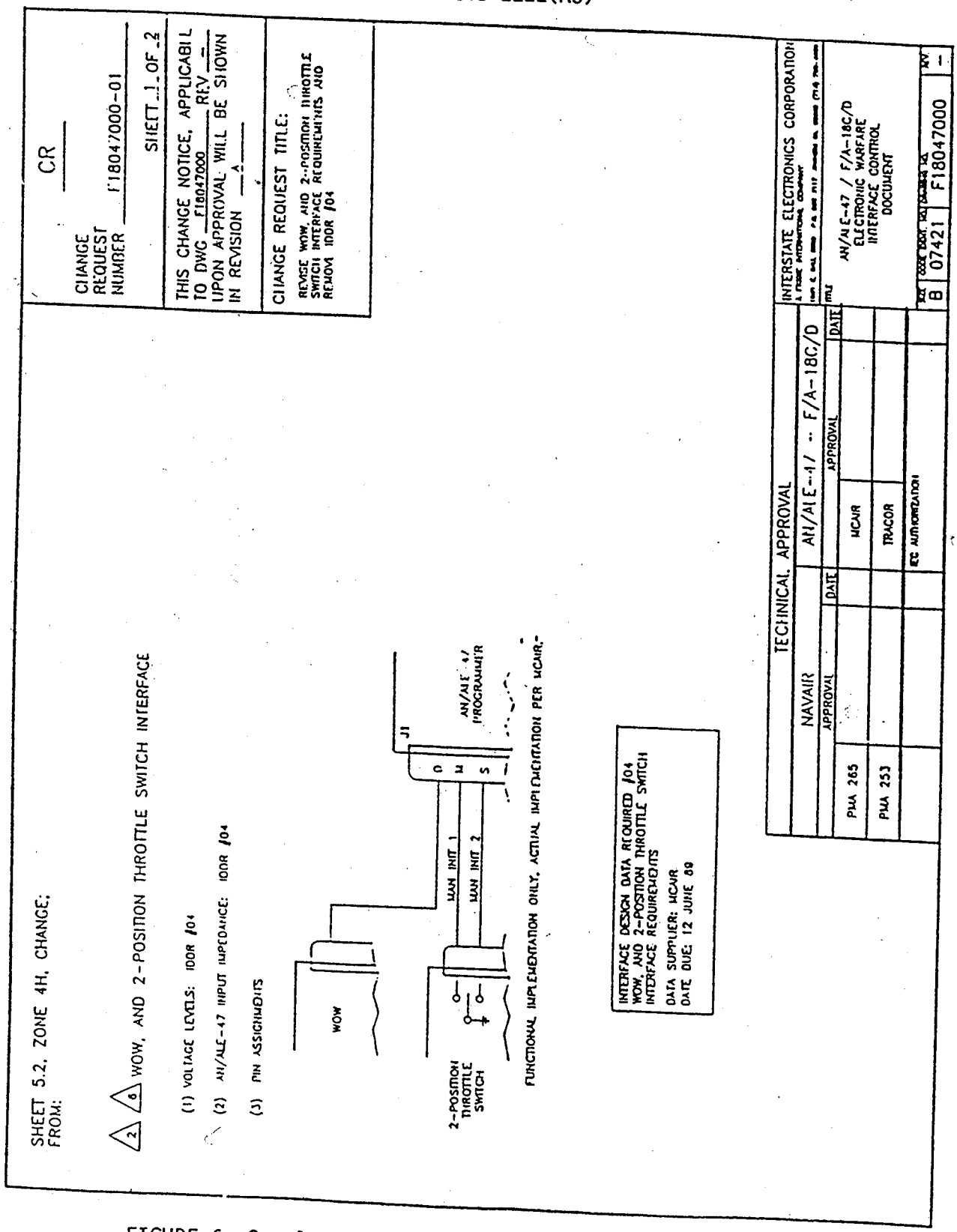


FIGURE 6. Sample of the first sheet of the change request.

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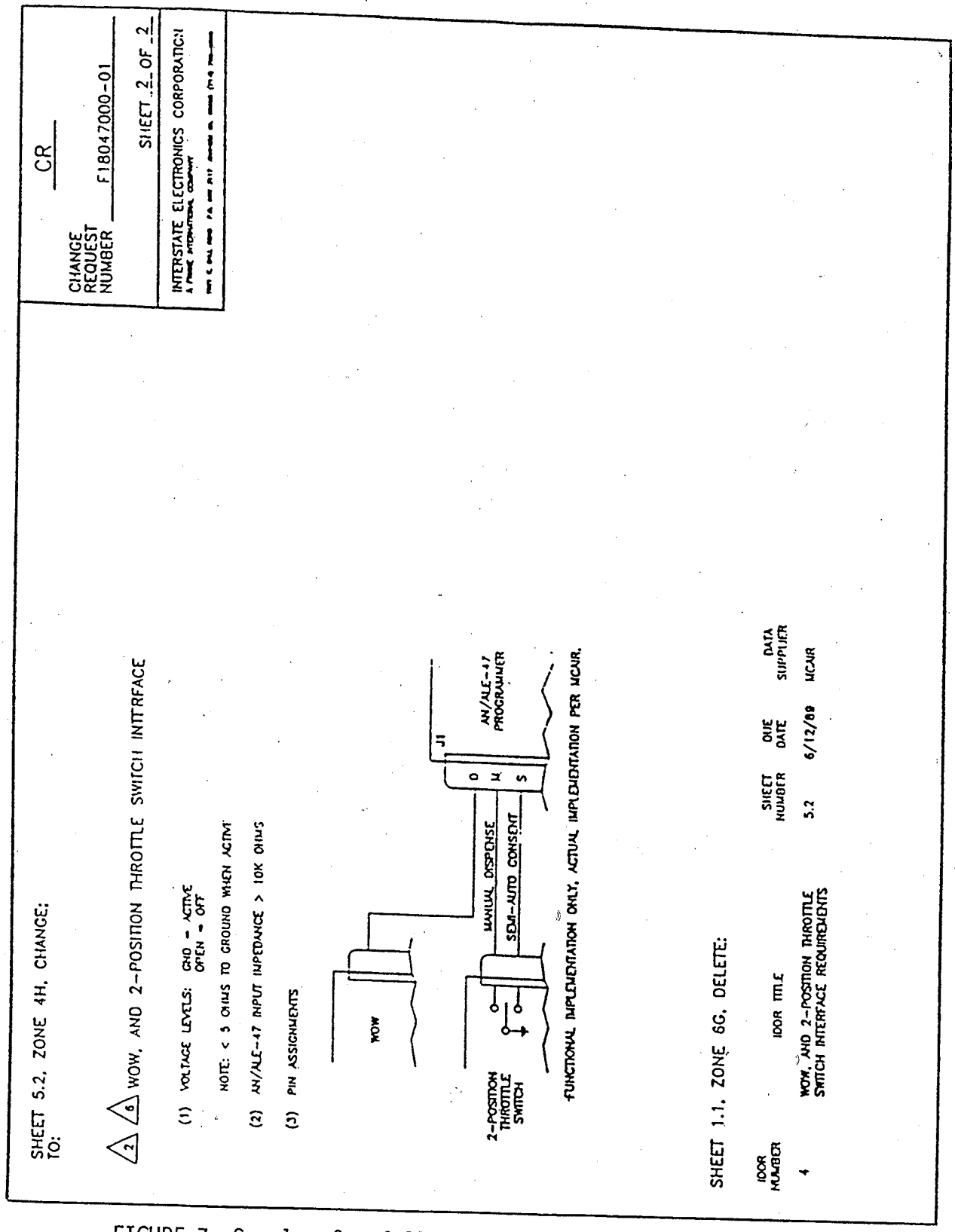


FIGURE 7. Sample of a follow-on sheet of the change request.

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5. DETAILED REQUIREMENTS

5.1 ICD process. In the initial stage of ICD preparation, the government field activity generates a proposed ICD as either a preliminary or comment issue, and submits it to the contractors and airframer for review and comment. Included in this issue are the IDDRs for the system interface being defined. After the ICD is sufficiently mature, the contractors or contractor and airframer sign and it becomes contractual. (See 6.3)

5.2 IDDR document structure. IDDRs identify incomplete or missing data required from each contractor or airframer with assigned due dates. IDDRs highlight potential interface conflicts and problems at an early stage.

5.2.1 IDDR format. The IDDR format allows an earlier ICD release date (agreed to and signed by all ICD parties) because incomplete interface requirements are delineated by IDDRs. The early release date facilitates effective interface control. (The change control procedures in 4.9 ensure that the recipient agrees with the data provided before the IDDR is closed.) All parties know the date each type of data will be supplied by the responsible contractor or airframer.

5.2.2 IDDR origin. Any contractor or airframer can initiate an IDDR by informing the government field activity. The government field activity will coordinate the proposed IDDR with both contractors or contractor and airframer prior to its inclusion. If the ICD is unreleased (unsigned), the next review copy will contain the IDDR. If the ICD is released (signed), a change request for the IDDR will be processed in accordance with 4.9.

5.2.3 Status reporting. The government field activity reports the status of the document, forthcoming IDDRs, and open issues that may require the contractor's or airframer's immediate attention.

5.3 ICD document structure.

5.3.1 ICD numbering method. The method for numbering ICDs is kept as simple and convenient as possible. The ICDs are numbered so that first-time users can usually determine the interfacing systems with only the ICD number. A partial list and samples of ICDs written in this unique format are included in appendix B.

The ICD numbering system uses nine alphanumeric characters. These nine characters actually consist of three contiguous groups of three characters each. The first group is used for specific types of platforms; this group consists of zeros for system-to-system ICDs (not peculiar to specific platforms). The second group is the last (usually) three characters of the military designation for one of the systems (e.g., 047 for the AN/ALE-47 Countermeasures Dispensing System [CMDS]). The third group is for the interfacing system. Therefore, ICD F18047000 contains all interfaces of the AN/ALE-47 installed on the F/A-18C/D. (If the installation varies between the F/A-18C/D and F/A-18E/F aircraft, the unused group of characters can be used for the specific installation, e.g., ICD F180470CD or F180470EF.)

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5.3.2 Document structure. The platform ICDs control the suite of systems' interfaces to the specific platform. The interface responsibility sheet of these ICDs contains a list of related ICDs for other avionics subsystems also contained in that particular aircraft suite.

The ICDs describing the interfaces between the various suites of equipment are common in that a particular ICD (e.g., ICD 000047067 for the AN/ALE-47 CMDS and the AN/ALR-67 Radar Warning Receiver [RWR]) is applicable for all aircraft with those two subsystems on board).

Each ICD is a stand-alone working document and is not dependent on any other ICD; however, a change in one ICD may have a direct impact on other ICDs. Therefore, "flags" are incorporated in all ICDs to identify these areas. Hence, if an area marked with a flag is desired to be changed, all affected ICDs must also be changed.

5.4 ICD data content. This describes the types of data to be incorporated in the ICDs. Data is not restricted to that described herein. Any other interfaces (e.g., hydraulic, thermodynamics, antenna patterns, RF interfaces, RF transmission specification, safety hazards, operational constraints) as required by either contractor or airframer will be incorporated.

The ICDs included in appendix B will be referred to throughout the remainder of this paragragh.

5.4.1 Common items. All "B" size sheets of the document are bordered with zone areas identifying sections on each sheet. The lower right-hand corner of each sheet contains the document number, the revision number and the sheet number. The lower right-hand corner also contains a description of the type of information found on that sheet.

5.4.2 IDDRs, technical approval, and design specification differences table. The first sheet of every ICD shall contain a table of all current IDDR items, a table of design specification differences, and a technical approval signoff block. This sheet and all other sheets listed in 5.4.2 through 5.4.6.7 shall be "A" size. The information described in 5.4.6.2 through 5.4.6.4 shall be contained on ICD sheets numbered i to ix. The information described in 5.4.6.5 through 5.4.6.7 shall be contained in the ICD sheets numbered 1.1 to 1.x.

5.4.2.1 IDDR table. The IDDR table provides a summary of the data needed to make the ICD a complete document. This table shall contain columns for each IDDR number, a short title of the missing data, the sheet number(s) containing details of the required data, the date by which the data must be supplied, and the name of the contractor or airframer supplying the data. The table may be expanded to as many sheets as necessary to contain the required information (see page 34).

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5.4.2.2 Technical approval. The technical approval block, located on the first sheet of each ICD, is the area where each party signs the ICD prior to release of the document as described in 4.5.6 (see page 34).

5.4.2.3 Revisions. The revision of the ICD shall appear on the right top or right bottom of each page for the established ICD document (see page 35).

5.4.2.4 Design specification differences table. This table (if applicable) shall highlight all the design specification differences that may cause incompatibilities between the two contractors or contractor and airframer. This table informs the user where additional information can be found within the ICD. The design specification differences are detailed in addendum B of the ICD (see page 123).

5.4.3 Completed IDDR table. The sheet(s) following the IDDR table shall be the completed IDDR table. This table provides information on IDDRs that have been satisfied and closed out. This table in effect provides a "history" of all IDDR items in the ICD. The table shall contain columns for each IDDR number, the title of the IDDR, the sheet number(s) containing details of the IDDR, the date the data was supplied, the name of the contractor or airframer supplying the data, and the acceptance of the data (e.g., satisfied, cancelled, etc.) (see page 35).

5.4.4 Effective sheets. This sheet shall contain columns that reflect changes made, the change in effect, the ICD preparer who made the changes, and the date changes were issued (see page 36).

5.4.5 Change request record. The change request record shall contain a list of the requested changes to the ICD. This sheet shall be incorporated with the effective sheets section of the ICD. Information on change procedures, including the change request, is given in 4.9. This sheet shall contain the change request number, the date of the change request, title and description, the sheet numbers in the ICD that are affected, and the ICD preparer who made the changes (see page 37).

5.4.6 Document structure.

5.4.6.1 Table of contents. The table of contents provides a paragraph number, title and a sheet number associated with each major section of the ICD. The table of contents documents the layout and order of the ICD (see page 38).

5.4.6.2 List of tables. The list of tables provides a listing of all of the tables contained in the ICD. This list shall be incorporated into the table of contents section of the ICD (see page 39).

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5.4.6.3 List of abbreviations and acronyms. The sheet(s) following the table of contents shall contain a list of abbreviations and acronyms. This list shall contain a column for each abbreviation and acronym used within the ICD along with another associated column containing the meaning of the abbreviation and acronym (see page 40).

5.4.6.4 Applicable documents. The sheet(s) following the list of abbreviations and acronyms shall contain a listing of all documents (e.g., current issue of Government and non-Government documents) which apply to the ICD (see page 42).

5.4.6.5 Introduction. The sheet(s) following the listing of applicable documents to the ICD shall be the introduction section to the ICD. This section shall contain three subsections, the first of which contains the scope of the ICD and a listing of all associated ICDs. The second subsection shall contain a description of the subsystem(s) or airframe composing the interface. The third subsection is for platform (airframe) ICDs only and shall contain the effectivity of the aircraft (see page 43).

5.4.6.6 Notation. The notation section shall contain information on how to read the symbolic and tabular nomenclature contained within the ICD (see page 43).

5.4.6.7 Requirements. The sheet(s) following the notation section shall contain the specific information on the requirements that apply to the interface. Platform ICDs will contain military requirement references for all environments the aircraft is expected to encounter. Weight and center of gravity requirements, if required, are included (see page 44).

5.4.6.8 Functional block diagram. Optional functional block diagrams may be included. These block diagrams are not contractual but are provided to supply bus architecture information. If there is no functional block diagram required for a particular ICD, the functional block diagram will be listed as not applicable (N/A) in the table of contents (see page 49).

5.4.6.9 System interface block diagram. The system interface responsibilities block diagram depicts the contractors' and airframer's responsibilities (see pages 51 and 139). The contractors' and airframer's systems are separated by vertical dotted lines to illustrate the two sides of the interface (see page 51). The systems (or platforms) are written at the top of the corresponding side of the sheet.

The interface block diagram is the "road map" to the rest of the document. It is used as a very quick means of getting to a particular interface and can be used in lieu of the table of contents.

All interfaces are assigned a unique number in a triangle (refer to the notation section for their definition). These interfaces are accompanied with rectangles containing the sheet number(s) and zone number(s) where additional information can be found. For two systems that interface indirectly through a third system, reference shall be made only to the appropriate ICD in which the information is contained for the direct system interface (see pages 225 and 227).

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5.4.6.10 Bus and data interfaces. Bus interfaces (e.g., MIL-STD-1553, RS-422, RS-485) are clearly depicted. All options and variables to these interfaces are identified, including bus controllers, remote terminals, data rates, poll rates and header word information. For clarity some of the bus interface parameters from the military standards are also included. Electrical pinouts are also provided (see pages 53 and 141).

When convenient, bus data formats may be included in addenda to the ICD on "A" size sheets. This option may be exercised pending the amount of data involved.

5.4.6.11 Discrete interfaces. Connectors and pinouts are depicted for all point-to-point wiring on the interface wiring diagram sheets. All required signal information pertaining to the interfaces is provided. This information includes the voltage levels, impedance, and signal meaning (see pages 57 and 145).

5.4.6.12 Operational description. The system operational functions for the particular system shall be clearly described in the ICD. It is not limited to the information that pertains to the interfaces or interaction between the system to another system or airframer (see page 65).

5.4.6.13 Software.

5.4.6.13.1 System software configuration. The system software configuration shall provide the version of the baseline software loaded and the various mission data files permissible for use. Built-in-Test (BIT) operation is also briefly described (see page 67).

5.4.6.13.2. Bus and data interface software messages. Bus and data interface software messages shall be clearly depicted in Addendum A of the ICD. This definition defines the avionics and electronic warfare bus interface command, data, and the status messages for a Bus Controller (BC) to a Remote Terminal (RT) (see page 89).

5.4.6.14 Mechanical interfaces. Mechanical interfaces are required on all platform ICDs. Information is provided to fully define the form factors including mounting methods and all tolerances. All platform ICDs that use common systems will have flags associated with the mechanical interfaces to indicate that if the system supplier makes a change, all platform ICDs are affected (see page 69).

5.4.6.15 Power. The power requirements will be documented for all applicable equipment on platform ICDs. These requirements will include the military standard reference and the power specifications for each Weapon Replaceable Assembly (WRA). Pinouts are also provided (see page 83).

The power requirements will include operating environment constraints including maximum power use limits, momentary power interruption operation, and the grounding and bonding requirements.

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5.4.6.16 RF transmission specifications. Platform ICDs shall contain the RF transmission requirements for all applicable installations. These requirements shall be included in the block diagram defining the RF transmission line, transmit or receiver line type, connector types, and length (see pages 243 and 245).

5.4.6.17 Equipment location. Platform ICDs will contain a sketch of the aircraft with the approximate location of each WRA (see page 85).

5.4.6.18 Cable and connector specifications. A table will show all connector part numbers and the wire types for that interface. The cables will be correlated with the interfaces identified in the system interface block diagram and with the connector designations (e.g., J1, J2) (see page 87).

The sheets listed in 5.4.6.8 through 5.4.6.18 shall be "B" size sheets (except 5.4.6.13.2 shall be an "A" size sheet).

5.5 Statement of work.

5.5.1 Applicability. This section provides information for equipment and aircraft manufacturers to define the scope of effort required to support the ICD-designated government field activity. ICD samples are included in appendix B.

5.5.2 ICD preparation. The government field activity will prepare all preliminary, comment issue and released ICDs. The contractors and airframer who sign the ICD and the changes to it, shall meet with the government field activity to initially define the interface requirements (including IDDRs), review the comment issues (which may require several iterations depending on the complexity), initiate the requests for changes to released ICDs (by informing the government field activity), and participate in meetings to accomplish the above.

5.5.3 Data for ICDs. Each contractor or airframer is obligated to supply the government field activity with data. The data required is all data needed to fully define the interfaces of 4.1. This data is not expected to be available immediately, especially for new system development. IDDRs (see 4.6) define the needed data for each specific type of unavailable data. Data is required for each interface. A clear determination of the ICDs for which data is expected should be obtained from the government Contracting Officer.

5.5.4 Method of transmittal. Data can be submitted via any practical means (e.g., telephone conversations, Fax, mail, FedEx, PC electronic mail system). The method of transmittal depends on the urgency and the extent of the change. Often a telephone call will suffice; at other times a red-lined markup is more efficient.

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5.5.5 ICD ICWG meetings. ICWG meetings will address several ICDs and are planned to be "agreement" meetings. The ICDs will have been previously reviewed by the contractors or airframer. The ICWG procedures are discussed in appendix A. The ICWG meeting can also be held via video television conferences.

5.5.6 Coordination cycle. Particular steps need to be taken once an ICD is determined to be required. A typical coordination cycle is provided in table I.

TABLE I. Typical coordination cycle.

Event	Time After ICD Go-Ahead
1. Meeting: A. Define Interface Requirements B. Identify IDDRs	4 to 8 weeks
2. Supply Data	8 to 12 weeks
3. Review Comment Issue #1	12 to 16 weeks
4. Repeat Items 1, 2 and 3 as required	4 to 16 weeks after prior item
5 Participate in ICWG meetings	As required in coordination cycle
6. Approve ICD	8 to 20 weeks

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Interface control documents conforming to the requirements of this standard are intended for use in NAVAIR applications. Samples I-IV in appendix B are samples of different types of ICDs and are included only for guidance. Addenda A and B apply only to sample I of appendix B.

6.2 Issue of DoDISS. When this standard is used in acquisition, the issue of the DoDISS to be applicable to this solicitation must be cited in this solicitation (see 2.1.1).

6.3 Data Requirements. The following Data Item Description (DID) must be listed as applicable, on the Contract Data Requirements List (DD Form 1423) when this standard is applied on a contract, in order to obtain data, except where DOD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
5.1	DI-CMAN-81248	Interface Control Drawing Documentation	

The above DID was cleared as of the date of this standard. The current issue of DOD 5010.12-L, Acquisitions Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared, DID's are cited on the DD Form 1423.

6.4 Subject term (key word) listing.

Avionics
Change request
Data exchanges
Design specification differences table
Interface coordination working group

Preparing Activity:
Navy-AS
(Project GDRQ-N109)

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MIL-STD-2222(AS)

APPENDIX A

ICWG PROCEDURES

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APPENDIX A

10. SCOPE

10.1 Scope. This appendix defines the role, responsibilities, and functions of the ICWG in the ICD process. This appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. Role of the ICWG. The ICWG is the official communications link between affected organizations in interface definition and problem resolution, documenting interface agreements, and coordinating ECPs, AFCs, CRs, and IDDRs. The ICWG shall consist of at least one member from each contractor and airframer, and government agency participating in the system development. A roster of all affected parties represented at the ICWG will be maintained by the government and this information will be provided to all participants. System interface control is maintained by the ICWG through the identification and definition of interfaces and the scheduling, preparation, approval, and control of the ICDs. The ICDs are maintained by the government field activity.

30.1 ICWG responsibilities. Contractors, airframers and the government agency responsibilities for the ICWG are shown in table II. The contractors and airframers have the prime responsibilities as indicated.

30.2 Coordination. The government will coordinate ICDs with all affected contractors and airframers through the ICWG. Copies of ICDs are provided to applicable parties prior to the ICWG meetings. All requests for information needed to develop, test, install, modify, or verify system interfaces will be referred to the government. Each contractor and airframer will designate a primary point of contact for ICWG activities and action items. The government field activity will maintain a list of all specified points of contact, including government counterparts, and provide this list to the government ICWG co-chairs.

30.3 Exchange of technical information.

30.3.1 Protection. The technical data necessary to define interfaces must be exchanged among affected ICWG members. At the same time, the classified and proprietary information (limited or restricted rights) must also be respected. Classified information, data, and documents will be handled and controlled in accordance with Industrial Security Manual (ISM) security directives and regulations, the provisions of DOD 5220.22-M, 5220.22-S, and provisions of all applicable contracts and security guidance (DD-254). Proprietary information will be controlled and safeguarded according to Federal Acquisition Regulation (FAR) 7-104 and the applicable contracts, and according to the Associate Contractor Agreements (ACAs) between ICWG members.

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30.3.2 Associate Contractor Agreements (ACA). ACAs will be established between airframers and manufacturers of the associated avionics, electronic warfare equipment, selected expendables, and support equipment (SE) including automated test equipment (ATE). These ACAs represent agreements to exchange technical information and data required to define interfaces.

30.3.3 Disclosure. Technical data and information generated under avionics programs may be subject to export control, including disclosure to foreign nationals, whether such data is provided orally or in written form. Therefore, contractors and airframers will obtain written government approval before assigning any foreign national to perform work or before granting foreign nationals access to avionics system information or data.

TABLE II. Matrix of interface control responsibilities.

Responsibilities	1	2
Provide interface design data required for ICD completion	X	
Assign responsibilities	X	X
Establish policy and procedures and chair ICWG		X
Administrative control and processing ICDs		X
Prepare ICD authorized by Acquisition Activity		X
Assist in resolution of interface problems	X	X
Establish complete interface control requirements	X	X
Use specifications and drawings to identify interfaces	X	X
Review interface recommendations with ICWG representatives	X	X
Request Acquisition Activity direction as required	X	
Assign ICD preparation responsibility		X
Obtain ICD agreements and signatures		X
Final ICD review and approval	X	X
Schedule and document ICWG actions and updates		X
Issue meeting agenda (10 working days before meeting) and minutes (10 working days after meeting)		X
Participate in and support ICWG meetings	X	X
Identify and evaluate CR changes for impact	X	X
Evaluate problems and define action items	X	X
Assign action items		X
Prepare and distribute approved ICDs		X
Maintain file on ICDs		X
Maintain ICWG activity status		X
Conduct fit checks and compatibility tests	X	

1 - Contractors and Airframer
2 - Government Agencies

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30.3.4 Publication. Publication or distribution of any information relating to avionics systems, either classified or unclassified, requires prior written clearance from the government. News releases, including photographs and films, public announcements, denial or confirmation of same on any part or phase regarding the subject matter of this document requires written government approval.

30.4 Organization.

30.4.1 Chair. The government shall chair all ICWG meetings. The government shall prepare the ICWG agenda with inputs from contractors and airframers. Contractors shall field and initiate interface pertinent dialogue in accordance with the meeting agenda.

30.4.2 Secretariat. The government field activity serves as the ICWG Secretariat. The Secretariat prepares agenda, coordinates meetings, prepares meeting minutes, and tracks action items.

30.4.3 Contractor members. Each contractor and airframer designates a senior representative with the authority to represent the organization's position at ICWG meetings. These individuals should also be the designated interface points of contact.

30.4.4 Customer representatives and other members. Customer representatives and other manufacturer and contractor ICWG members attend ICWG meetings at the invitation of the Chair.

30.5 Change approval and disapproval. Change procedures for released ICDs are described in 4.9. The status of CRs and CNs are summarized at each ICWG meeting.

30.5.1 Disagreements. If an ICWG member disagrees with the document, that member should immediately notify the government. If disagreements cannot be resolved by consensus (and within scope), the ICWG chair analyzes the issue and makes a ruling. Copies of the approved decision are provided to the affected parties for implementation.

30.5.2 Change implementation. Once a proposed CR is approved by the ICWG, the affected ICWG members have the responsibility to implement the resulting CN.

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APPENDIX B

PARTIAL LIST AND SAMPLES OF U.S. NAVY ELECTRONIC WARFARE SYSTEMS ICDs

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APPENDIX B

10. SCOPE.

10.1 Scope. This appendix gives examples and samples of different types of ICDs and is not a mandatory part of this standard. The information contained herein is for information only.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30.1 ICDs for different types of aircraft. ICDs are being written for three classes of aircraft: tactical, helicopter and patrol. Each specific aircraft should have an ICD associated with it for each avionics system onboard. ICDs also exist between all avionics systems that have interfaces between them.

30.1.1 ICDs for tactical aircraft. Examples of ICDs for integration of avionics systems in different types of aircraft are given below.

F18047000, AN/ALE-47 CMDS to F/A-18C/D (see sample I)
F18067000, AN/ALR-67 RWR to F/A-18C/D
F14047000, AN/ALE-47 CMDS to F-14D
F14165000, AN/ALQ-165 ASPJ to F-14D
F14BOL000, BOL to F-14A/A+/D
F14ASR000, AN/ALR-67 (ASR) to F-14D
E6B047000, AN/ALE-47 CMDS to EA-6B
60H047000, AN/ALE-47 CMDS to HH-60H
V22047000, AN/ALE-47 CMDS to MV-22

30.1.2 ICDs for general avionics systems. Examples of ICDs for general avionics systems are given below. These ICDs are applicable to all aircraft as appropriate.

000047067, AN/ALE-47 CMDS to AN//ALR-67 RWR (see sample II)
000047126, AN/ALE-47 CMDS to AN/ALQ-126B Jammer (see sample III)
000047165, AN/ALE-47 CMDS to AN/ALQ-165 ASPJ
000047E50, AN/ALE-47 CMDS to AN/ALE-50
000047CLC, AN/ALE-47 CMDS to HARM CLC
000047BOL, AN/ALE-47 CMDS to BOL
000047039, AN/ALE-47 CMDS to AN/APR-39A(XE-2) RWR
000047R47, AN/ALE-47 CMDS to AN/AAR-47 MWS
000047144, AN/ALE-47 CMDS to AN/ALQ-144
000047157, AN/ALE-47 CMDS to AN/ALQ-157
000047GNX, AN/ALE-47 CMDS to GEN-X
000067165, AN/ALR-67 RWR to AN/ALQ-165ASPJ

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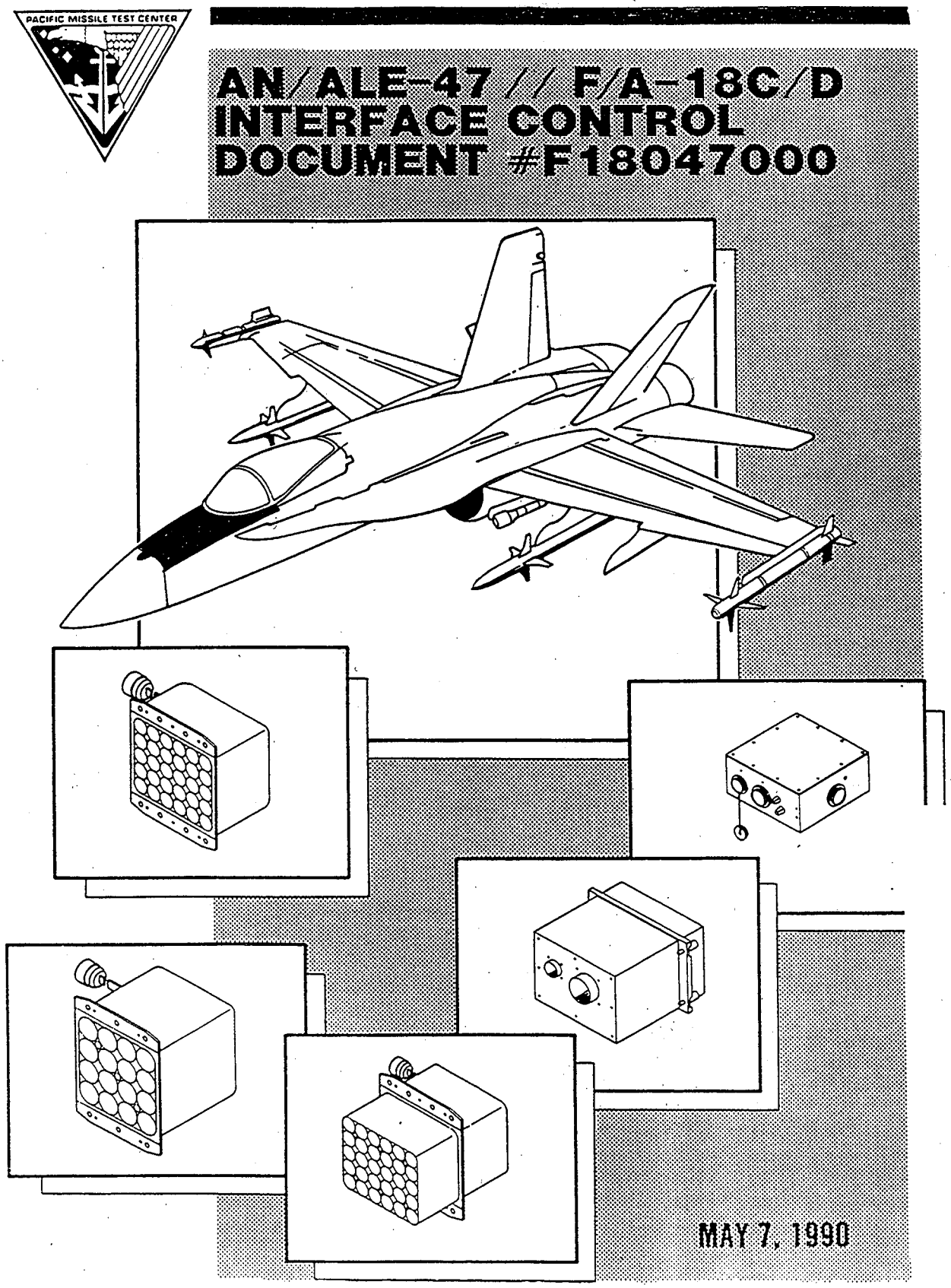
APPENDIX B

SAMPLE I. PLATFORMS-TO-SYSTEM ICD

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APPENDIX B



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APPENDIX B

TABLE IA

INTERFACE DESIGN DATA REQUIRED (IDDR) TABLE					
IDDR NUMBER	IDDR TITLE	SHEET NUMBER	DUE DATE	SUPPLIER	ACCEPTANCE
8	REFERENCE LIST OF INST. DWGS.	11.1	6/30/90	MCAIR	
27	MINI DISPENSER LOCATION	11.1	6/30/90	MCAIR	

TABLE IB

DESIGN SPECIFICATION DIFFERENCES (DETAILS PROVIDED IN ADDENDUM B)	
DESCRIPTION	SHEET
DYNAMICALLY ALTERABLE AVIONICS MUX BUS COMMANDS	B-1

AVIONICS INTERFACE CONTROL DOCUMENT (AVICD)

TECHNICAL APPROVAL					
NAVAIR			AN/ALE-47 // F/A-18C/D		
APPROVAL		DATE	APPROVAL		DATE
PMA 265			MCAIR	René Kranlage	5/7/90
PMA 253			TRACOR		5/10/90

INTERSTATE ELECTRONICS CORPORATION
A FIGGIE INTERNATIONAL COMPANY
1001 E. BALL ROAD P.O. BOX 3117 ANAHEIM CA. 92803 (714) 758-0500

TITLE: AN/ALE-47 // F/A-18C/D AVIONICS INTERFACE CONTROL DOCUMENT

SIZE	CODE IDENT. NO.	DRAWING NO.	REV.
A	07421	F18047000	--

SCALE NONE | REV IN-FILE DATE | SHEET 1

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APPENDIX B

TABLE I.1A

COMPLETED IDDR's					
IDDR NUMBER	IDDR TITLE	SHEET NUMBER	DUE DATE	SUPPLIER	ACCEPTANCE
1	AIRCRAFT BLOCK NO. AND BuNos				CANCELLED
2	ICM CONTROL PANEL PINOUTS	5.1		MCAIR	SATISFIED
3	SYSTEM SOFTWARE CONFIGURATION	ADD. A		TRACOR/MCAIR	PER ADD. A
4	ALE-47 PROGRAMMER FASTENER TYPE	8.1		TRACOR/MCAIR	SATISFIED
5	OPERATIONAL VIBRATION REQ.	8.1		MCAIR	CANCELLED
6	MAG-N, MAG-N1, AND MAG-N2 MAX	8.1		TRACOR	CANCELLED
	FULLY LOADED WEIGHT				
7	AN/ALE-47 EQUIPMENT LOCATION	11.1		MCAIR	SATISFIED
9	CABLE/CONNECTOR SPECIFICATIONS	12.1		MCAIR	SATISFIED
10	REFERENCE LIST OF CABLING DIAGRAMS	12.1		MCAIR	SATISFIED
11	CROSS REFERENCE TABLE	12.1		TRACOR/MCAIR	CANCELLED
12	ERD CONNECTION	3.1		MCAIR	SATISFIED
13	PROGRAMMER GROUNDS (PIN 43)	4.1		TRACOR	SATISFIED
14	INTERFACE UNIT PART NUMBER	4.1		TRACOR	SATISFIED
15	MASCII	4.2		TRACOR	SATISFIED
16	PART NUMBERS	4.1		MCAIR	SATISFIED
17	AIRCRAFT IDENTIFICATION CODES	5.4		TRACOR	SATISFIED
18	IMPLEMENTATION OF EXT. SAFETY SW.	9.1		MCAIR	SATISFIED
19	MLV INTERFACE	5.4		TRACOR	SATISFIED
20	CORRECT CHASSIS GROUNDS	5.3		TRACOR	SATISFIED
21	AN/ALE-47 UNIT TOLERANCES	8.1		TRACOR	SATISFIED
22	LATEST DISPENSER DIMENSIONS	8.3		TRACOR	SATISFIED
23	MAG-N2 DIMENSIONS	8.5		TRACOR	SATISFIED
24	UPDATE SHEETS 5.1 AND 5.2	5.1, 5.2		TRACOR/MCAIR	SATISFIED
25	CMDS PROGRAM RESPONSE	6.1		TRACOR/MCAIR	SATISFIED
26	ACCELERATION: LSB, MSB, RANGE	A-14		MCAIR	SATISFIED

AVIONICS INTERFACE
CONTROL DOCUMENT (AVICD)

SIZE	CODE IDENT. NO.	DRAWING NO.	REV.
A	07421	F18047000	—
SCALE NONE	REV IN-FILE DATE		SHEET I.1

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EFFECTIVE SHEETS

SHEET NUMBERS	CHANGE IN EFFECT	DATE ISSUED
i - viii	ORIGINAL ISSUE	May 7, 1990
1.1 - 1.6	ORIGINAL ISSUE	May 7, 1990
2.1	ORIGINAL ISSUE	May 7, 1990
3.1	ORIGINAL ISSUE	May 7, 1990
4.1 - 4.2	ORIGINAL ISSUE	May 7, 1990
5.1 - 5.4	ORIGINAL ISSUE	May 7, 1990
6.1	ORIGINAL ISSUE	May 7, 1990
7.1	ORIGINAL ISSUE	May 7, 1990
8.1 - 8.7	ORIGINAL ISSUE	May 7, 1990
9.1	ORIGINAL ISSUE	May 7, 1990
11.1	ORIGINAL ISSUE	May 7, 1990
12.1	ORIGINAL ISSUE	May 7, 1990
ADDENDUM A	ORIGINAL ISSUE	May 7, 1990
ADDENDUM B	ORIGINAL ISSUE	May 7, 1990
ADDENDUM C	ORIGINAL ISSUE	May 7, 1990

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CHANGE REQUEST (CR) RECORD

CR NUMBER	DATE	TITLE/DESCRIPTION	SHEETS AFFECTED

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LIST OF ABBREVIATIONS/ACRONYMS

AAED	ADVANCED AIRBORNE EXPENDABLE DECOY
ACI	AIRCRAFT IDENTIFICATION
ASPJ	AIRBORNE SELF-PROTECTION JAMMER
AV	AVIONICS
AVICD	AVIONICS INTERFACE CONTROL DOCUMENT
AVMUX	AVIONICS MULTIPLEX
BIT	BUILT-IN-TEST
BPS	BITS PER SECOND
CASC	CASCADE
CBIT	CONTINUOUS BUILT-IN-TEST
CG	CENTER OF GRAVITY
CH	CHAFF
CSK	COUNTERSINK
CM	COUNTERMEASURES
CMDS	COUNTERMEASURES DISPENSING SYSTEM
DA-N	DISPENSER ASSEMBLY - NAVY (STANDARD DISPENSER)
DA-N1	DISPENSER ASSEMBLY - NAVY 1 (MINI DISPENSER)
DCP	DISPENSER CONTROL PANEL
DISP	DISPENSER
ECM	ELECTRONIC COUNTERMEASURES
EMC	ELECTRO-MAGNETIC COMPATIBILITY
EMI	ELECTRO-MAGNETIC INTERFERENCE
EMP	ELECTRO-MAGNETIC PROTECTION
ERD	EQUIPMENT READY DISCRETE
FL	FLARE
FPS	FEET PER SECOND
FWD	FORWARD
GEN-X	GENERIC EXPENDABLE
GND	GROUND
IBIT	INITIATED BUILT-IN-TEST
ICM CP	INTEGRATED COUNTERMEASURES CONTROL PANEL
IDAP	INTEGRATED DEFENSE AVIONICS PROGRAM
IDDR	INTERFACE DESIGN DATA REQUIRED
IN/HR	INCHES/HOUR
INIT	INITIATED
JMR	JAMMER
LMG	LEFT MAIN GEAR

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LIST OF ABBREVIATIONS/ACRONYMS - Continued

MA	MILLIAMPERES
MAG	MAGAZINE
MAG-N	MAGAZINE-NAVY (STANDARD MAGAZINE)
MAG-N1	MAGAZINE-NAVY 1 (MINI MAGAZINE)
MAG-N2	MAGAZINE-NAVY 2 (EXTENDED MAGAZINE)
MAN	MANUAL
MASCII	MODIFIED ASCII
MDC	McDONNELL DOUGLAS CORPORATION
MDF	MISSION DATA FILE
ME	MESSAGE ERROR
MLV	MEMORY LOADER VERIFIER
MRP	MISCELLANEOUS RELAY PANEL
MOI	MOMENT OF INERTIA
MUX	MULTIPLEX
MWS	MISSILE WARNING SYSTEM
OFP	OPERATIONAL FLIGHT PROGRAM
P	PARITY
PBIT	POWER-UP BUILT-IN-TEST
PN	PART NUMBER
PRGMR	PROGRAMMER
PWR	POWER
RLY	RELAY
RMG	RIGHT MAIN GEAR
RT	REMOTE TERMINAL
RTN	RETURN
RWR	RADAR WARNING RECEIVER
SDL	SEQUENCER DATA LINK
SQ	SQUIB
SW	SWITCH
T/F	TERMINAL FLAG
UDM	USER DATA MEMORY
WOW	WEIGHT ON WHEELS

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APPLICABLE DOCUMENTS

The following documents of the exact issue indicated form a part of this EWICD to the extent specified herein.

MILITARY DOCUMENTS

MIL-A-87221	Aircraft Structures, General Specifications for, Amend 1, 15 Apr '86
MIL-B-5087B	Bonding, Electrical and Lightning Protection, for Aerospace Systems, Int Amend 3, 24 Dec '84
MIL-C-38999H	General Specification for Connectors, Electrical, Circular, Miniature, High Density, Quick- Disconnect, Environment Resistant, Removable Crimp and Hermetic Solder Contacts, Amend 2, 30 Sept '86
MIL-E-5400T	Electronic Equipment, Aircraft, General Specification for, Rev T, 16 Nov '79, Amend 2, 9 May '86
MIL-E-6051D	Systems Electromagnetic Compatibility Requirements, Notice 1, 26 Feb '88
MIL-STD-210C	Climatic Extremes for Military Equipment, 9 Jan '87
MIL-STD-461C	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference, Notice 2, 15 Oct '87
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of, Notice 3, 9 Feb '71 and Notice 6, 15 Oct '87
MIL-STD-704A	Aircraft Electric Power Characteristics, Notice 3
MIL-STD-810D	Environmental Test Methods, Notice 1, 31 Jul '86

McDONNELL DOUGLAS DOCUMENTS

MDC A3818, Rev B	F/A-18 Avionics Multiplex Design Specification

TRACOR DOCUMENTS

CP64314	System Specification for the Prime Mission Equipment, Dispensing System, Countermeasures, AN/ALE-47, 8 March 1989
CP64345	Interface Specification AN/ALE-47 Countermeasures Dispenser System (CMDS) to Aircraft and Associated Equipment

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APPENDIX B

1.0 INTRODUCTION

1.1 Scope

This document is an Avionics Interface Control Document (AVICD) between the AN/ALE-47 Countermeasures Dispensing System (CMDS) and the F/A-18C/D. This document establishes the requirements for data, electrical, and mechanical interfaces.

For information only, additional AN/ALE-47 interface requirements for the F/A-18C/D will be contained in the following documents:

000047025 (AN/ALE-47 // AN/AWG-25 HARM CLC)
000047067 (AN/ALE-47 // AN/ALR-67(V)2 RWR)
000047126 (AN/ALE-47 // AN/ALQ-126B DECM JAMMER)
000047165 (AN/ALE-47 // AN/ALQ-165 ASPJ)
000047687 (AN/ALE-47 // AN/ASM-687 MLV)

1.2 Description

The F/A-18C/D proposed AV suite consists of the AN/ALR-67(V)2 Radar Warning Receiver (RWR), the AN/ALQ-126B DECM JAMMER, the AN/ALQ-165 ASPJ, the AN/AWG-25 HARM CLC, and the AN/ALE-47 Countermeasures Dispensing System (CMDS). The AN/ALE-47 CMDS is a threat-adaptive dispenser system designed to interface with an aircraft's EW suite, determine the nature of incoming threats, and automatically dispense appropriate expendables. The CMDS is designed to be compatible with MDC A3818. The CMDS is compatible with the same payload types as the ALE-39 with greater integration and automation for the pilot. With its data fusion capability from other EW suite sensors and aircraft avionics, threat adaptive dispensing will provide the optimum and most efficient countermeasures deployment, thus freeing the pilot to concentrate on maneuvering techniques and mission goals.

1.3 Effectivity

Applicable to F/A-18C/D flight test aircraft.

2.0 NOTATION

2.1 Symbols

△1 Indicates interface 1. Interface numbers located within triangles throughout this document are referenced to interfaces on sheet 3.1.

2.1 / 4C Indicates additional data shown on sheet 2.1, zone 4C. In the event that related information is spread out over more than one sheet, only the first sheet will be indicated in this manner.

2.2 Table Nomenclature

Table 4.2A indicates the first table on sheet 4.2.

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3.0 REQUIREMENTS

3.1 EMI/EMC Requirements

Electromagnetic compatibility (EMC) requirements shall conform to MIL-E-6051D. Electromagnetic interference (EMI) requirements shall conform to MIL-STD-461C Notice 2 and testing per MIL-STD-462 Notices 3 and 6.

The following MIL-STD-462 test methods are required for installation of the ALE-47 on the F/A-18C/D:

CE03: Conducted emissions, power and interconnecting leads, 0.015 to 50 MHz
CE07: Conducted emissions, power leads, spikes, time domain
CS01: Conducted susceptibility, power leads, 30 Hz to 50 KHz
CS02: Conducted susceptibility, power and interconnecting control leads, 0.05 to 400 MHz
RE02: Radiated emissions, electric field, 14 KHz to 10 GHz
RS02: Radiated susceptibility, magnetic induction field, spikes and power frequencies
RS03: Radiated susceptibility testing shall be IAW Table 1.2A.

Table 1.2A. RS03 Signals

Frequency (MHz)	Modulation	Power Density (mw/cm2) Peak	Ave	v/m(6)
0.014-850	AM 90% 1KHz Sine Wave	-	.106	20
215-225	(1) PAM 10 (2) PW(3) 200 pps(4)	.317	-	35
215-225	PAM 200 (2) PW 200 pps	.317(5)	-	35
400-450	PAM 15 (2) PW 300 pps	.106	-	20
450-850	PAM 10 (2) PW 1000 pps	.106	-	20
850-940	PAM 2 (2) PW 1000 pps	.106	-	20
850-940	PAM 125 (2) PW 300 pps	.106	-	20
940-1215	PAM 10 (2) PW 1000 pps	.106	-	20
1215-1365	PAM 10 (2) PW 3000 pps	.106	-	20
1365-2900	PAM 1 (2) PW 1000 pps	.106	-	20
2900-3100	PAM 3 (2) PW 1000 pps	1.6	-	78
2900-3100	PAM 70 (2) PW 200 pps	1.6	-	78
3100-9000	PAM 1 (2) PW 1000 pps	.106	-	20
9000-14000	PAM 0.2 (2) PW 1000 pps	.10	-	20
14000-18000	PAM 0.2 (2) PW 50000 pps	.106	-	20

Notes: See next page

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Notes for Table 1.2A:
(1) Pulse amplitute modulated
(2) Microseconds
(3) Pulse width
(4) Pulses per second
(5) Peak power density for PAM signals is defined as the power density measured by an average power measuring device if the generator is run CW; i.e., average power density within the pulse envelope.
(6) Volts per meter equivalent shown for reference only.
(7) Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and/or field strength shown.

3.2 Physical Characteristics

The following table summarizes the physical characteristics of each unit of the AN/ALE-47 CMDS installed on the F/A-18C/D.

Table 1.3A CMDS Physical Characteristics

Unit	Quantity/ Aircraft	Maximum Weight (Pounds)
Programmer	1	7
Sequencers	2	5 ea
Dispensers	2/Seq	
DA-N Standard (a)	(2 Total)	5 ea
DA-N1 Mini (b)	(2 Total)	3.5 ea
Magazines	1/Disp	
MAG-N Standard	(2 Total)	10 ea (c)
MAG-N1 Mini	(2 Total)	5.5 ea (c)
MAG-N2 Extended	(2 Total)	17 ea (c)

Notes: (a) Holds 30 expendables/dispenser
(b) Holds 16 expendables/dispenser
(c) Unloaded weight

3.3 Operating Environment

The following subparagraphs describe the various environments in which the CMDS is operational. The environmental parameters along with the requirements associated with each parameter are discussed and referenced to applicable Military Standards.

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3.3.1 Temperature-Altitude
-60°C to +100°C, Non-Operation
-54°C to +71°C, Continuous
+71°C to +95°C, Intermittent (30 min), Non-Cockpit
+71°C to +85°C, Intermittent, Cockpit

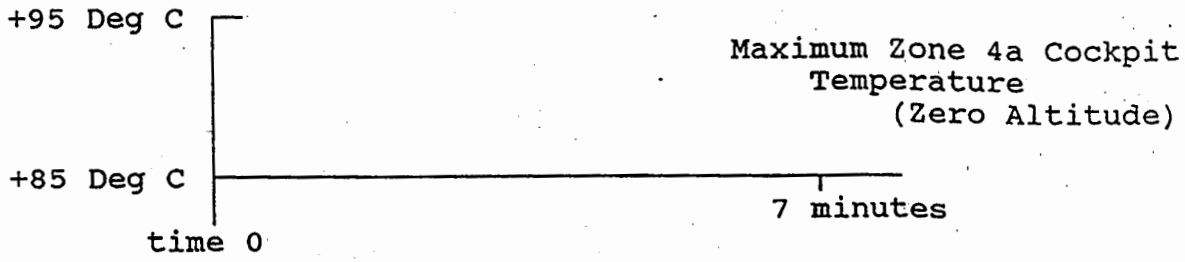


Figure 1.3A. Programmer Temperature - Altitude Operating Environment

3.3.2 Temperature Shock
The CMDS shall withstand exposure to temperature changes of 1.7°C per second throughout the operational temperature range. Test methods shall conform to MIL-STD-810D, Method 503.2.

3.3.3 Explosive Atmosphere
The CMDS shall be designed to operate in an ambient explosive gaseous mixture of air and JP-4/JP-5 without causing ignition, with the exclusion of impulse cartridges.

3.3.4 Humidity
The humidity factor shall conform to MIL-E-5400T, Class 2, and the 45 degree drip-proof requirement of MIL-STD-108E.

3.3.5 Salt-Sea Atmosphere
Salt-sea atmosphere conditions shall conform to MIL-E-5400T, Class 2.

3.3.6 Fungus
The CMDS shall be designed with non-nutrient materials.

3.3.7 Shock
Shock resistance for the system has a designed shock of 20G (shock response spectrum) and crash safety of 40G (shock response spectrum).

3.3.8 Sand and Dust
Sand and dust requirements shall conform to MIL-STD-210.

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3.3.9 Altitude
The CMDS is operational 280 feet below sea level up to and including 50,000 feet.

3.3.10 Salt Fog
Salt fog testing shall conform to MIL-STD-810D, Method 809.2.

3.3.11 Acceleration
The CMDS equipment shall operate at +/- 10G individually along all 3 axes.

3.3.12 Vibration
Vibration specifications shall conform to MIL-STD-810D, Procedure 1A.

3.3.13 Acoustic Noise
Acoustic noise testing shall conform to MIL-STD-810D, Category A, and shall consist of the following:

Programmer: 140 dB
Sequencers: 145 dB
Dispensers and Magazines: 160 dB

3.3.14 Normal Design Loads
The CMDS shall meet performance requirements without any damage or set during normal operating load factors in the range between 10.0G download and 4.1G upload combined with 3.0G forward and 2.0G aft acting loads, and during application of side loads of +/- 0.5G combined with 1.0G download.

3.3.15 Limit Load Factors
The CMDS shall be capable of withstanding the following acceleration loads without structural damage. Minor degradation of performance may be permitted during the application of these loads, but full performance shall be immediately and automatically restored upon removal of the following loads:

11.5G down
10.0G up
10.0G down, 3.0G fwd
10.0G fwd and aft
10.0G side to side

3.3.16 Ultimate Load Factors
The CMDS shall have sufficient static strength to withstand at least 1.5 times the limit loads. The CMDS shall not be required to operate under these loads, however, inadvertent operation under these loaded conditions shall not create a safety hazard during or after such loading.

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3.3.17 Crash Landing Load Factor
In regard to the programmer, the craft landing load factor shall conform to MIL-A-008865, Amendment 3, Paragraph 3.3.2B.

3.3.18 Rain
In regard to the programmer, the CMDS shall be designed to withstand exposure, without deterioration of performance or finish, to rainfall of 5.2 in./hr. at wind speeds of 110 fps.

3.3.19 Solar Radiation
In regard to the programmer, the CMDS shall be designed to withstand exposure to radiant rates of 355 BTU/sq.ft./hr.

3.3.20 Fluids
The CMDS shall be designed to withstand exposure, without damage or degradation
of performance, to the following fluids:

a) JP-4 or JP-5 (MIL-T-5624M) and NATO equivalent fuels. Fuel temperatures shall be within the range of 18°C. to 80°C.
b) Hydraulic fluids (MIL-H-5606 and MIL-H-83282). The hydraulic fluid shall be within the temperature range of 20°C. to 135°C.
c) Lubricating oil (MIL-L-7808). The lubricating oil shall be within the temperature range of 18°C. to 135°C.
d) Coolants - Fluorocarbons, silicon, silicon ester, and glycol families.
e) Deicing fluid (MIL-A-8243). Glycol deicing fluid which may be applied to the aircraft under high pressure as a mixture of 50 percent glycol and 50 percent water at a temperature of 65°C.
f) Water - Normal cleaning fluids applied with high pressure spray (sequencers and dispensers).

3.3.21 Explosive Decompression
Cockpit items must be able to withstand explosive decompression.

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APPENDIX B

ADDENDUM A

AVIONICS MUX BUS 1553 INTERFACE COMMAND, DATA, AND STATUS MESSAGES

This protocol definition defines the 1553 avionics bus interface between the CMDS (a remote terminal) and the F-18 Mission Computer (the bus controller). The header words are defined on sheet 4.2 of F18Ø47ØØØ.

AV MUX BUS MESSAGES

Message Name	Message Subaddress		Source	Page
CMDS CONTROL COMMAND	1ØØØØ	(1Øh)	MC	A-4
CMDS STATUS RESPONSE	1ØØØ1	(11h)	CMDS	A-7
A/C NAVIGATION DATA	1ØØ1Ø	(12h)	MC	A-12
MANUAL PROGRAM DATA	11ØØØ	(18h)	MC	A-17
MANUAL PROGRAM DATA REPORT	11ØØ1	(19h)	CMDS	A-22
CMDS BINGO MODIFICATION	11Ø1Ø	(1Ah)	MC	A-24
CMDS BINGO LEVELS	11Ø11	(1Bh)	CMDS	A-26
MEMORY INSPECT COMMAND	111ØØ	(1Ch)	MC	A-28
MEMORY INSPECT RESPONSE	111Ø1	(1Dh)	CMDS	A-29
BIT COMMAND	1111Ø	(1Eh)	MC	A-3Ø
BIT RESPONSE	Ø1111	(ØFh)	CMDS	A-32

This addendum consists of 34 pages

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APPENDIX B
SUBADDRESS UTILIZATION

The following subaddresses (message numbers) are utilized by this protocol. Subaddresses marked "RESERVED" are not currently available for growth messages, while subaddresses not listed are available.

Message	Hexadecimal Subaddress	CMDS transmit/receive
RESERVED	Ø1h	RECEIVE
RESERVED	Ø1h	TRANSMIT
RESERVED	Ø2h	RECEIVE
RESERVED	Ø2h	TRANSMIT
RESERVED	Ø3h	RECEIVE
RESERVED	Ø4h	RECEIVE
RESERVED	Ø5h	RECEIVE
RESERVED	Ø6h	RECEIVE
RESERVED	Ø7h	RECEIVE
RESERVED	Ø7h	TRANSMIT
RESERVED	Ø8h	RECEIVE
CMDS CONTROL COMMAND	1Øh	RECEIVE
CMDS STATUS RESPONSE	11h	TRANSMIT
A/C NAVIGATION DATA	12h	RECEIVE
RESERVED	13h	TRANSMIT
RESERVED	14h	RECEIVE
RESERVED	14h	TRANSMIT
RESERVED	15h	RECEIVE
RESERVED	15h	TRANSMIT
RESERVED	16h	RECEIVE
RESERVED	16h	TRANSMIT
RESERVED	17h	RECEIVE
RESERVED	17h	TRANSMIT
MANUAL PROGRAM DATA	18h	RECEIVE
MANUAL PROGRAM DATA REPORT	19h	TRANSMIT
CMDS BINGO LEVELS	1Bh	TRANSMIT
MEMORY INSPECT COMMAND	1Ch	RECEIVE
RESERVED	1Dh	RECEIVE
MEMORY INSPECT RESPONSE	1Dh	TRANSMIT
BIT COMMAND	1Eh	RECEIVE
BIT RESPONSE	ØFh	TRANSMIT

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APPENDIX B

DATA REPRESENTATION

All fields specified as "n-bit fixed point numbers" are in fact represented as 2's complement integers of the prescribed length in bits, with a sign bit as the highest order bit, if the number is declared signed. They may be masked, shifted (with sign extension where appropriate), etc., to obtain standard (16 bit) integer quantities representing multiples of the LSB remaining after any low-order masking.

> To convert negative values of two's complement data to straight binary, complement the word, and add 1 to the result. Sign bits (denoted by SGN) values are + = Ø, - = 1. (Positive two's complement data is straight binary.) In an "n-bit fixed point number" data field, the highest-ordered bit is the sign bit; the MSB value provided with these fields is the value of the bit adjacent to the sign bit.

Spare bits (denoted by S) are set equal to "Ø". Reserved bits (denoted by RSV) are not guaranteed to be Ø.

Bits marked "Spare" on input messages from BC should be masked out before message parsing, as they can't be 1ØØ% trusted to be Ø. Bits marked "RSV" are used for other purposes, and contain values undefined by this interface. Spare bits transmitted by the CMDS are expected to be Ø, but future modifications may result in utilization of these bits; accordingly, they should also be masked out until/unless explicitly included in this interface.

A bit value of "1" means the discrete or flag is asserted (true/on), unless otherwise stated. The bit value may be defined in greater detail in some messages.

Subaddresses followed by an "h" indicate a hexadecimal subaddress.

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Message: CMDS CONTROL COMMAND
Subaddress: 1Øh
Frequency: 2Ø/sec
Source: MC (AV Bus Master)
Destination: CMDS (RT)

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
C	terminal address						T/R	subaddress					word count			
	Ø	1	Ø	1	Ø		Ø	1	Ø	Ø	Ø	Ø	Ø	Ø	Ø	1	1
D1	CMDS MODE			MAN PROG			CHF ON	FLR ON	O1 ON	O2 ON	RWR ON	JMR ON	MWS ON	WOW	ABL	ABR
D2	PGM DSP	M5 DSP	M6 DSP	SEM DSP	RESERVED										RUN SIL	RPG CMP
D3	PGM CMP	M5 CMP	M6 CMP	SEM CMP	RESERVED											

DATA WORD 1

BIT	FIELD	DESCRIPTION
15-13	CMDS MODE	Mode Switch Selection Code Format: 3-bit code (one octal digit) Ø = STANDBY 1 = MANUAL 2 = SEMI-AUTO 3 = AUTO 4-7 = Undefined
12-1Ø	MAN PROG	Manual Program Selection Code Format: 3-bit code (one octal digit) Ø = Undefined 1 = Manual Program # 1 2 = Manual Program # 2 3 = Manual Program # 3 4 = Manual Program # 4 5 = Manual Program # 5 6 = Manual Program # 6 7 = Undefined
		Enable / Inhibit Switches
9	CHF ON	CHAFF DISPENSING ENABLED
8	FLR ON	FLARE DISPENSING ENABLED
7	O1 ON	OTHER 1 DISPENSING ENABLED
6	O2 ON	OTHER 2 DISPENSING ENABLED
5	RWR ON	RWR DISPENSING ENABLED
4	JMR ON	JAMMER DISPENSING ENABLED
3	MWS ON	MWS DISPENSING ENABLED

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DATA WORD 1 (Continued)

BIT	FIELD	DESCRIPTION
		Platform Status Flags
2	WOW	WEIGHT ON WHEELS
1	ABL ON	LEFT AFTERBURNER ON
Ø	ABR ON	RIGHT AFTERBURNER ON

DATA WORD 2

BIT	FIELD	DESCRIPTION
		Dispense Initiate Flags
15	PGM DSP	DISPENSE SELECTED MANUAL PROGRAM (from MAN PGM)
14	M5 DSP	DISPENSE MANUAL PROGRAM #5
13	M6 DSP	DISPENSE MANUAL PROGRAM #6
12	SEM DSP	CONSENT TO DISPENSE (SEMI-AUTO mode)
11-2	RSV	Reserved
		Command flags
1	RUN SIL	AUTOMATIC DISPENSE DISABLE (RUN SILENT) (1 = disabled)
		Manual Reprogramming Complete
Ø	RPG CMP	MANUAL PROGRAM PARAMETER LOAD COMPLETED

DATA WORD 3

BIT	FIELD	DESCRIPTION
15	PGM CMP	Complement of PGM DSP of Data Word 2
14	M5 CMP	Complement of M5 DSP of Data Word 2
13	M6 CMP	Complement of M6 DSP of Data Word 2
12	SEM CMP	Complement of SEM DSP of Data Word 2
11-Ø		Reserved

NOTES: Typically, at most one (1) of the Dispense Initiate flags is asserted. If more than one is asserted simultaneously, the CMDS will execute the appropriate response according to its normal threat processing prioritization methods. A dispense initiate is triggered by each transition from de-asserted (Ø) to asserted (1), providing that the corresponding complement of the flag (in Data Word 3) reads as (Ø).

The RPG CMP flag is asserted after the MC has completely cycled through the six manual programs with the MANUAL PROGRAM DATA message, in response to the assertion of the DAT LOS power-up flag in the CMDS STATUS message. The CMDS de-asserts DAT LOS upon assertion of RPG CMP.

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When asserted, the category dispense enables (CHF ON, FLR ON, O1 ON, O2 ON) allow CMDS dispense programs to include the respective expendable category. The source enables (RWR ON, JMR ON, MWS ON) allow the CMDS to consider input (sensor data and/or dispense commands) from a particular source, when enable is asserted. These source enables affect dispense program selection only in SEMI-AUTO and AUTO modes.

When the RUN SILENT flag is set, the AN/ALE-47 does the following:

If the AN/ALE-47 is in auto mode or semi-auto mode when RUN SILENT is asserted, no automatic dispensing is allowed - dispense authorization by the pilot is required, regardless of sensor/source or MDF flags.

Upon deassertion of the RUN SILENT flag, the AN/ALE-47 resumes operation according to the mode setting and MDF flags.

The RUN SILENT flag does not affect operation in any other modes.

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Message: CMDS STATUS RESPONSE
Subaddress: 11h
Frequency: 2Ø/sec
Source: CMDS (RT)
Destination: MC (AV Bus Master)

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
C	terminal address					T/R	subaddress					word count				
	Ø	1	Ø	1	Ø	1	1	Ø	Ø	Ø	1	Ø	1	Ø	Ø	Ø

Appropriate Status Word

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
D1	CMDS MODE			MAN PROG			CHF ON	FLR ON	O1 ON	O2 ON	RWR ON	JMR ON	MWS ON	WOW	ABL	ABR
D2	PGM DSP	M5 DSP	M6 DSP	SEM DSP	RESERVED						JMR AUT	MWS AUT	RSV		RUN SIL	RPG CMP
D3	CHF DSP	FLR DSP	O1 DSP	O2 DSP	RWR DSP	JMR DSP	MWS DSP	JET DSP	RESERVED						BIT	SEM RDY
D4	CHF LOW	FLR LOW	O1 LOW	O2 LOW	RESERVED						MSF DET	MAN DEG	SEM DEG	AUT DEG	S Ø	DAT LOS
D5	RESERVED						CHAFF INVENTORY									
D6	RESERVED						FLARE INVENTORY									
D7	RESERVED						OTHER 1 INVENTORY									
D8	RESERVED						OTHER 2 INVENTORY									

NOTE: The settings in words D1, D2 are the CMDS current settings of the corresponding bits in the CMDS CONTROL COMMAND Message; in the presense of discrete inputs and mode changes for other reason than the CMDS CONTROL COMMAND settings, these may not necessarily echo the settings from that message. A more complete description of these fields is given in the notes for the CMDS CONTROL COMMAND message.

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DATA WORD 1

BIT	FIELD	DESCRIPTION
15-13	CMDS MODE	Mode Switch Selection Code Format: 3-bit code (one octal digit) Ø = STANDBY 1 = MANUAL 2 = SEMI-AUTO 3 = AUTO 4-7 = Undefined
12-1Ø	MAN PROG	Manual Program Selection Code Format: 3-bit code (one octal digit) Ø = Undefined 1 = Manual Program # 1 2 = Manual Program # 2 3 = Manual Program # 3 4 = Manual Program # 4 5 = Manual Program # 5 6 = Manual Program # 6 7 = Undefined
		Enable/Inhibit Switches
9	CHF ON	CHAFF DISPENSING ENABLED
8	FLR ON	FLARE DISPENSING ENABLED
7	O1 ON	OTHER 1 DISPENSING ENABLED
6	O2 ON	OTHER 2 DISPENSING ENABLED
5	RWR ON	RWR DISPENSING ENABLED
4	JMR ON	JAMMER DISPENSING ENABLED
3	MWS ON	MWS DISPENSING ENABLED
		Platform Status flags
2	WOW	WEIGHT ON WHEELS
1	ABL ON	LEFT AFTERBURNER ON
Ø	ABR ON	RIGHT AFTERBURNER ON

DATA WORD 2

BIT	FIELD	DESCRIPTION
		Dispense Initiate flags
15	PGM DSP	DISPENSE SELECTED MANUAL PROGRAM (from MAN PGM)
14	M5 DSP	DISPENSE MANUAL PROGRAM #5
13	M6 DSP	DISPENSE MANUAL PROGRAM #6
12	SEM DSP	CONSENT TO DISPENSE (SEMI-AUTO mode)

NOTE: Typically, at most one (1) of the Dispense Initiate flags is asserted. If more than one is asserted simultaneously, the CMDS will execute the appropriate response according to its normal threat processing prioritization methods. These flags, however, will show the CMDS perception of the current states of these control inputs.

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DATA WORD 2 (Continued)

BIT	FIELD	DESCRIPTION
11-6	RSV	Reserved
		SEMI AUTO Interpretation (AUTO = 1, SEMI AUTO = ø)
5	JMR AUT	Jammer Dispense Requests AUTOmatic, without SEMI AUTO Initiate required in SEMI AUTO mode
4	MWS AUT	MWS Dispense Requests AUTOmatic, without SEMI AUTO Initiate required in SEMI AUTO mode
3-2	RSV	Reserved
		Command flags
1	RUN SIL	AUTOMATIC DISPENSE DISABLE (RUN SILENT) (1 = disabled)
		Manual Reprogramming Complete
ø	RPG CMP	MANUAL PROGRAM PARAMETER LOAD COMPLETED

DATA WORD 3

BIT	FIELD	DESCRIPTION
		Dispense action in progress within last 1øø msec
15	CHF DSP	CHAFF DISPENSE
14	FLR DSP	FLARE DISPENSE
13	O1 DSP	OTHER 1 DISPENSE
12	O2 DSP	OTHER 2 DISPENSE
11	RWR DSP	RWR INITIATED DISPENSE
1ø	JMR DSP	JAMMER INITIATED DISPENSE
9	MWS DSP	MWS INITIATED DISPENSE
8	JET DSP	JETTISON
7-2		Reserved
1	BIT	CMDS BIT IN PROCESS
ø	SEM RDY	SEMI-AUTO PROGRAM READY FOR DISPENSE INITIATION

DATA WORD 4

BIT	FIELD	DESCRIPTION
		Low Stores Status
15	CHF LOW	CHAFF LOW STORES LEVEL REACHED
14	FLR LOW	FLARE LOW STORES LEVEL REACHED
13	O1 LOW	OTHER 1 LOW STORES LEVEL REACHED
12	O2 LOW	OTHER 2 LOW STORES LEVEL REACHED

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DATA WORD 4 (Continued)

BIT	FIELD	DESCRIPTION
11-6		Reserved
		Degraded Dispense Mode (disabled or misfired = 1)
5	MSF DET	MISFIRE DETECTED (since power-up)
4	MAN DEG	MANUAL DISPENSE UNAVAILABLE
3	SEM DEG	SEMI-AUTO DISPENSE UNAVAILABLE
2	AUT DEG	AUTO DISPENSE UNAVAILABLE
1		SPARE
Ø	DAT LOS	USER-INPUT MANUAL DISPENSE PROGRAMS UNAVAILABLE (asserted at CMDS PROGRAMMER power-up/reset)

NOTE: The degraded/unavailable modes of operational flags (MAN DEG, SEM DEG, and AUT DEG) are asserted when the CMDS determines that the respective dispensing mode is not functioning/unavailable, and stays asserted for as long as that mode remains unavailable.

The DAT LOS flag is asserted after power-up of the CMDS PROGRAMMER. Upon assertion of this flag the MC cycles through the six manual programs with the MANUAL PROGRAM DATA message, and then asserts the RPG CMP flag in the CMDS CONTROL COMMAND message.

The CMDS de-asserts DAT LOS upon assertion of RPG CMP.

DATA WORD 5

BIT	FIELD	DESCRIPTION
15-1Ø		Reserved
9-Ø	CHAFF	INVENTORY QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

DATA WORD 6

BIT	FIELD	DESCRIPTION
15-1Ø		Reserved
9-Ø	FLARE	INVENTORY QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

DATA WORD 7

BIT	FIELD	DESCRIPTION
15-1Ø		Reserved
9-Ø	OTHER 1	INVENTORY QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

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DATA WORD 8

BIT	FIELD	DESCRIPTION
15-1Ø		Reserved
9-Ø	OTHER 2	INVENTORY QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

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Message: A/C NAVIGATION DATA
Subaddress: 12h
Frequency: 1Ø/sec
Source: MC (AV Bus Master)
Destination: CMDS (RT)

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
	terminal address					T/R	subaddress					word count				
	Ø	1	Ø	1	Ø	Ø	1	Ø	Ø	1	Ø	1	1	Ø	1	Ø
D1	LAT	LNG	ALT MSL	ALT AGL	VVL	PCH	ROL	DFT	PCH RAT	ROL RAT	YAW RAT	THD	GTK	GSP	IAS	S Ø
D2	SGN	LATITUDE (HIGH ORDER WORD)														
D3	LATITUDE (LOW ORDER WORD)															
D4	SGN	LONGITUDE (HIGH ORDER WORD)														
D5	LONGITUDE (LOW ORDER WORD)															
D6	SGN	BAROMETRIC CORRECTED ALTITUDE (ALT MSL) (HIGH ORDER WORD)														
D7	ALT MSL LOW				RESERVED											
D8	SGN	RADAR ALTITUDE (ALT AGL)													ZRO Ø	ZRO Ø
D9	SGN	VERTICAL VELOCITY														
D1Ø	SGN	PITCH														
D11	SGN	ROLL														
D12	SGN	DRIFT														
D13	SGN	PITCH RATE														

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APPENDIX B

D14	ROLL RATE																
	SGN																
D15	YAW RATE																
	SGN																
D16	TRUE HEADING																
	SGN																
D17	GROUND TRACK																
	SGN																
D18	GROUND SPEED																
	SGN																
D19	INDICATED AIRSPEED																
	SGN																
D2Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	ACC	TMP	
D21	ACCELERATION X COMPONENT																
	SGN																
D22	ACCELERATION Y COMPONENT																
	SGN																
D23	ACCELERATION Z COMPONENT																
	SGN																
D24	TRUE FREESTREAM AIR TEMPERATURE																
	SGN																
D25	TIME DATA SENT TO CMDS (REFERENCED TO SYSTEM CLOCK)																
D26	TIME DATA VALID (REFERENCED TO SYSTEM CLOCK)																

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DATA WORD 1

BIT	FIELD	DESCRIPTION
15	LAT	LATITUDE VALID
14	LNG	LONGITUDE VALID
13	ALT MSL	BAROMETRIC ALTITUDE ("ALT MSL") VALID
12	ALT AGL	RADAR ALTITUDE ("ALT AGL") VALID
11	VVL	VERTICAL VELOCITY VALID
1Ø	PCH	PITCH VALID
9	ROL	ROLL VALID
8	DFT	DRIFT VALID
7	PCH RAT	PITCH RATE VALID
6	ROL RAT	ROLL RATE VALID
5	YAW RAT	YAW RATE VALID
4	THD	TRUE HEADING VALID
3	GTK	GROUND TRACK VALID
2	GSP	GROUND SPEED VALID
1	IAS	INDICATED AIRSPEED VALID
Ø		SPARE

NOTE: Validity flags in DATA WORD 1 (valid/TRUE = 1, not valid/FALSE = Ø). Signs in DATA WORDS 2 through 26 (SGN: + = Ø, - = 1)

DATA WORDS 2 THROUGH 8

LATITUDE:
- Format: 32-bit signed fixed point number (in two words)
- High word: LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
  MSB = 1/2 semi-circle (9Ø.Ø degrees)
- Low word: LSB = 1/2,147,483,648 semi-circle (Ø.ØØØØØØØ83819 degrees)
  MSB = 1/65,536 semi-circle (Ø.ØØ2746582 degrees)
- Range: -1 to +1 semi-circle (-18Ø.Ø to +18Ø.Ø degrees)

LONGITUDE:
- Format: 32-bit signed fixed point number (in two words)
- High word: LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
  MSB = 1/2 semi-circle (9Ø.Ø degrees)
- Low word: LSB = 1/2,147,483,648 semi-circle (Ø.ØØØØØØØ83819 degrees)
  MSB = 1/65,536 semi-circle (Ø.ØØ2746582 degrees)
- Range: -1 to +1 semi-circle (-18Ø.Ø to +18Ø.Ø degrees)

BAROMETRIC CORRECTED ALTITUDE (ALT MSL):
- Format: 19-bit signed fixed point number (in two words)
- High word: LSB = 4 feet; MSB = 65,536 feet
- Low word: LSB = 1/2 feet; MSB = 2 feet
- Range: -15ØØ to +8Ø,ØØØ feet

RADAR ALTITUDE (ALT AGL):
- Format: 14-bit signed fixed point number
- LSB = 1 feet; MSB = 4Ø96 feet
- Range: Ø to +8191 feet (Valid up to 5ØØØ feet)

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DATA WORDS 9 THROUGH 16

VERTICAL VELOCITY:
Format: 16-bit signed fixed point number
LSB = 1/8 ft/sec; MSB = 2Ø48 ft/sec
Range: -4Ø96 to +4Ø95 ft/sec

PITCH:
Format: 16-bit signed fixed point number
LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
MSB = 1/2 semi-circle (9Ø.Ø degrees)
Range: -11/18 to +11/18 semi-circle (-11Ø.Ø to +11Ø degrees)
(positive nose up)

ROLL:
Format: 16-bit signed fixed point number
LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
MSB = 1/2 semi-circle (9Ø.Ø degrees)
Range: -1 to +1 semi-circle (-18Ø.Ø to +18Ø.Ø degrees)
(positive right wing down)

DRIFT:
Format: 16-bit signed fixed point number
LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
MSB = 1/2 semi-circle (9Ø.Ø degrees)
Range: -1 to +1 semi-circle (-18Ø.Ø to +18Ø.Ø degrees)
(positive nose right)

PITCH RATE:
Format: 16-bit signed fixed point number
LSB = 1/64 degree/sec; MSB = 256 deg/sec
Range: -512 to 511 deg/sec (positive nose up)

ROLL RATE:
Format: 16-bit signed fixed point number
LSB = 1/64 degree/sec; MSB = 256 deg/sec
Range: -512 to 511 deg/sec (positive right wing down)

YAW RATE
Format: 16-bit signed fixed point number
LSB = 1/64 degree/sec; MSB = 256 deg/sec
Range: -512 to 511 deg/sec (positive nose right)

TRUE HEADING:
Format: 16-bit signed fixed point number
LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
MSB = 1/2 semi-circle (9Ø.Ø degrees)
Range: -1 to +1 semi-circle (-18Ø.Ø to +18Ø.Ø degrees)
(positive clockwise from north)

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DATA WORDS 17 THROUGH 26

GROUND TRACK:
Format: 16-bit signed fixed point number
LSB = 1/32,768 semi-circle (Ø.ØØ5493164 degrees)
MSB = 1/2 semi-circle (9Ø.Ø degrees)
Range: -1 to +1 semi-circle (-18Ø.Ø to +18Ø.Ø degrees)
(positive clockwise from north)

GROUND SPEED:
Format: 16-bit signed fixed point number
LSB = 1/8 ft/sec; MSB = 2Ø48 ft/sec
Range: -4Ø96 to +4Ø95 ft/sec

INDICATED AIRSPEED:
Format: 16-bit signed fixed point number
LSB = Ø.Ø3125 knots; MSB = 512 knots
Range: -1Ø24 to +1Ø23 knots

Validity flags (valid/TRUE = 1, not valid/FALSE = Ø)
ACC: ACCELERATION VALID
TMP: TEMPERATURE VALID

ACCELERATION X COMPONENT:
Format: 16-bit signed fixed point number
LSB = Ø.Ø15625 ft/sec; MSB = 256 ft/sec
Range: -512 to +511 ft/sec

ACCELERATION Y COMPONENT:
Format: 16-bit signed fixed point number
LSB = Ø.Ø15625 ft/sec; MSB = 256 ft/sec
Range: -512 to +511 ft/sec

ACCELERATION Z COMPONENT:
Format: 16-bit signed fixed point number
LSB = Ø.Ø15625 ft/sec; MSB = 256 ft/sec
Range: -512 to +511 ft/sec

TRUE FREESTREAM AIR TEMPERATURE:
Format: 16-bit unsigned integer number
LSB = 1 degrees Kelvin; MSB = 32768 degrees Kelvin
Range: Ø to +5ØØ degrees

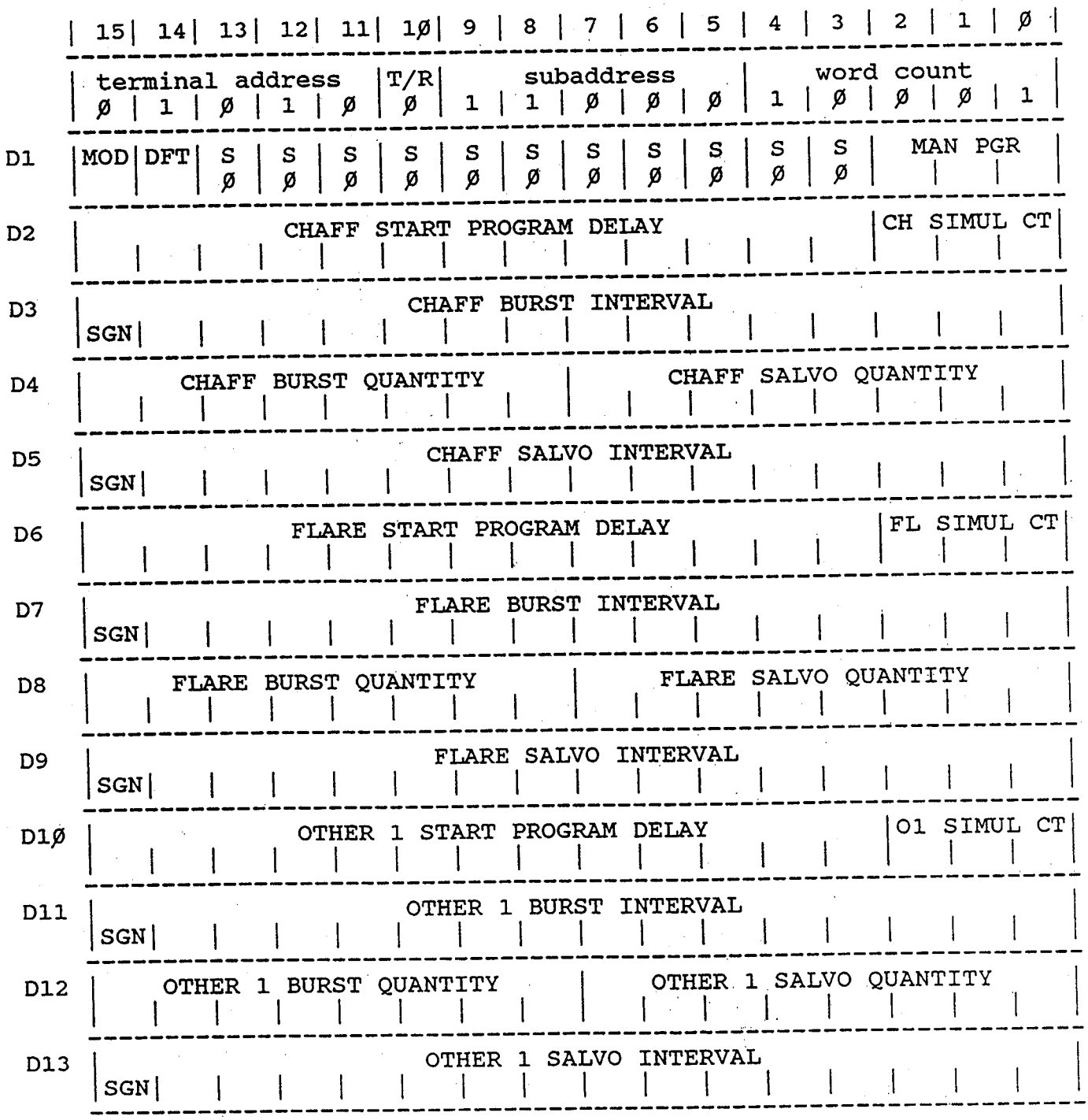
TIME DATA SENT TO CMDS
Format: 16-bit unsigned integer number
LSB = 64 microseconds; MSB = 2,Ø97,152 microseconds
Range: Ø to 4,194,24Ø microseconds, with rollover possible

TIME DATA VALID
Format: 16-bit unsigned integer number
LSB = 64 microseconds; MSB = 2,Ø97,152 microseconds
Range: Ø to 4,194,24Ø microseconds, with rollover possible

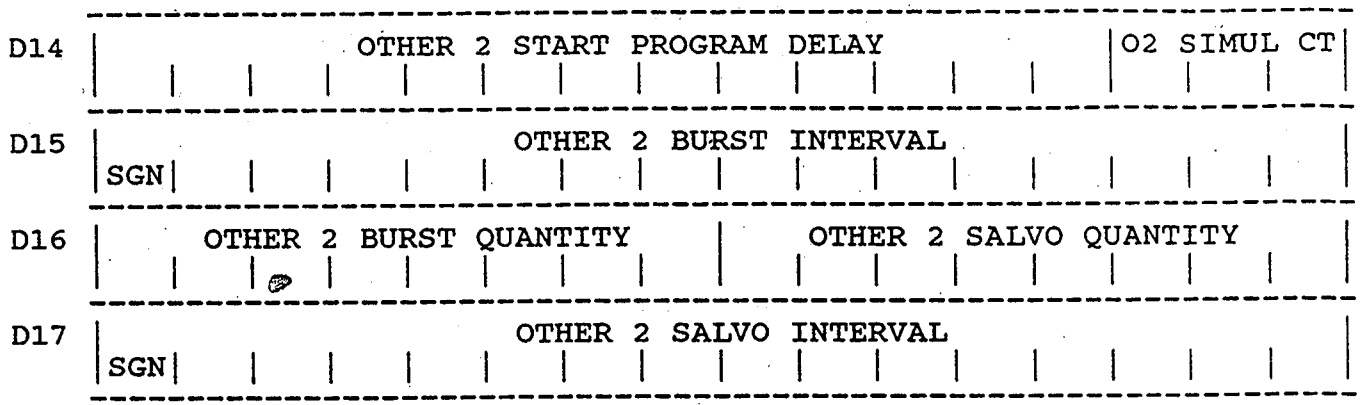
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
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APPENDIX B

Message: MANUAL PROGRAM DATA
Subaddress: 18h
Frequency: 1Ø/sec
Source: MC (AV Bus Master)
Destination: CMDS (RT)



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DATA WORD 1

BIT	FIELD	DESCRIPTION
15	MOD	DATA MODIFIED (Ø indicates no change from previous data for the specified manual program)
14	DFT	DEFAULT DATA REQUESTED (Ø indicates use current working data for the specified manual program)
13-3		SPARES
2-Ø	MAN PGM	MANUAL PROGRAM NUMBER Format: 3-bit code (one octal digit) Ø = No program specified 1-6 = Manual Program with the corresponding number 7 = Undefined

NOTE: The CMDS must be in STANDBY mode before it will process ANY fields of this message. The given values in words D2 through D17 will be ignored unless the MOD flag indicates a change has been made. If the DFT flag is asserted, the parameter values of the default manual program from the MDF will replace the current working values, if the MAN PGM field has a value of 1 through 6, specifying an actual manual program.

DATA WORD 2

CHAFF START PROGRAM DELAY:
Format: 13-bit unsigned integer number
LSB = 1 millisecond; MSB = 4Ø96 millisecond
Range: Ø milliseconds to 8ØØØ milliseconds (nominal)

CH SIMUL CT: CHAFF SIMULTANEOUS COUNT
Format: 3-bit unsigned integer number (one octal digit)
Ø = Undefined
1-4 = number of simultaneous payloads to dispense
5-7 = Undefined

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DATA WORDS 3 THROUGH 9

CHAFF BURST INTERVAL:
Format: 16-bit signed fixed point number
LSB = 1 millisecond; MSB = 16,384 millisecond
Range: 5 milliseconds to 1Ø.ØØØ sec (nominal)

CHAFF BURST QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 expendable; MSB = 128 expendables
Range: Ø to 2Ø expendables (nominal)

CHAFF SALVO QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 salvo; MSB = 128 salvoes
Range: Ø to 2Ø salvos (nominal)

CHAFF SALVO INTERVAL:
Format: 16-bit signed fixed point number
LSB = 5Ø milliseconds; MSB = 819.2 sec
Range: 5ØØ milliseconds to 15Ø.ØØØ sec (nominal)

FLARE START PROGRAM DELAY:
Format: 13-bit unsigned integer number
LSB = 1 millisecond; MSB = 4Ø96 millisecond
Range: Ø milliseconds to 8ØØØ milliseconds (nominal)

FL SIMUL CT: FLARE SIMULTANEOUS COUNT
Format: 3-bit unsigned integer number (one octal digit)
Ø = Undefined
1-4 = number of simultaneous payloads to dispense
5-7 = Undefined

FLARE BURST INTERVAL:
Format: 16-bit signed fixed point number
LSB = 1 millisecond; MSB = 16,384 millisecond
Range: 5 milliseconds to 1Ø.ØØØ sec (nominal)

FLARE BURST QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 expendable; MSB = 128 expendables
Range: Ø to 2Ø expendables (nominal)

FLARE SALVO QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 salvo; MSB = 128 salvoes
Range: Ø to 2Ø salvos (nominal)

FLARE SALVO INTERVAL:
Format: 16-bit signed fixed point number
LSB = 5Ø milliseconds; MSB = 819.2 sec
Range: 5ØØ milliseconds to 15Ø.ØØØ sec (nominal)

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Check the source to verify that this is the current version before use.

DATA WORDS 1Ø THROUGH 15

OTHER 1 START PROGRAM DELAY:
Format: 13-bit unsigned integer number
LSB = 1 millisecond; MSB = 4Ø96 millisecond
Range: Ø milliseconds to 8ØØØ milliseconds (nominal)

OTHER 1 SIMULTANEOUS COUNT (O1 SIMUL CT):
Format: 3-bit unsigned integer number (one octal digit)
Ø = Undefined
1-4 = number of simultaneous payloads to dispense
5-7 = Undefined

OTHER 1 BURST INTERVAL:
Format: 16-bit signed fixed point number
LSB = 1 millisecond; MSB = 16,384 millisecond
Range: 5 milliseconds to 1Ø.ØØØ sec (nominal)

OTHER 1 BURST QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 expendable; MSB = 128 expendables
Range: Ø to 2Ø expendables (nominal)

OTHER 1 SALVO QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 salvo; MSB = 128 salvoes
Range: Ø to 2Ø salvos (nominal)

OTHER 1 SALVO INTERVAL:
Format: 16-bit signed fixed point number
LSB = 5Ø milliseconds; MSB = 819.2 sec
Range: 5ØØ milliseconds to 15Ø.ØØØ sec (nominal)

OTHER 2 START PROGRAM DELAY:
Format: 13-bit unsigned integer number
LSB = 1 millisecond; MSB = 4Ø96 millisecond
Range: Ø milliseconds to 8ØØØ milliseconds (nominal)

OTHER 2 SIMULTANEOUS COUNT (O2 SIMUL CT):
Format: 3-bit unsigned integer number (one octal digit)
Ø = Undefined
1-4 = number of simultaneous payloads to dispense
5-7 = Undefined

OTHER 2 BURST INTERVAL:
Format: 16-bit signed fixed point number
LSB = 1 millisecond; MSB = 16,384 millisecond
Range: 5 milliseconds to 1Ø.ØØØ sec (nominal)

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DATA WORDS 16 THROUGH 17

OTHER 2 BURST QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 expendable; MSB = 128 expendables
Range: Ø to 2Ø expendables (nominal)

OTHER 2 SALVO QUANTITY:
Format: 8-bit unsigned integer
LSB = 1 salvo; MSB = 128 salvoes
Range: Ø to 2Ø salvos (nominal)

OTHER 2 SALVO INTERVAL:
Format: 16-bit signed fixed point number
LSB = 5Ø milliseconds; MSB = 819.2 sec
Range: 5ØØ milliseconds to 15Ø.ØØØ sec (nominal)

NOTES: Values outside the valid range for a given quantity are "clamped" to the nearest valid value in the ALE-47; that is, too-small (or negative) values are taken as meaning the minimum in-range value, while too-large values are considered to mean the maximum in-range value.

At initialization of this link, the MAN PGM field of this message is set to zero.

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APPENDIX B

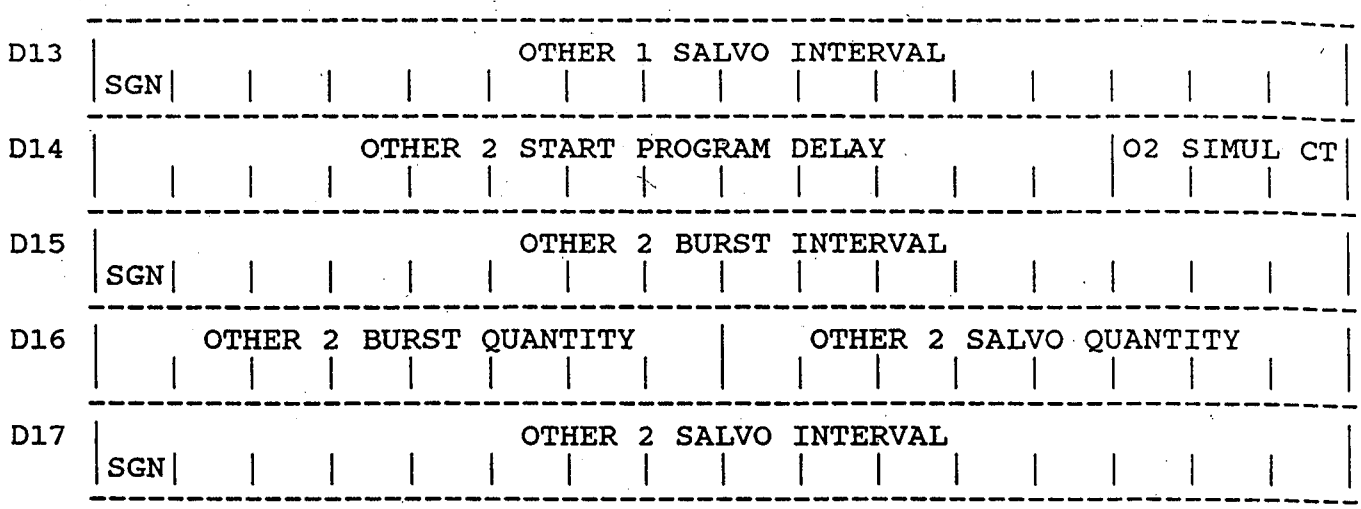
Message: MANUAL PROGRAM DATA REPORT
Subaddress: 19h
Frequency: 1Ø/sec
Source: CMDS (RT)
Destination: MC (AV Bus Master)

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
	terminal address					T/R	subaddress						word count			
	Ø	1	Ø	1	Ø	1	1	1	Ø	Ø	1	1	Ø	Ø	Ø	1

APPROPRIATE STATUS WORD

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
D1	DAT VLD	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	MAN PGM		
D2	CHAFF START PROGRAM DELAY													CH SIMUL CT		
D3	SGN	CHAFF BURST INTERVAL														
D4	CHAFF BURST QUANTITY								CHAFF SALVO QUANTITY							
D5	SGN	CHAFF SALVO INTERVAL														
D6	FLARE START PROGRAM DELAY													FL SIMUL CT		
D7	SGN	FLARE BURST INTERVAL														
D8	FLARE BURST QUANTITY								FLARE SALVO QUANTITY							
D9	SGN	FLARE SALVO INTERVAL														
D1Ø	OTHER 1 START PROGRAM DELAY													O1 SIMUL CT		
D11	SGN	OTHER 1 BURST INTERVAL														
D12	OTHER 1 BURST QUANTITY								OTHER 1 SALVO QUANTITY							

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
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DATA WORD 1

BIT	FIELD	DESCRIPTION
15	DAT VLD	Data Valid (asserted = 1) indicates that the CMDS believes that the message contains a complete set of data for the specified manual program.
14-3		SPARES
2-Ø	MAN PGM	MANUAL PROGRAM NUMBER Format: 3-bit code (one octal digit) Ø = No program specified 1-6 = Manual Program with the corresponding number 7 = Undefined

DATA WORDS 2 THROUGH 17

SAME FIELD DESCRIPTIONS AS THE MANUAL PROGRAM DATA MESSAGE

NOTE: The values of this message represent the current CMDS perception of the manual program parameters for the specified program. In any mode other than STANDBY, this message will NOT have the DAT VLD flag asserted. In STANDBY mode, after a latency of up to 8Ø milliseconds, if the MAN PGM of the latest MANUAL PROGRAM DATA message has a value of from 1 through 6, and after incorporation of any changes specified by that message, the MANUAL PROGRAM DATA REPORT message data fields are updated to reflect the current parameters for the specified manual program, and then the DAT VLD flag is asserted. AT CMDS initialization (at power-up, or after power loss, the MAN PGM field is initialized to Ø and DAT VLD is initialized as de-asserted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

Message: CMDS BINGO MODIFICATION
Subaddress: 1Ah
Frequency: 1Ø/sec
Source: MC (AV Bus Master)
Destination: CMDS (RT)

15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
terminal address					T/R	subaddress					word count				
Ø	1	Ø	1	Ø	Ø	1	1	Ø	1	Ø	Ø	Ø	1	Ø	Ø

Appropriate Status Word

D1	HON	RESERVED					CHAFF BINGO LEVEL									
D2	RESERVED						FLARE BINGO LEVEL									
D3	RESERVED						OTHER 1 BINGO LEVEL									
D4	RESERVED						OTHER 2 BINGO LEVEL									

DATA WORD 1

BIT	FIELD	DESCRIPTION
15	HON	Honor message values (words D1 - D4)
14-1Ø		RESERVED
9-Ø	CHAFF	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

DATA WORD 2

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	FLARE	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

DATA WORD 3

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	OTHER 1	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

DATA WORD 4

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	OTHER 2	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

NOTE: Category BINGO LEVEL QUANTITIES are the Low Stores values which are initially defined in the AN/ALE-47 MDF. A value of zero in any category effectively overrides the previous low stores level for that category. Upon initialization the CMDS assigns to the BINGO levels the values from the MDF. Upon activation of the Honor bit, the CMDS uses (and reports) the values contained herein. If the Honor bit is reset, the CMDS uses the MDF values.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

Message: CMDS BINGO LEVELS
Subaddress: 1Bh
Frequency: 1Ø/sec
Source: CMDS (RT)
Destination: MC (AV Bus Master)

15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
terminal address					T/R	subaddress					word count				
Ø	1	Ø	1	Ø	1	1	1	Ø	1	1	Ø	Ø	1	Ø	Ø

Appropriate Status Word

D1	RESERVED						CHAFF BINGO LEVEL									
D2	RESERVED						FLARE BINGO LEVEL									
D3	RESERVED						OTHER 1 BINGO LEVEL									
D4	RESERVED						OTHER 2 BINGO LEVEL									

DATA WORD 1

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	CHAFF	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

DATA WORD 2

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	FLARE	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

DATA WORD 3

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	OTHER 1	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

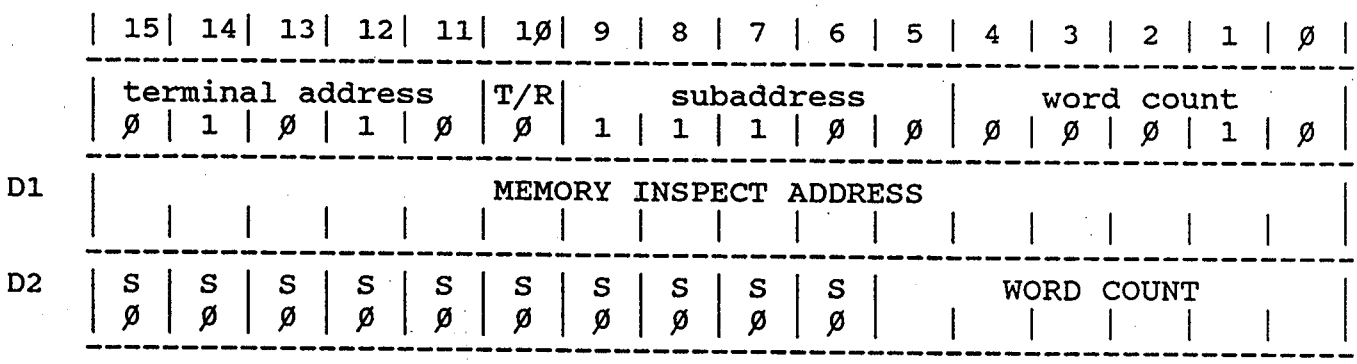
DATA WORD 4

BIT	FIELD	DESCRIPTION
15-1Ø		RESERVED
9-Ø	OTHER 2	BINGO LEVEL QUANTITY: Format: 1Ø-bit unsigned integer LSB = 1 expendable; MSB = 512 expendables Range: Ø to 999 expendables (nominal)

NOTE: Category BINGO LEVEL QUANTITIES are the Low Stores values below which automatic dispensing cannot continue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
Check the source to verify that this is the current version before use.

Message: MEMORY INSPECT COMMAND
Subaddress: 1Ch
Frequency: 1Ø/sec, when selected by operator
Source: MC (AV Bus Master)
Destination: CMDS (RT)



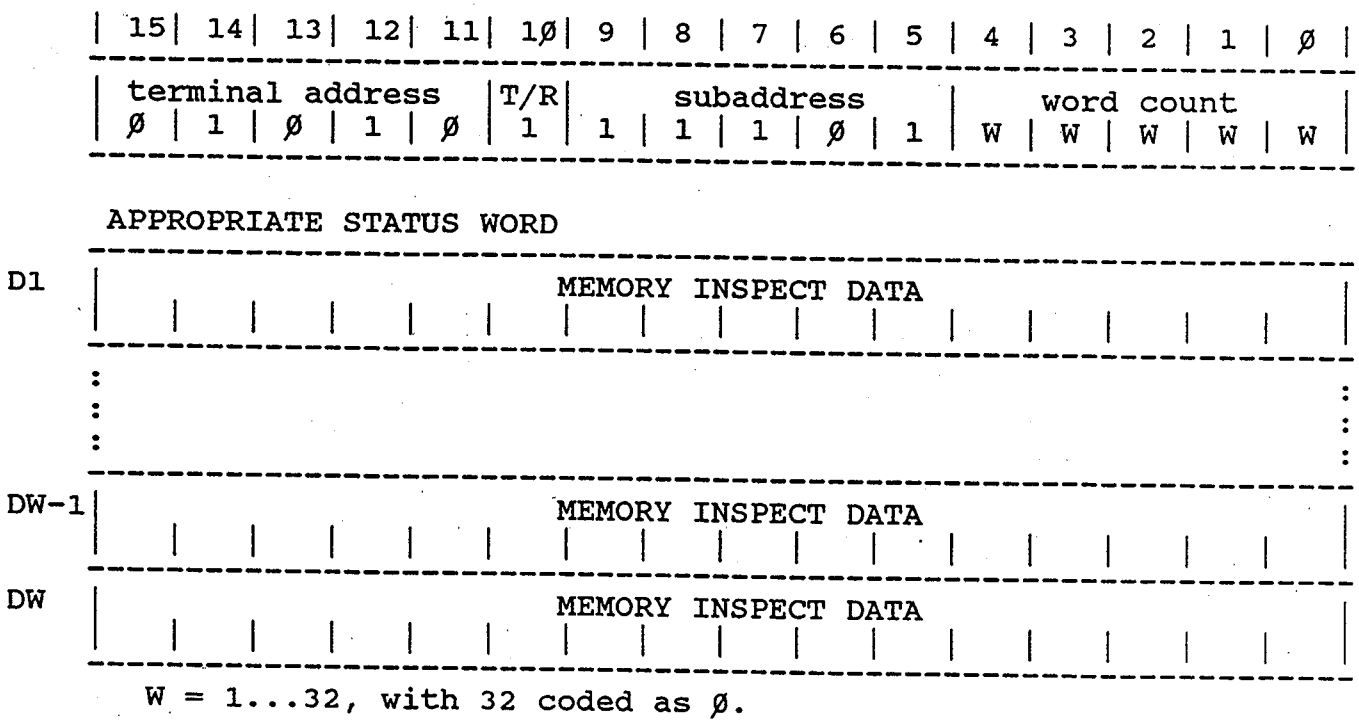
DATA WORD 1
MEMORY INSPECT ADDRESS:
Format: 16 bit unsigned integer representing the address to fetch data from

DATA WORD 2
WORD COUNT: NUMBER OF WORDS TO RETURN
Format: 6 bit unsigned integer
Range: 1...32; out of range values will be treated as if they had been a "1"

NOTE: A minimum of 8Ø milliseconds is expected between issuing the inspect memory command and reading the response, to allow for CMDS latency in reading the addressed data into the response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
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Message: MEMORY INSPECT RESPONSE
Subaddress: 1Dh
Frequency: 1Ø/sec, when selected by operator
Source: CMDS (RT)
Destination: MC (AV Bus Master)



W = 1...32, with 32 coded as Ø.

NOTES: After a latency of up to 8Ø milliseconds, the data contents of this message are updated to reflect the content of the memory addresses specified in the previous MEMORY INSPECT COMMAND message from the MC. The content of data words corresponding to non-existant memory addresses is indeterminate and undefined, as is the content of words transmitted by the CMDS in excess of the "WORD COUNT" field in word D2 of the MEMORY INSPECT COMMAND.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:58Z
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Message: BIT COMMAND
Subaddress: 1Eh
Frequency: 1/sec
Source: MC (AV Bus Master)
Destination: CMDS (RT)

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	Ø
	terminal address					T/R	subaddress					word count				
	Ø	1	Ø	1	Ø	Ø	1	1	1	1	Ø	Ø	Ø	Ø	1	Ø
D1	I BIT	MAT	CLR MFL	INV RST	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	S Ø	IN FLT
D2	TERMINAL TEST WORD															

DATA WORD 1

BIT	FIELD	DESCRIPTION
15	IBIT	PERFORM INITIATED BIT FLAG (run IBIT = 1). Assertion initiates the CMDS's non-interfering IBIT, which then runs to completion.
14	MAT	MAINTENANCE MODE FLAG (maintenance mode = 1, operational mode = Ø). Assertion causes more detailed maintenance fault codes to be reported following the completion of IBIT, in addition to the summary pilot-oriented data otherwise reported.
13	CLR MFL	CLEAR MASTER FAULT LIST (clear = 1, do not clear = Ø). Assertion of this bit causes the ROM based master fault list to have its fault bits cleared.
12	INV RST	INVENTORY RESET (reset = 1, do not reset = Ø). Assertion of this bit causes the sequencer inventory to be set to full magazine status, in the configuration specified in the Mission Data File, for each magazine installed. (If pollable squibs are used, INV RST will result in an inventory based on polling.)
11-1		SPARES
Ø	IN FLT	IN FLIGHT INDICATOR (in flight = 1)

DATA WORD 2

BIT	FIELD	DESCRIPTION
15-Ø	TERMINAL TEST WORD:	Format: 16 bit unsigned integer Range: N/A

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NOTE: A minimum of 8Ø milliseconds is expected between issuing the BIT command and reading the response, to allow for CMDS latency in wrapping the terminal test word into the response.

The IBIT, MAT, CLR MFL, and INV RST flags are all detected as they transition from Ø to 1, and will not cause additional initiation of actions without another such transition. MAT, CLR MFL and INV RST are ignored if the CMDS is not in STANDBY mode or IBIT is not asserted. CLR MFL and INV RST are ignored unless MAT is asserted.

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APPENDIX B

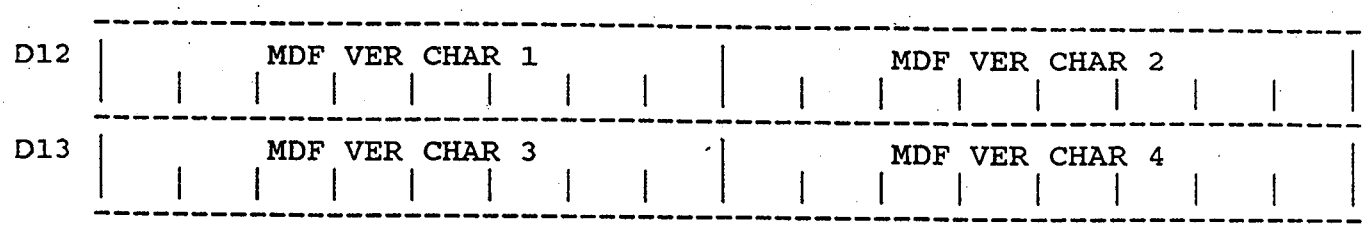
Message: BIT RESPONSE
Subaddress: ØFh
Frequency: 1/sec
Source: CMDS (RT)
Destination: MC (AV Bus Master)

	15	14	13	12	11	1Ø	9	8	7	6	5	4	3	2	1	Ø
	terminal address					T/R	subaddress					word count				
	Ø	1	Ø	1	Ø	1	Ø	1	1	1	1	Ø	1	1	Ø	1

APPROPRIATE STATUS WORD

D1	ALE-47 CONFIGURATION WORD															
D2	TERMINAL TEST WORD ECHO															
D3	BIT ON	EQP FLT	BIT CPL	EQP DEG	PGM WRA	S Ø	PGM SR1	PGM SR2	PGM SR3	PGM SR4	PGM SR5	PGM SR6	RESERVED (CCU)			
D4	RESERVED												S1 DE1	S1 DE2	S1 SR1	S1 SR2	S1 WRA
D5	RESERVED												S2 DE1	S2 DE2	S2 SR1	S2 SR2	S2 WRA
D6	C1 VLD	RESERVED					CMDS FAULT CODE 1										
D7	C2 VLD	RESERVED					CMDS FAULT CODE 2										
D8	Ø	Ø	1	1	1	Ø	Ø	1	Ø	Ø	1	1	Ø	Ø	Ø	1	
D9	Ø	Ø	Ø	Ø	Ø	Ø	1	1	Ø	Ø	1	Ø	Ø	1	Ø	1	
D10	OFP - 1								OFP - 2								
D11	OFP - 3								Ø	Ø	Ø	Ø	Ø	Ø	Ø	Ø	

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DATA WORD 1
ALE-47 CONFIGURATION WORD:
Format: 16 bit unsigned integer
Range: N/A

DATA WORD 2
TERMINAL TEST WORD ECHO:
Format: 16 bit unsigned integer
Range: N/A

DATA WORD 3

BIT	FIELD	DESCRIPTION
15	BIT ON	PBIT/IBIT IN PROGRESS (1 = BIT in progress)
14	EQP FLT	EQUIPMENT FAULT (1 = fault). Asserted whenever one or more BIT faults are detected.
13	BIT CPL	PBIT/IBIT COMPLETED (1 = BIT completed). Asserted whenever the previous power-up or initiated BIT is completed and reset when IBIT command is lowered.
12	EQP DEG	EQUIPMENT DEGRADE (1 = CMDS functionally degraded). Asserted whenever at least one of MAN DEG, SEM DEG, or AUT DEG is asserted in the CMDS STATUS RESPONSE message.
11	PGM WRA	PROGRAMMER WRA FAILURE FLAG (1 = failed)
1Ø		SPARE
9	PGM SR1	PROGRAMMER SRA 1 FAILURE FLAG (1 = failed)
8	PGM SR2	PROGRAMMER SRA 2 FAILURE FLAG (1 = failed)
7	PGM SR3	PROGRAMMER SRA 3 FAILURE FLAG (1 = failed)
6	PGM SR4	PROGRAMMER SRA 4 FAILURE FLAG (1 = failed)
5	PGM SR5	PROGRAMMER SRA 5 FAILURE FLAG (1 = failed)
4	PGM SR6	PROGRAMMER SRA 6 FAILURE FLAG (1 = failed)
3-Ø		Reserved

DATA WORD 4

BIT	FIELD	DESCRIPTION
15-5		Reserved
4	S1 DE1	Sequencer 1 Dispenser 1 FAILURE FLAG (1 = failed)
3	S1 DE2	Sequencer 1 Dispenser 2 FAILURE FLAG (1 = failed)
2	S1 SR1	SEQUENCER 1 SRA 1 FAILURE FLAG (1 = failed)
1	S1 SR2	SEQUENCER 1 SRA 2 FAILURE FLAG (1 = failed)
Ø	S1 WRA	Sequencer 1 WRA FAILURE FLAG (1 = failed)

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DATA WORD 5

BIT	FIELD	DESCRIPTION
15-5		Reserved
4	S2 DE1	Sequencer 2 Dispenser 1 FAILURE FLAG (1 = failed)
3	S2 DE2	Sequencer 2 Dispenser 2 FAILURE FLAG (1 = failed)
2	S2 SR1	SEQUENCER 2 SRA 1 FAILURE FLAG (1 = failed)
1	S2 SR2	SEQUENCER 2 SRA 2 FAILURE FLAG (1 = failed)
Ø	S2 WRA	Sequencer 2 WRA FAILURE FLAG (1 = failed)

DATA WORD 6

BIT	FIELD	DESCRIPTION
15	C1 VLD	CMDS FAULT CODE 1 VALID
14-1Ø		Reserved
9-Ø		CMDS FAULT CODE 1 Range: Ø...1Ø23 Codes: The codes correspond to fault messages displayed on the CMDS CCU. These messages are documented in the IFS.

DATA WORD 7

BIT	FIELD	DESCRIPTION
15	C2 VLD	CMDS FAULT CODE 2 VALID
14-1Ø		Reserved
9-Ø		CMDS FAULT CODE 2 Range: Ø...1Ø23 Codes: The codes correspond to fault messages displayed on the CMDS CCU. These messages are documented in the IFS.

DATA WORDS 8 AND 9
Contain the fixed MASCII character sequence: '91C-'

DATA WORD 1Ø
Contains the first two characters of the 3 character MASCII code for the ALE-47 OFP ID loaded at the time of transmission.

DATA WORD 11
Highbyte - Contains the third character of the OFP ID.
Lowbyte - Contains the fixed MASCII "space" character (blank).

DATA WORDS 12 AND 13
Contain the AN/ALE-47 MASCII character codes for the AN/ALE-47 MDF loaded at the time of transmission.

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APPENDIX B

ADDENDUM B

DESIGN SPECIFICATION DIFFERENCES BETWEEN MCAIR AND TRACOR

The differences described in this addendum are design differences which may cause incompatibilities between the AN/ALE-47 and the F-18 aircraft.

B.1 AVIONICS MUX BUS COMMANDS

The AN/ALE-47 MDF contains two bits (one for the jammer and one for the MWS) that permit the jammers and MWS to auto-dispense.

B.1.1 McAir's Position

McAir requires the capability to dynamically alter, via the Avionics BUS the control bit that may permit auto-dispensing by the jammer and MWS. McAir desires to insure that expendables will not be dispensed when in the semi-auto mode.

B.1.2 Tracor's Position

The RUN SILENT flag (bit 1 of word 2 of the CMDS Control Command message) has been redefined to assure that automatic dispensing is prohibited in the auto and semi-auto modes whenever this bit is set. Dispense can still be initiated by the pilot. Therefore this flag can be used to assure auto-dispensing does not take place.

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APPENDIX B

ADDENDUM C

CHANGE REQUEST PROPOSAL SHEETS

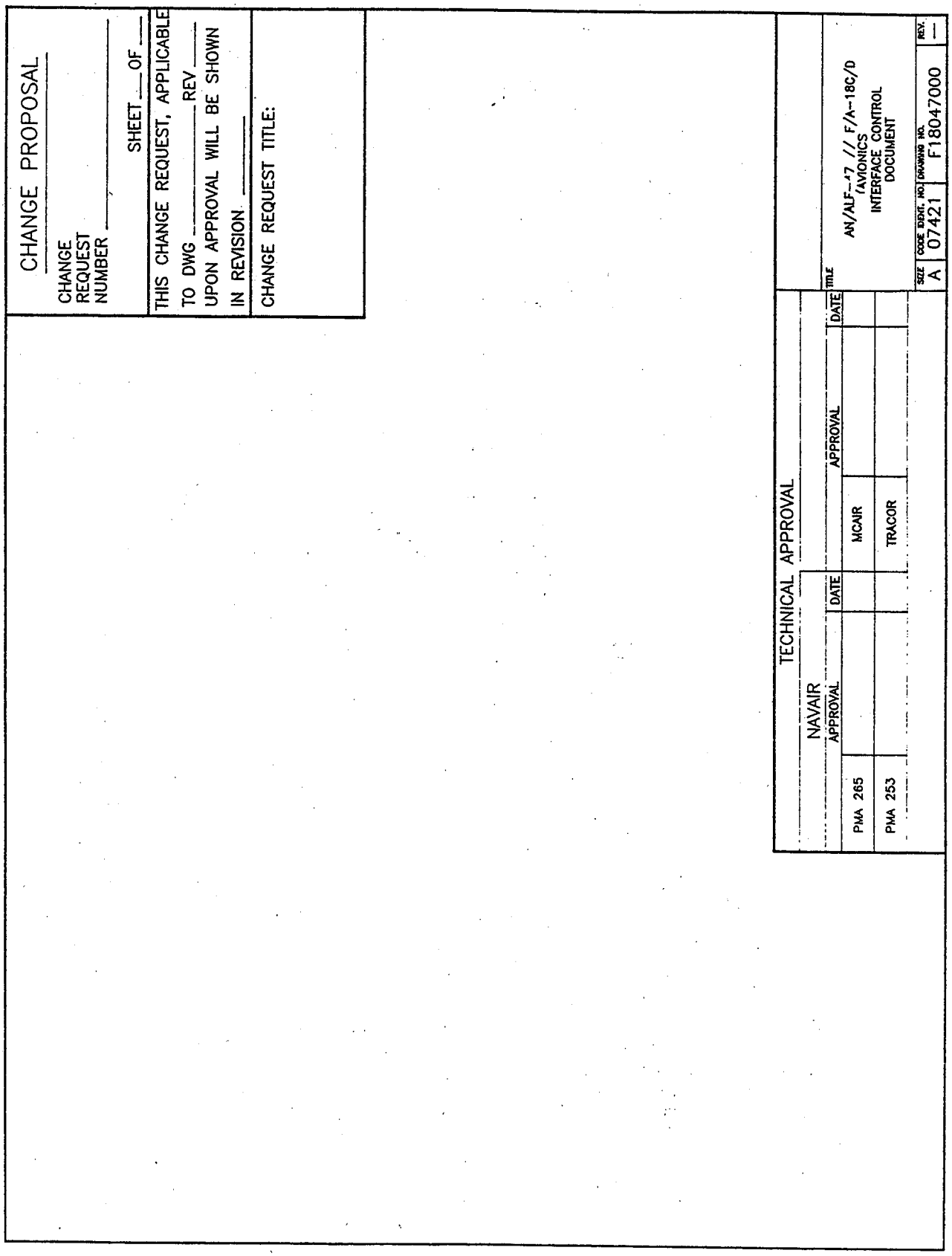
Changes to this document will be made using the Change Request Proposal Sheets contained in this Addendum. Forward proposed changes to:

Commander, Naval Air Warfare Center
Weapons Division - Point Mugu
Electronic Warfare System Integration Code P1343 (4043)
Point Mugu, CA 93042-5001

PH (805) 989-1343/1344
AV 351-1343/1344

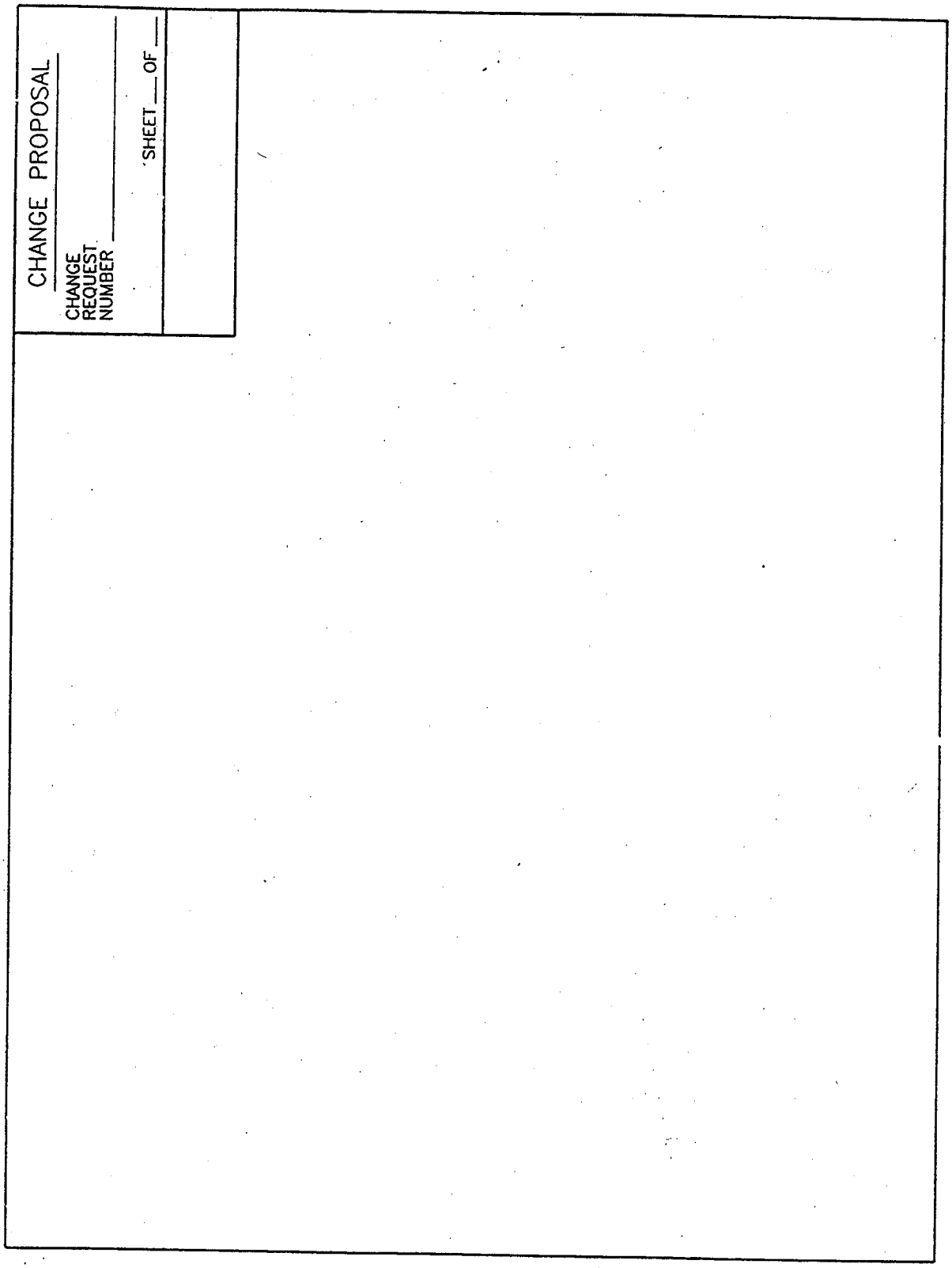
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APPENDIX B



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APPENDIX B



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APPENDIX B

SAMPLE II. SYSTEM-TO-SYSTEM ICD

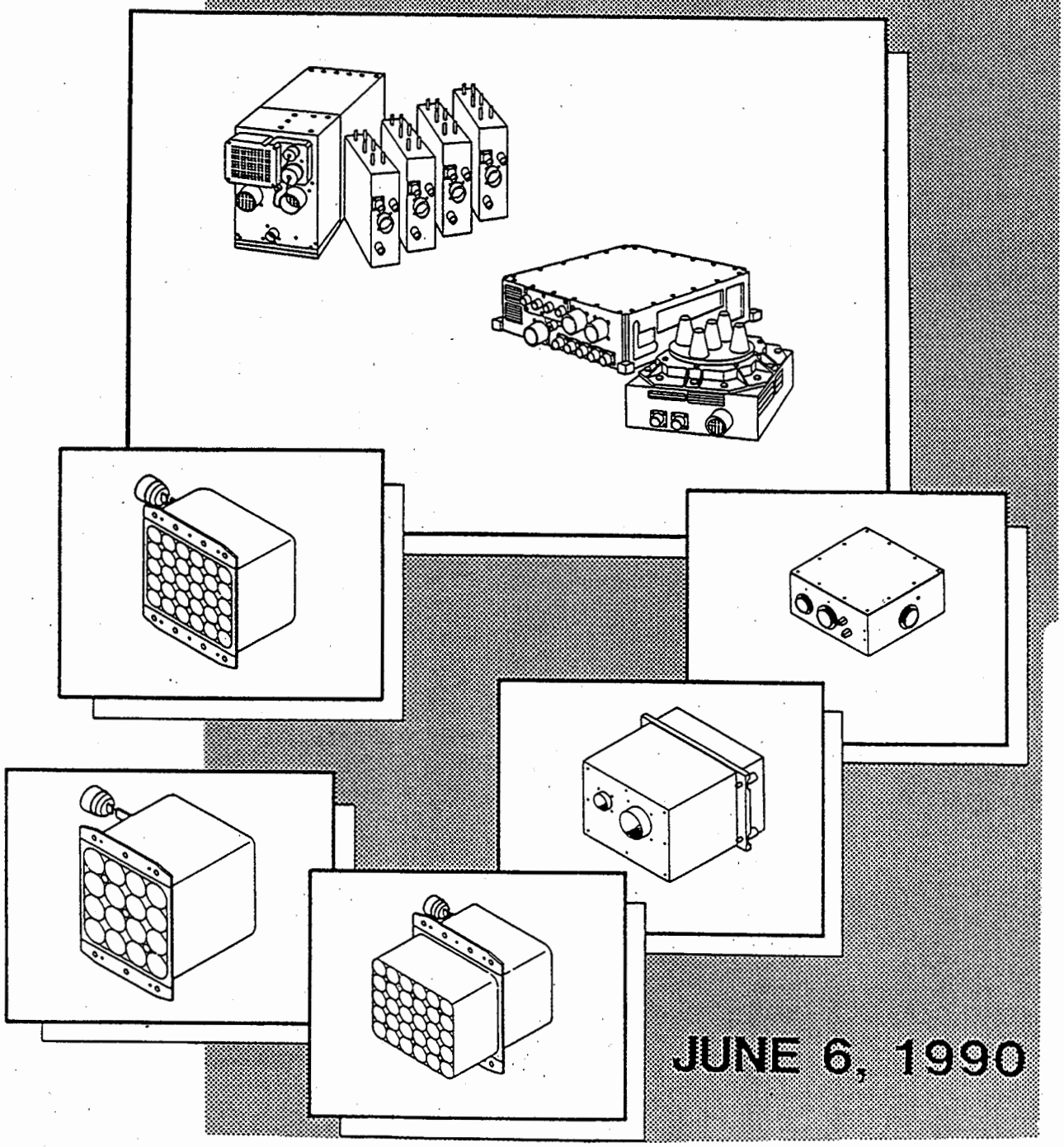
(DIRECT INTERFACE)

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PACIFIC MISSILE TEST CENTER

AN/ALE-47 // AN/ALR-67B (V)2 INTERFACE CONTROL DOCUMENT #000047067



JUNE 6, 1990

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APPENDIX B

TABLE iA

INTERFACE DESIGN DATA REQUIRED (IDDR) TABLE					
IDDR NUMBER	IDDR TITLE	SHEET NUMBER	DUE DATE	SUPPLIER	ACCEPTANCE
3	AN/ALE-47 AND AN/ALR-67(V)2 SOFTWARE CONFIGURATION	7.1		TRACOR/LITTON	

TABLE iB

COMPLETED IDDR's					
IDDR NUMBER	IDDR TITLE	SHEET NUMBER	DUE DATE	SUPPLIER	ACCEPTANCE
1	INTERFACE UNIT PART NUMBER	5.1		LITTON	SATISFIED
2	PERIOD OF INTERMESSAGE GAP	4.1		LITTON	SATISFIED

TABLE iC

DESIGN SPECIFICATION DIFFERENCES (DETAILS PROVIDED IN ADDENDUM B)	
DESCRIPTION	SHEET
MESSAGE UPDATE CRITERIA FOR AOA CHANGES	B-1
INCLUSION OF SIGNAL UPDATE MESSAGE	B-1
SELECTED SIGNAL REQUEST MESSAGE	B-2
JAMMER DECOYS MESSAGE	B-4
BIT DISPLAY MESSAGE	B-7

AVIONICS INTERFACE
CONTROL DOCUMENT (AVICD)

TECHNICAL APPROVAL					
NAVAIR			AN/ALE-47 // AN/ALR-67(V)2		
APPROVAL		DATE	APPROVAL		DATE
PMA 253			TRACOR	Jerri Sanders	6/21/90
			LITTON		

INTERSTATE ELECTRONICS CORPORATION
A FIGGIE INTERNATIONAL COMPANY
1001 E. BALL ROAD P.O. BOX 3117 ANAHEIM CA. 92803 (714) 758-0500

TITLE: AN/ALE-47 // AN/ALR-67(V)2 AVIONICS INTERFACE CONTROL DOCUMENT

SIZE	CODE IDENT. NO.	DRAWING NO.	REV.
A	07421	000047067	-

SCALE NONE | REV BY—FILE DATE | SHEET

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APPENDIX B

EFFECTIVE SHEETS

SHEET NUMBERS	CHANGE IN EFFECT	DATE ISSUED
i - vii	ORIGINAL ISSUE	June 6, 1990
1.1 - 1.2	ORIGINAL ISSUE	June 6, 1990
3.1	ORIGINAL ISSUE	June 6, 1990
4.1 - 4.2	ORIGINAL ISSUE	June 6, 1990
5.1	ORIGINAL ISSUE	June 6, 1990
7.1	ORIGINAL ISSUE	June 6, 1990
ADDENDUM A	ORIGINAL ISSUE	June 6, 1990
ADDENDUM B	ORIGINAL ISSUE	June 6, 1990
ADDEMDUM C	ORIGINAL ISSUE	June 6, 1990

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APPENDIX B

CHANGE REQUEST (CR) RECORD

CR NUMBER	DATE	TITLE/DESCRIPTION	SHEETS AFFECTED	COGNIZANT CONTRACTOR
01	7/90	Clean-up pages to obtain approval of Rev -	1.1, 4.1, 4.2, and A-6	I.E.C

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APPENDIX B

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LIST OF ABBREVIATIONS/ACRONYMS

ASPJ	AIRBORNE SELF-PROTECTION JAMMER
AV	AVIONICS
AVICD	AVIONICS INTERFACE CONTROL DOCUMENT
CMDS	COUNTERMEASURES DISPENSING SYSTEM
DECM	DECEPTIVE ELECTRONIC COUNTERMEASURES
DISP	DISPENSER
IDDR	INTERFACE DESIGN DATA REQUIRED
MAX	MAXIMUM
MDC	MCDONNELL DOUGLAS CORPORATION
ME	MESSAGE ERROR
MLV	MEMORY LOADER VERIFIER
MUX	MULTIPLEXER
P	PARITY
RT	REMOTE TERMINAL
RWR	RADAR WARNING RECEIVER
SDL	SEQUENCER DATA LINK
SW	SWITCH
SYNC	SYNCHRONOUS
T/R	TRANSMIT/RECEIVE
UDF	USER DATA FILE
WOW	WEIGHT ON WHEELS

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APPLICABLE DOCUMENTS

The following documents of the exact issue indicated form a part of this AVICD to the extent specified herein.

MILITARY DOCUMENTS

MIL-C-85332(AS)	AN/ALR-67, COMPUTER

ITT DOCUMENTS

2662616	CLASSIFIED SUPPLEMENT FOR HARM/JAMMER INTERFACE CONTROL DOCUMENT, REV. A, 10/86
2662613	HARM TO JAMMER INTERFACE CONTROL DOCUMENT, REV. A, 10/86
2661728	RWR TO AN/ALQ-165 (ASPJ) WORD FORMATS

TRACOR DOCUMENTS

CP64345	System Specification for the Prime Mission Equipment, Dispensing System, Countermeasures, AN/ALE-47
CP64345	Interface Specification AN/ALE-47 Countermeasures Dispensing System (CMDS) to Aircraft and Associated Equipment

McDONNELL DOUGLAS DOCUMENTS

MDC A3818, Rev B	F/A-18 Avionics Multiplex Design Specification

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1.0 INTRODUCTION

1.1 Scope

This document is an Avionics Interface Control Document AV ICD) between the AN/ALE-47 Countermeasures Dispensing System (CMDS) and the AN/ALR-67B(V)2 Radar Warning Receiver (RWR). This document establishes the requirements for data and electrical interfaces.

For information only, additional interface requirements will be contained in the following documents:

000047004 (AN/ALE-47 // HARM CLC)
000047126 (AN/ALE-47 // AN/ALQ-126B DECM JAMMER)
000047165 (AN/ALE-47 // AN/ALQ-165 ASPJ)
000047687 (AN/ALE-47 // AN/ASM-687 MLV)

1.2 Description

The AN/ALR-67B(V)2 Countermeasures Receiving Set (commonly referred to as an RWR) is a digitally controlled system on fighter and attack aircraft designed to intercept, detect, evaluate and display emitters in the high and low radio frequency bands of a dense radar signal environment. It is designed to hand off specified threat signals and their status to on-board jammers and the HARM missile system. The AN/ALR-67B(V)2 is the bus controller that controls the electronic warfare serial data bus and works cooperatively with on-board jammers, missile systems, the fire control radar, and the aircraft mission computer.

The AN/ALE-47 CMDS is a threat-adaptive dispenser system designed to interface with an aircraft's AV suite, determine the nature of incoming threats, and automatically dispense appropriate expendables. The CMDS is compatible with the same payload types as the AN/ALE-39 with greater integration and automation for the pilot. With its data fusion capability from other AV suite sensors and aircraft avionics, threat adaptive dispensing will provide the optimum and most efficient countermeasures deployment, thus freeing the pilot to concentrate on maneuvering techniques and mission goals. The AN/ALE-47 CMDS acts as a remote terminal.

1.3 Effectivity

This ICD is applicable to any platform containing both the AN/ALE-47 CMDS and the AN/ALR-67B(V)2 RWR.

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2.0 NOTATION

2.1 Symbols



Indicates interface 1. Interface numbers located within triangles throughout this document are referenced to interfaces on sheet 3.1.



Indicates additional data shown on sheet 4.1, zone 4C. In the event that related information is spread out over more than one sheet, only the first sheet will be indicated in this manner.

2.2 Table Nomenclature

Table 4.2A indicates the first table on sheet 4.2.

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The digital copy of MIL-STD-2222 w/ Appendix "B", dated 12 AUG 1992 does not include foldout pages 49 thru 88, 139 thru 148, 225 thru 230 and 243 thru 246. Please call the ASSIST SERVICE DESK at 215-737-8000 for copies of these pages.

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ADDENDUM A
to ICD 000047067, AN/ALE-47 // AN/ALR-67B(V)2 EWICD

AV**MUX BUS INTERFACE COMMAND, DATA, AND STATUS MESSAGES**

This protocol definition defines the Avionics bus interface between the CMDS (a remote terminal) and the AN/ALR-67B(V)2 RWR or the AN/ALR-45F (except as noted in the remarks of the self-test message). The header words are defined on sheet 4.2 of ICD 000047067.

CONTENTS OF ADDENDUM A

	PAGE
DATA MESSAGE DIRECTORY	
Sub-Contents for AN/ALR-67B(V)2 to AN/ALE-47 AV MUX Bus Messages	A-2
Sub-Contents for AN/ALE-47 to AN/ALR-67B(V)2 AV MUX Bus Messages	A-3
Receive Messages (to AV Equipments)	A-4
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INTERDIGITAL PROCESSOR COMMUNICATION	
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Data Transfer	A-6
Handshake	A-7
AV Bus Message	A-8
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DATA MESSAGE FORMATS	
AN/ALR-67B(V)2 to AN/ALE-47 AV MUX Bus Messages	A-11 to A-36
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SUB-CONTENTS OF AN/ALR-67B(V)2 to AN/ALE-47 AV MUX BUS MESSAGES

Message Name	Message Subaddress		Page
Signal Message	00001	(01h)	A-11
Reserved (for Signal Update)	00010	(02h)	
Signal Delete Message	00011	(03h)	A-15
	00100	(04h)	
Ambiguity Message	00101	(05h)	A-17
HARM Launch Message	00110	(06h)	A-19
Invalid Request Message	00111	(07h)	A-20
Reserved (for Part Track File)	01000	(08h)	
Reserved	01001	(09h)	
Configuration Message	01010	(0Ah)	A-21
	01011	(0Bh)	
AN/ALR-45F or AN/ALR-67B(V)2 Self Test	01100	(0Ch)	A-23
Jamming Assignments Message	01101	(0Dh)	A-25
HARM Status Message	01110	(0Eh)	A-28
	01111	(0Fh)	
	10000	(10h)	
Reserved	10001	(11h)	
Reserved	10010	(12h)	
Reserved	10011	(13h)	
Reserved	10100	(14h)	
RT to RT Message [Growth]	10101	(15h)	
Reserved	10110	(16h)	
Reserved	10111	(17h)	
Reserved	11000	(18h)	
Reserved	11001	(19h)	
Reserved	11010	(1Ah)	
Reserved	11011	(1Bh)	
Reserved	11100	(1Ch)	
Reserved (MLV)	11101	(1Dh)	
Dispense Request Message	11110	(1Eh)	A-30
	11111	(1Fh)	

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SUB-CONTENTS OF AN/ALE-47 to AN/ALR-67B(V)2 AV MUX BUS MESSAGES

Message Name	Message Subaddress		Page
Reserved	00001	(01h)	
	00010	(02h)	
Reserved (Selected Signal Request Message)	00011	(03h)	*
	00100	(04h)	
	00101	(05h)	
Reserved (Jammer Decoys Message)	00110	(06h)	*
Poll Command and Response Messages	00111	(07h)	A-37
Reserved	01000	(08h)	
Reserved	01001	(09h)	
	01010	(0Ah)	
Reserved	01011	(0Bh)	
Reserved	01100	(0Ch)	
Reserved	01101	(0Dh)	
Reserved	01110	(0Eh)	
Emitter ID Preselect Message	01111	(0Fh)	A-39
Reserved	10000	(10h)	
Reserved	10001	(11h)	
Reserved	10010	(12h)	
Reserved (for MLV Configuration)	10011	(13h)	
Reserved	10100	(14h)	
RT-RWR-RT pass-through (Growth)	10101	(15h)	
Reserved	10110	(16h)	
Reserved (MLV)	10111	(17h)	
Reserved (Bit Display Message)	11000	(18h)	*
	11001	(19h)	
	11010	(1Ah)	
	11011	(1Bh)	
Expendable Set-up Message	11100	(1Ch)	A-40
	11101	(1Dh)	
CMDS Status/Inventory Message	11110	(1Eh)	A-43
	11111	(1Fh)	

* Refer to Addendum B

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********** RECEIVE MESSAGES (TO AV EQUIPMENTS) -- Provided for Information Only **********

	ALE-47	ALQ-126B	ALQ-162	ALQ-165	IDAP	JTIDS TACTS
01	Signal	New Signal	New Signal		New Signal	
02	Reserved	Signal Update	Signal Update		Signal Update	
03	Signal Delete	Signal Delete	Signal Delete	Confirmation 74	Signal Delete	
04		TCV Status				
05	Ambiguity List	Ambiguity List				
06	HARM Launch	HARM Launch	HARM Launch	HARM Launch	HARM Launch	
07	Invalid Request	Invalid Request	Invalid Request	Invalid Request		
08	Reserved	Part Track File	Part Track File		Part Track File	
09	Reserved	Sel Signal Dump			Sel Signal Dump	
0A	Configuration	Configuration	Configuration		Configuration	
0B		Radar Data	Radar Data			
0C	45F/67 Self Test		45F/67 Self Test			
0D	Jam Assignments	Jam Assignments				
0E	HARM Status	HARM Status	HARM Status	HARM Status		
0F			TACTS Data			
10			Blanking Setup			
11	Reserved			Messages 74		
12	Reserved			Invalid Request	Emit Shut-Down Req	
13	Reserved	Expendable Set-up	Expendable Set-up	Expendable Set-up		
14	Reserved			Lookthru Setup (02)		
15	[Growth RT - RT]			TCV Status		
16	Reserved			67 Test Data (09)		
17	Reserved			New Signal (07)		
18	Reserved	(Nav Data)	(Nav Data)	(Nav Data)		
19	Reserved			Req 165 Test Data		
1A	Reserved			RWR Mode/Pri (0C)		
1B	Reserved			Signal Update		
1C	Reserved			Signal Delete		
1D	Reserved MLV	Reserved MLV	Reserved MLV	Reserved MLV	Reserved MLV	
1E	Dispense Request	(Status/Inventory)	(Status/Inventory)	(Status/Inventory)		
1F						

Parentheses indicate suggested messages.
Brackets indicate intent of reserved sub-addresses

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********** TRANSMIT MESSAGES (FROM AVEQUIPMENTS) -- Provided for Information Only **********

	ALE-47	ALQ-126B	ALQ-162	ALQ-165	IDAP	JTIDSTACTS
01	Reserved	TCV Request		Jam Assignment		
02	Reserved	Part Track Req	Part Track Req			
03	Sel Signal Req	Sel Signal Req				
04		Interface Test	Interface Test			
05		Self Test Data	Self Test Data			
06	Jammer Decoys	126B Jam Assign				
07	Poll	Poll	Poll	Track File Req		
08	Reserved	Data Dump				
09	Reserved	Correlation		67 Test Data Req		
0A		Radar Request				
0B	Reserved	Freq Meas Req				
0C	Reserved	126B Config	162 Config	RWR Mode Req		
0D	Reserved		162 Jam Assign	(JLB Setup)		
0E	Reserved		TACTS Response	(Jam Chan Status)		
0F	ID Preselect		ID Preselect			
10	Reserved			Poll		
11	Reserved					
12	Reserved					
13	MLV Config	MLV Config	MLV Config	MLV Config	(MLV Config)	
14	Reserved					
15	[Growth RT - RT]	[Growth RT - RT]	[Growth RT - RT]	[Growth RT - RT]		
16	Reserved					
17	Reserved (MLV)					
18	(Bit Display)					
19						
1A						
1B						
1C	Expendable Set-Up					
1D						
1E	Status/Inventory	(Dispense Req)	(Dispense Req)	(Dispense Req)		
1F						

Parentheses indicate suggested messages.
Brackets indicate intent of reserved sub-addresses

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INTERDIGITAL PROCESSOR COMMUNICATION

Communication Control - Interdigital processor communication on the multiplex bus shall be controlled by a poll and poll response method. The AN/ALR-67B(V)2 acting as AV bus controller shall periodically poll to determine if an AN/ALE-47 is present and to determine if it wishes to send a message. If the AN/ALE-47 is present it shall respond with a poll response. The AN/ALE-47 Terminal Address shall be 01010 (0A Hex).

Once communication between the AN/ALR-67B(V)2 and the AN/ALE-47 is established, the AN/ALR-67B(V)2 shall normally poll once every 20 milliseconds. Excluding the poll and transmit command messages, no more than a total of 256 words, which includes two words for overhead, may be sent from the AN/ALR-67B(V)2 to the AN/ALE-47 per polling period.

If the AN/ALE-47 cannot process more data it shall respond to a poll with the busy bit set (bit 3 in the status word). The AN/ALR-67B(V)2 shall not send any messages other than a poll until the busy bit is cleared in a subsequent poll response.

Should the AN/ALE-47 expend its entire inventory of expendables, the AN/ALR-67B(V)2 will change to the slow poll rate for the AN/ALE-47 (consisting of one group of one to three poll words during a one second period).

Data transfer - Data transfers on the EW bus can be between the AN/ALE-47 and the AN/ALR-67B(V)2. The protocols to be used for different types of data transfers are described in this section.

AN/ALR-67B(V)2 to AN/ALE-47

Automatic AN/ALR-67B(V)2 to AN/ALE-47 - If the AN/ALR-67B(V)2 wishes to send data to the AN/ALE-47 it shall automatically send the message with the transmit/receive bit of the command word set to 0 for receive. If the busy bit of the last AN/ALE-47 poll response was set to 1, the AN/ALR-67B(V)2 shall wait to send the data until the AN/ALE-47 sends a poll response with the busy bit cleared. The AN/ALR-67B(V)2 will automatically send the Signal message on all emitters of interest to the AN/ALE-47 (as determined by the ID Preselect message from the AN/ALE-47) whenever detectable criteria exceed those listed below. The Signal message will be sent on initial detection on all emitters, and the Signal message will be sent whenever any emitter transitions to a lethal status (such an emitter will then be reported as changes occur as listed below, if it had not been previously).

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Data requested from AN/ALR-67B(V)2 - If the AN/ALE-47 requires data from the AN/ALR-67B(V)2 it will respond to an AN/ALR-67B(V)2 poll command with the service request bit set to one (bit 8 of the status word). It shall also use the appropriate fields of the poll response data word to indicate which message it wishes the AN/ALR-67B(V)2 to send. The AN/ALR-67B(V)2 shall then send the appropriate message to the AN/ALE-47.

AN/ALE-47 to AN/ALR-67B(V)2

AN/ALR-67B(V)2 requested data - The AN/ALR-67B(V)2 requests data from the AN/ALE-47 by sending a message to the AN/ALE-47 specifying what data it wants sent. The AN/ALE-47 will respond to an AN/ALR-67B(V)2 poll command with the service request bit set to one (bit 8 of the status word). It shall also use the appropriate fields of the poll response data word to indicate which message it wishes the AN/ALR-67B(V)2 to send. The AN/ALR-67B(V)2 shall then send a transmit command to the AN/ALE-47 telling it to send the data. The service request bit is not set to one in the status word for any response other than the poll response; it has no meaning for other responses, and is ignored.

Once communication between the AN/ALR-67B(V)2 and the AN/ALE-47 is established, if the AN/ALR-67B(V)2 fails to poll the AN/ALE-47 within a two second period, the AN/ALE-47 determines that the AN/ALR-67B(V)2 is not functioning correctly and operates independently.

Handshake

On power-up or reset of the AN/ALR-67B(V)2:

AN/ALR-67B(V)2 will poll the AN/ALE-47 at the slow rate (1 poll per second) and command receipt of the Configuration Message with the first-time bit (flag) set to one.

AN/ALE-47 responds with a valid status word. Communication is then considered established. The data of any poll response received by the AN/ALR-67B(V)2 prior to receipt of a valid status word for the configuration message is ignored.

Then begins the 20 millisecond polling cycle. AN/ALE-47 requests to send CMDS Status/Inventory, Expendable Set-up, and ID Preselect messages. The Self Test message may also be requested at this time.

When all the required message transmissions are completed, the Initialization Handshake Sequence is considered complete. When the Initialization Handshake is completed, the AN/ALR-67B(V)2 will relay the CMDS Status/Inventory and Expendable Set-up Messages to the jammer(s), indicating that the AN/ALE-47 is working with the AN/ALR-67B(V)2. If the Initialization Handshake is not completed, the AN/ALR-67B(V)2 will not service any of the AN/ALE-47's requests and the AN/ALE-47 will not be able to communicate with any other

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equipment via the AV mux bus. The poll rate drops back to the slow poll rate of 1/second.

On power-up or reset of the AN/ALE-47:
If the AN/ALE-47 is reset (detected by RST bit in Expendable Set-up Message) then the above sequence will be completed again; the first time bit reflects the appropriate power-up status (or lack thereof).

In the event that another system on the AV bus resets (e.g. Jammer), the RTM bit in a Jamming Assignments Message will cause the AN/ALE-47 to send an Expendable Set-up and a CMDS Status/Inventory message for retransmission to the reset remote terminal.

Detailed Initialization Handshake (after establishment of communications)

a) Optional: AN/ALE-47 transmits Expendable Set-up Message with RST set (in case the AN/ALE-47 reset)
b) AN/ALR-67B(V)2 sends Configuration Message (First Time bit set if the AN/ALR-67B(V)2 has reset)
c) Expendable Set-up Message from AN/ALE-47
d) ID Preselect Message from AN/ALE-47
e) CMDS Status/Inventory from AN/ALE-47

At this point, the link is considered initialized. The AN/ALR-67B(V)2 subsequently reports on any found emitters requested by the ID Preselect messages in its current track file, but this is not part of the initialization handshake. Similarly, the AN/ALE-47 may request the reception of the Self Test Message from the AN/ALR-67B(V)2 at any time after the handshake is complete, while the AN/ALR-67B(V)2 may send this message at any time.

Bus Message

Consists of command, data, and status word(s).

Details

Additional details of communications between the AN/ALE-47 and the AN/ALR-67B(V)2 which are not covered above are included here.

Mode commands: **The AN/ALE-47 will not lock up when Mode Commands are received across the AV mux bus.**

Broadcast messages: **There are currently no broadcast messages on the AV bus.**

Reserved subaddresses are identified in the message lists.

The AN/ALE-47 decides that the AN/ALR-67B(V)2 is no longer available or is in a degraded mode based on the AN/ALR-67B(V)2 Self-Test and a (reprogrammable) 2 second time-out.

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The AN/ALR-67B(V)2 decides that the AN/ALE-47 has died or gone to a degraded mode per 1553 standard bit usage and the Status message as defined in MDC A3818 Rev B.

The AN/ALR-67B(V)2's minimum intermessage gap on the AV bus is one msec.

The message retransmission time is approximately 170 usec from detection of error to command word retransmission.

In the event of a data transmission error, three tries (attempts) for each error will occur. Timing error, parity error, and message error protocols will drive retransmission attempts.

Under the Signal message, if the AN/ALE-47 was busy and the AN/ALR-67B(V)2 has data waiting but the AN/ALE-47 then decides it needs to transmit data, the priority scheme for transmission is: AN/ALR-67B(V)2 transmits, polls; then the AN/ALE-47 transmits.

The AN/ALE-47 terminal address is determined by external pins which determine aircraft installation. It is specified by an internal value, reprogrammable by MLV download. The terminal address used during MLV download is determined by the external pins and an internal value not modifiable by MLV download.

Generic Expendable (GEN-X) reprogramming occurs internally to the AN/ALE-47 based on the emitter ID passed to it from the AN/ALR-67B(V)2. This also occurs in a similar manner for expendables requested via the Dispense Request Message; the GEN-X SPeCial handling field of this message causes a Signal Message to be sent by the AN/ALR-67B(V)2 prior to the relay of the Dispense Request.

The AN/ALR-67B(V)2 will automatically transmit a Signal Message to the AN/ALE-47 when:

- Emitter ID has changed
- Track is lost on an Emitter
- Emitter is redetected
- Magnitude of DF is changed $\geq \pm$ 10 degrees
- Amplitude is changed $\geq \pm$ 10 dB
- HARM designated status has changed
- Missile Alert/Missile Launch status has changed
- GEN-X is requested for a threat not identified in the ID Preselect Message

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Additional Information

The following data transfers are not currently implemented on the bus. The following is provided for information only.

Remote terminal to AN/ALE-47 transfer

If an bus remote terminal wishes to send a message to the AN/ALE-47 it shall respond to an AN/ALR-67B(V)2 poll command with a transmit request message. The AN/ALR-67B(V)2 shall send a transmit command message to the remote terminal allowing it to send the message. After receiving the data from the remote terminal, the AN/ALR-67B(V)2 shall send the data message automatically to the AN/ALE-47. The AN/ALE-47 shall send data to other remote terminals in the same manner.

AN/ALE-47 data requested by remote terminal

If an bus remote terminal wishes to receive data from the AN/ALE-47, it shall respond to an AN/ALR-67B(V)2 poll command with a transmit request message. The AN/ALR-67B(V)2 shall send a transmit command message to the remote terminal. After receiving the request from the remote terminal the AN/ALR-67B(V)2 shall send the message to the AN/ALE-47; the data in this message indicates the nature of the request to the AN/ALE-47. When the AN/ALE-47 has the data requested by the remote terminal, it shall respond to a poll command with a transmit request. The AN/ALR-67B(V)2 shall then send a transmit command to the AN/ALE-47. After receiving the data from the AN/ALE-47, the AN/ALR-67B(V)2 shall send the data to the requesting remote terminal. The AN/ALE-47 shall request data from other remote terminals in the same manner.

Data Representation

Spare bits (denoted by SPR) are set equal to "0". Reserved bits (denoted by RSV) are not guaranteed to be 0. Sign bits (denoted by SGN) values are + = 0, - = 1.

Bits marked "Spare" on input messages from BC should be masked out before message parsing, as they can't be 100% trusted to be 0. Bits marked "RSV" are used for other purposes, and contain values undefined by this interface. Spare bits transmitted by the CMDS are expected to be 0, but future modifications may result in utilization of these bits; accordingly, they should also be masked out until/unless explicitly included in this interface.

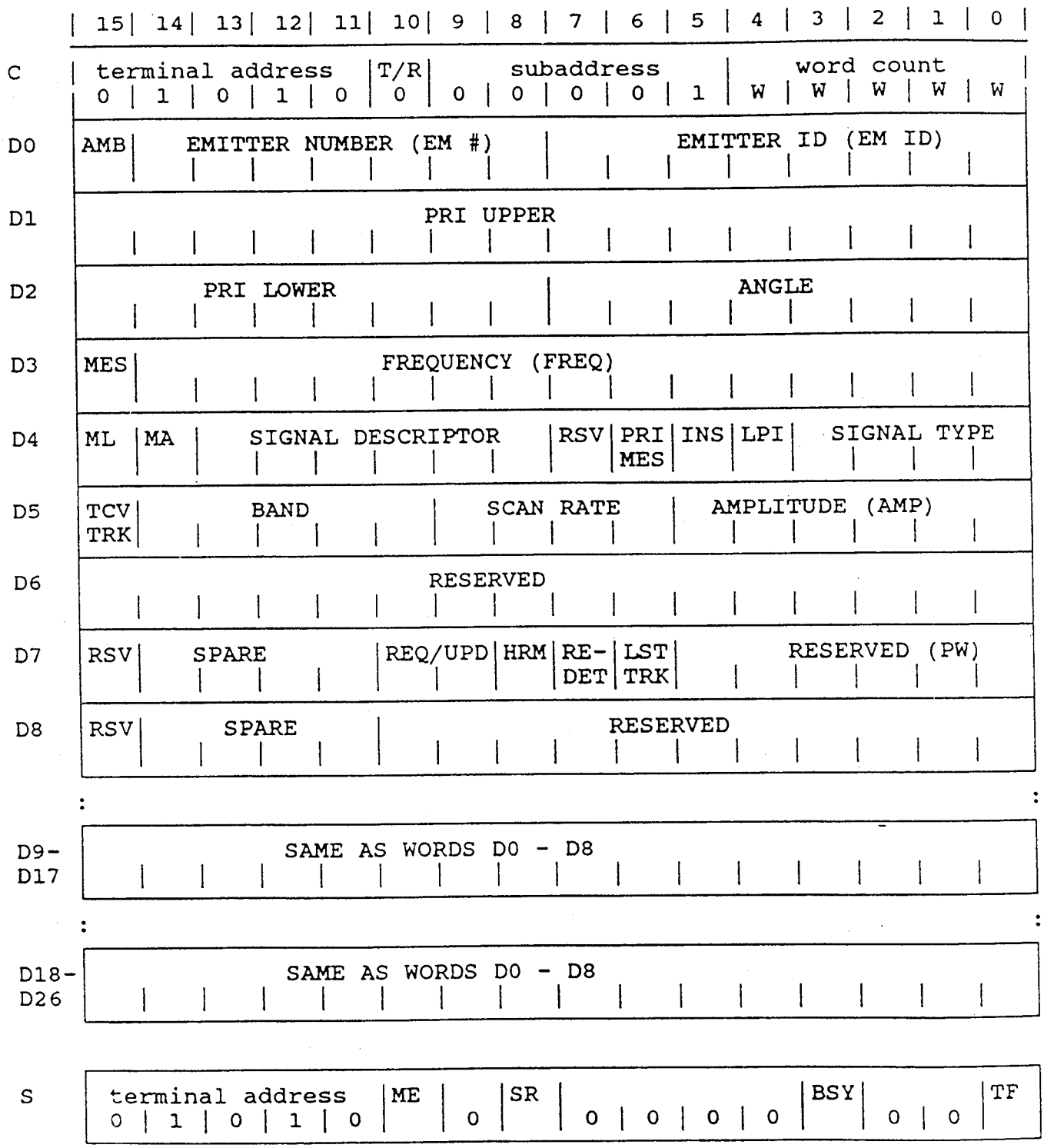
A bit value of "1" means the discrete or flag is asserted (true/on), unless otherwise stated. The bit value may be further defined in greater detail in some messages.

Subaddresses followed by an "h" indicate a hexadecimal subaddress.

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SIGNAL MESSAGE

Sub-Address: 00001



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APPENDIX B

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive. Set to 0 for Receive.
9-5	SA	SubAddress. Signal Message = 00001.
4-0	W	Word count. 01001 = Nine data words, one signal. 10010 = Eighteen data words, two signals. 11011 = Twenty-seven data words, three signals.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	AMB	1 = AMBiguous
14-8	EM#	AN/ALR-67B(V)2 Track List Reference number
7-0	EM ID	AN/ALR-67B(V)2 Signal Type identifier

DATA WORD 1

BIT	FIELD	DESCRIPTION
15-0	PRI UP	16 MSBs of PRI (LSB = 1 usec)

DATA WORD 2

BIT	FIELD	DESCRIPTION
15-8	PRI LO	8 LSBs of PRI (LSB = 3.9 nsec)
7-0	ANGLE	Average angle (LSB = 1.4°) (Aircraft referenced bearing)

DATA WORD 3

BIT	FIELD	DESCRIPTION
15	MES	1 = MEaSured frequency 0 = Not Measured
14-0	FREQ	Absolute FREQuency (LSB = 1 MHz)

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DATA WORD 4

BIT	FIELD	DESCRIPTION
15	ML	1 = Missile Launch
14	MA	1 = Missile Alert
13-8	SIG DES	SIGnal DEScriptor. The interpretation of this field depends on the Signal Type field contained in this data word (bits 0-3). If the Signal Type is Stable, then bits 8-11 of the Signal Descriptor field describe a Guidance Pulse Group as follows: 0 = No Guidance Pulse Group 1 = Type 1 Guidance Pulse Group 2 = Type 2 Guidance Pulse Group 3 = Not Assigned If the Signal Type is Jitter, then bits 12-13 of the Signal Descriptor field describe the amount of Jitter as follows: 0 = 0 to .5 usec 1 = .5 to 2 usec 2 = 2 to 10 usec 3 = Greater than 10 usec If the Signal Type is Stagger, then bits 8-13 of the Signal Descriptor field give the Stagger Code as defined in the Emitter Library (See Classified Addendum). If the Signal Type is Crystal uniquely, then bits 8-13 of the Signal Descriptor field give the Crystal Code as defined in the Emitter Library (See Classified Addendum).
7	RSV	ReSerVed
6	PRI MES	0 = PRI not MEaSured 1 = PRI MEaSured
5	INS	1 = RWR in INS mode
4	LPI	1 = Low Probability of Intercept Emitter
3-0	SIG TYP	SIGnal TYPe (See Classified Addendum)

DATA WORD 5

BIT	FIELD	DESCRIPTION
15	TCV TRK	See Classified Addendum
14-10	BAND	AN/ALR-67B(V)2 BAND number (See Classified Addendum)
9-6	SCAN	See Classified Addendum
5-0	AMP	See Classified Addendum

DATA WORD 6

Reserved for format compatibility between RWRs and AN/ALQ-165.

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DATA WORD 7

BIT	FIELD	DESCRIPTION
15	RSV	ReSerVed
14-11	SPR	SPaRe
10-9	REQ/UPD	00 = New Emitter 01 = UPDated Emitter 10 = Undefined 11 = REQuested by CMDS
8	HRM	1 = HARM designated emitter
7	RE-DET	1 = Emitter previously reported lost (bit 6 set = 1) has been RE-DETected (i.e., age out counter is incrementing)
6	LST TRK	1 = AN/ALR-67B(V)2 age out counter is decrementing

***

Note: when bits 7-6 = 00, emitter is currently active

***

5-0	RSV	ReSerVed (used by AN/ALR-56M for Pulse Width)

DATA WORD 8

BIT	FIELD	DESCRIPTION
15	RSV	ReSerVed
14-11	SPR	SPaRe
10-0	RSV	ReSerVed (for format compatibility between RWRs and AN/ALQ-165.)

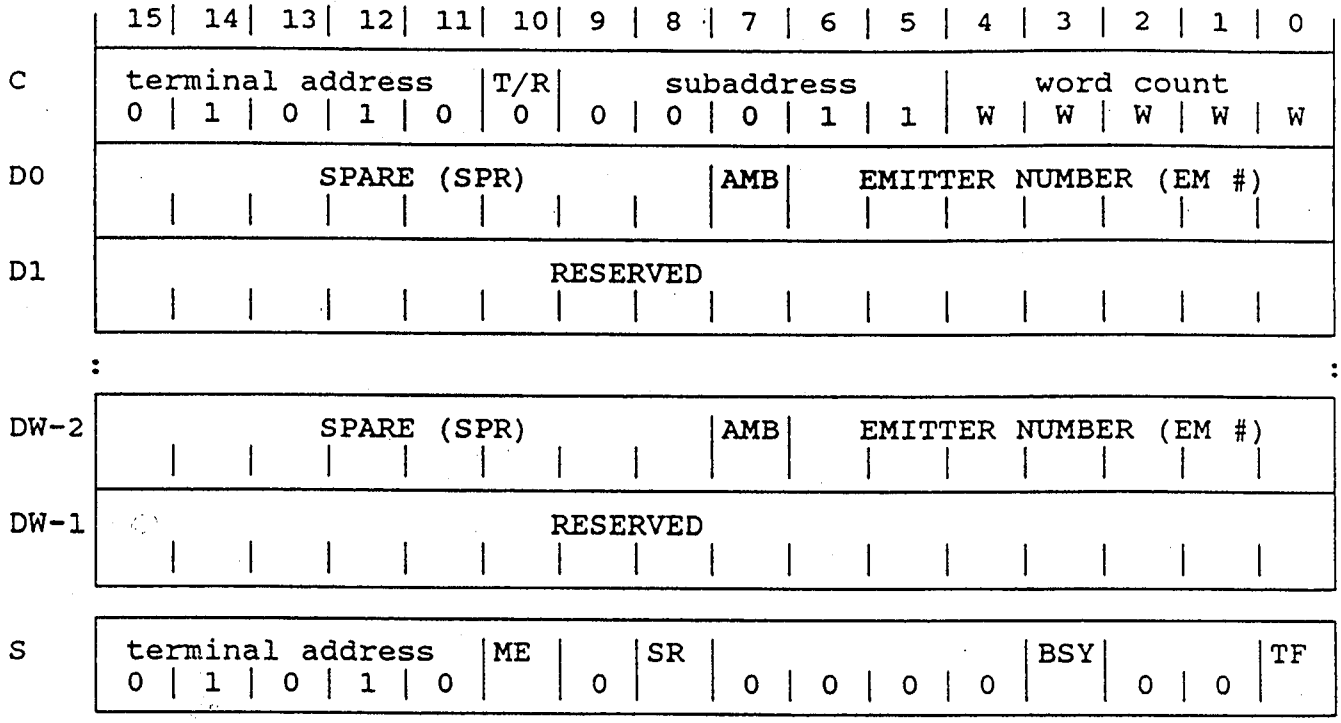
Remarks - The AN/ALR-67B(V)2 shall automatically transmit a Signal Message to the AN/ALE-47 via the AV bus whenever it detects a new signal (automatic signal updates for signals within the priority range of the AN/ALE-47, as determined by the ID Preselect message). If the busy bit in the last AN/ALE-47 poll response was set, the AN/ALR-67B(V)2 shall wait until that bit is cleared in a subsequent AN/ALE-47 poll response before it sends the Signal Message.

The AN/ALR-67B(V)2 will automatically transmit a Signal Message to the AN/ALE-47 when:

- Emitter ID has changed
- Track is lost on an Emitter
- Emitter is redetected
- Magnitude of DF is changed by at least ± 10 degrees
- Amplitude is changed by at least ± 10 dB
- HARM designated status has changed
- Missile Alert/Missile Launch status has changed
- GEN-X is requested for a threat not identified in the ID Preselect Message

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SIGNAL DELETE MESSAGE Sub-Address: 00011



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive. Set to 0 for Receive.
9-5	SA	SubAddress. Signal Delete = 00011.
4-0	W	Word count. 00010 = One signal deleted. 00100 = Two signals deleted. : 11110 = Fifteen signals deleted.

DATA WORD 0 (all EVEN DATA WORDS)

BIT	FIELD	DESCRIPTION
15-8	SPR	SPaRe. This field is not used.
7	AMB	AMBiguous. If this bit is set the given emitter number was associated with a list of ambiguous emitter identification codes.
6-0	EM #	EMitter Number. This is the track list reference number assigned by the AN/ALR-67B(V)2 to the signal being deleted.

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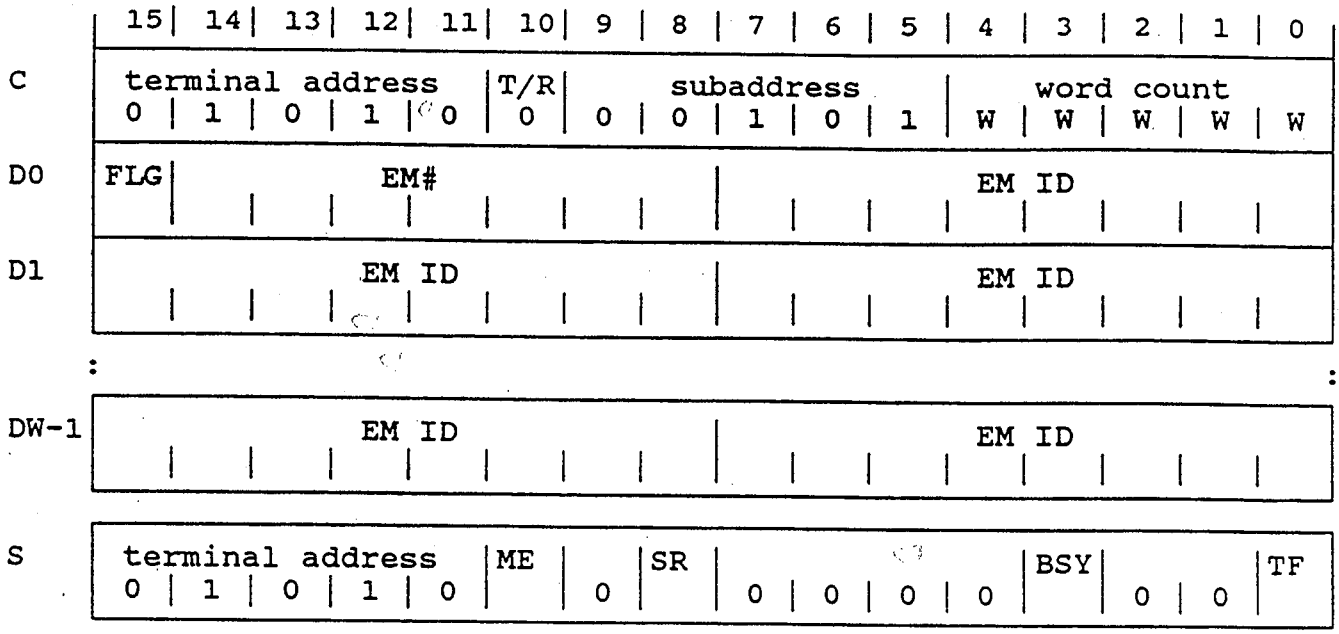
DATA WORD 1 (all ODD DATA WORDS)

BIT	FIELD	DESCRIPTION
15-0	RSV	ReSerVed (used by AN/ALR-56M for transaction codes with ASPJ)

Remarks - The AN/ALR-67B(V)2 shall automatically send a signal delete message to the AN/ALE-47 via the AV bus whenever the AN/ALR-67B(V)2 determines that a previously reported signal is no longer present in the environment or is no longer within the priority range selected by the AN/ALE-47. Each signal deleted message may contain data on up to 15 emitters.

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AMBIGUITY MESSAGE Sub-Address: 00101



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. Ambiguity message = 00101.
4-0	W	Word count. Depending on the number of ambiguities the data word count shall be anywhere from 1 to 30 words.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	FLG	FLaG. If this bit is set, the EM#/EM ID field contains an emitter number.
14-8	EM#	EMitter Number. This is the track list reference number assigned by the AN/ALR-67B(V)2.
7-0	EM ID	EMitter IDentification. Specifies possible emitter type ambiguous with the one listed in Signal Message.

DATA WORD 1 to W-1

BIT	FIELD	DESCRIPTION
15-8	EM ID	EMitter IDentification. Specifies possible emitter type ambiguous with the one listed in Signal Message.
7-0	EM ID	EMitter IDentification. Specifies possible emitter type ambiguous with the one listed in Signal Message.

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Remarks - The AN/ALR-67B(V)2 shall automatically send an ambiguity message to the AN/ALE-47 specifying possible ambiguities between a previously sent emitter and other signal types. The message shall consist of the emitter number from the previous message and the emitter identification numbers for ambiguous signals.

Field Interdependency - When the flag in bit 15 is set to 1 in the data word, it means that the upper byte of that word is the emitter number of a threat which has an ambiguous emitter identification. The lower byte shall contain one of these ambiguous emitter identifications. The upper byte of the next data word contains another ambiguous emitter identification. The lower byte of the second word could contain a third ambiguous emitter identification, and so on. Only one emitter number, with associated ambiguous emitter identifications, may be reported in any instance of this message. When EM ID = 00h, it shall be ignored. This would normally only occur in the lower byte.

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APPENDIX B

HARM LAUNCH MESSAGE Sub-Address: 00110

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress					word count				
	0	1	0	1	0	0	0	0	1	1	0	0	0	0	0	1
D0	SPARE (SPR)					BAND CODE (BND COD)										0
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. HARM Launch message = 00110.
4-0	W	Word count. 00001 = One data word.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-1	BND COD	HARM Launch Sub-Bands. See classified addendum for description of which HARM sub-bands are active. (1 = HARM band active for search or launch.)
0	0	Zero

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INVALID REQUEST MESSAGE Sub-Address: 00111

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress					word count				
	0	1	0	1	0	0	0	0	1	1	1	0	0	0	0	1
D0	ERR	BSY	RSV			MR	ISA	IEN	RSV	EMITTER NUMBER (EM#)						
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. Invalid Request = 00111.
4-0	W	Word count. 00001 = One data word.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	ERR	ERRor. If this bit is set there was a transmission error in the last message sent by the AN/ALE-47.
14	BSY	BuSY. If this bit is set the AN/ALR-67B(V)2 is too busy to process the request at this time. Try the request again during the next poll cycle.
13-12	RSV	ReSerVed (for compatibility with AN/ALQ-126B use)
11	MR	Multiple Request. 1 = More than one message request per POLL period.
10	ISA	Invalid Sub-Address. 1 = Error in Sub-Address
9	IEN	Invalid Emitter Number. 1 = Undefined Emitter Number specified.
8	RSV	ReSerVed
7-0	EM#	Emitter Number. An undefined Emitter Number specified by the AN/ALE-47.

Remarks - The AN/ALR-67B(V)2 shall send an invalid request message to the AN/ALE-47 via the bus whenever the AN/ALE-47 makes a request of the AN/ALR-67B(V)2 to which it cannot respond correctly. This could be due to an error in transmission, or the AN/ALR-67B(V)2 is too busy to process the request, multiple message requests within a polling period or the AN/ALE-47 specified an undefined emitter number to the AN/ALR-67B(V)2.

Field Interdependency - The value in the Emitter Number field is meaningful whenever IEN = 1.

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APPENDIX B

CONFIGURATION MESSAGE Sub-Address: 01010

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress					word count				
	0	1	0	1	0	0	0	1	0	1	0	0	0	0	1	0
D0	STD RET	F-18	AV-8B	A-6E	F-14A	F-14D	A-6F	TF-18	TAV 8B	SPARE						
D1	STD RET	SPARE				1ST TIM	126 AFT	ALQ 162	ALE 47	MMW	APR 43	ALR 50	126 FWD	126 A	ALQ 165	HRM CLC
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. Configuration = 01010.
4-0	W	Word count. 00010 = Two word message.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	STD RET	STanDard RETrofit. If this bit is set the AN/ALR-67B(V)2 is a standard system. Otherwise it is a retrofit system.
14	F-18	If this bit is set the system is on an F/A-18 aircraft.
13	AV-8B	If this bit is set the system is on an AV-8B aircraft.
12	A-6E	If this bit is set the system is on an A-6E aircraft.
11	F-14A	If this bit is set the system is on an F-14A aircraft.
10	F-14D	If this bit is set the system is on an F-14D aircraft.
9	A-6F	If this bit is set the system is on an A-6F aircraft.
8	TF-18	If this bit is set the system is on a TF-18 aircraft.
7	TAV-8B	If this bit is set the system is on a TAV-8B aircraft.
6-0	SPR	SPaRe. This field is not used.

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DATA WORD 1

BIT	FIELD	DESCRIPTION
15	STD RET	STanDard RETrofit. If this bit is set the AN/ALR-67B(V)2 is a standard system. Otherwise it is a retrofit system.
14-11	SPR	SPaRe. This field is not used.
10	1ST TIM	The first-time flag shall be set to 1 only during the power-up sequence and reinitialization. Upon sensing this flag set, the AN/ALE-47 shall know that no prior track list information is valid. If the AN/ALR-67B(V)2 resets and reinitializes during one polling period, this flag will be the only indication to the AN/ALE-47 that it should also reinitialize.
9	ALQ-126B AFT	If this bit is set there is an aft AN/ALQ-126B system present.
8	ALQ-162	If this bit is set there is an AN/ALQ-162 system present.
7	ALE-47	If this bit is set there is an AN/ALE-47 system present.
6	MMW	If this bit is set there is a millimeter wave system present.
5	APR-43	If this bit is set there is an AN/APR-43 system present.
4	ALR-50	If this bit is set there is an AN/ALR-50 system present.
3	ALQ-126B FWD	If this bit is set there is a forward AN/ALQ-126B system present.
2	ALQ-126A	If this bit is set there is an AN/ALQ-126A system present.
1	ALQ-165	If this bit is set there is an AN/ALQ-165 system present.
0	HRM CLC	If this bit is set there is a HARM system present.

Remarks - During the power-up handshake and whenever it changes thereafter, the AN/ALR-67B(V)2 shall automatically send this message to the AN/ALE-47 via the AV bus.

Field Interdependency - The first-time flag in bit 10 of data word 1 shall be set to 1 only during the power-up sequence. Upon sensing this flag set, the AN/ALE-47 shall know that no prior track list information is valid. If the AN/ALR-67B(V)2 resets and reinitializes during one polling period, this first time flag will be the only indication that the AN/ALE-47 has that it should also reinitialize.

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APPENDIX B

SELF-TEST MESSAGE Sub-Address: 01100

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress					word count				
	0	1	0	1	0	0	0	1	1	0	0	0	0	1	0	1
D0	EXT	APR 43	T	CSU	LB 315	LB 225	LB 135	LB 045	315	225	135	045	L	S	A	R
D1	126 AFT	ALE 47	126 FWD	FWD DGR	I DAP	MBE	ALQ 162	162 DGR	APR 43	MMW	ALQ 126	RSV	ALQ 165	IBU	CLC	N/A
D2-4	RESERVED															
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. AN/ALR-67B(V)2 Self-Test Data = 01100.
4-0	W	Word count. 00101 = Five data words.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	EXT	1 = failure detected in interface to EXTernal equipment (identified by data word 1)
14	APR 43	1 = interface failure detected in AN/APR-43 (details in data word 4; not used by AN/ALE-47)
13	T	1 = overtemperature detected in CP-1293 or R-2055
12	CSU	1 = failure detected in C-10250 Control Status Unit
11	LB 315	Low Band 315 degrees
10	LB 225	Low Band 225 degrees
9	LB 135	Low Band 135 degrees
8	LB 045	Low Band 045 degrees
7	315	1 = Quadrant 315° failure
6	225	1 = Quadrant 225° failure
5	135	1 = Quadrant 135° failure
4	45	1 = Quadrant 45° failure
3	L	1 = Low band integrated array failure
2	S	1 = BIT failure detected in the R-2055 Special Receiver (type of failures listed in data word 3; not used by AN/ALE-47)

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DATA WORD 0 (Continued)

BIT	FIELD	DESCRIPTION
1	A	1 = BIT failure detected in the CP-1293 Analyzer (type of failures listed in data word 2; not used by AN/ALE-47)
0	R	1 = Receiver failure, no more emitter data

DATA WORD 1

BIT	FIELD	DESCRIPTION
15	AFT 126	1 = AFT AN/ALQ-126B interface failure
14	ALE 47	1 = AN/ALE-47 interface failure
13	FWD 126	1 = ForWarD AN/ALQ-126B interface failure
12	FWD DGR	1 = DeGRaded mode failure in FWD AN/ALQ-126B interface
11	IDAP	1 = IDAP interface failure
10	MBE	1 = Mux Bus Emulator interface failure
9	ALQ 162	1 = AN/ALQ-162 interface failure
8	162 DEG	1 = DeGRaded mode failure in AN/ALQ-162 interface
7	APR 43	1 = AN/APR-43 interface failure
6	MMW	1 = MilliMeter-Wave (MMW) receiver interface failure
5	ALQ 126	1 = AN/ALQ-126B interface failure
4	RSV	ReSerVed (for Laser Threat Warning (LTW))
3	ALQ 165	1 = AN/ALQ-165(V) (ASPJ) interface failure
2	IBU	1 = Interference Blanker Unit failure
1	CLC	1 = HARM CLC interface Failure
0	N/A	Not Assigned

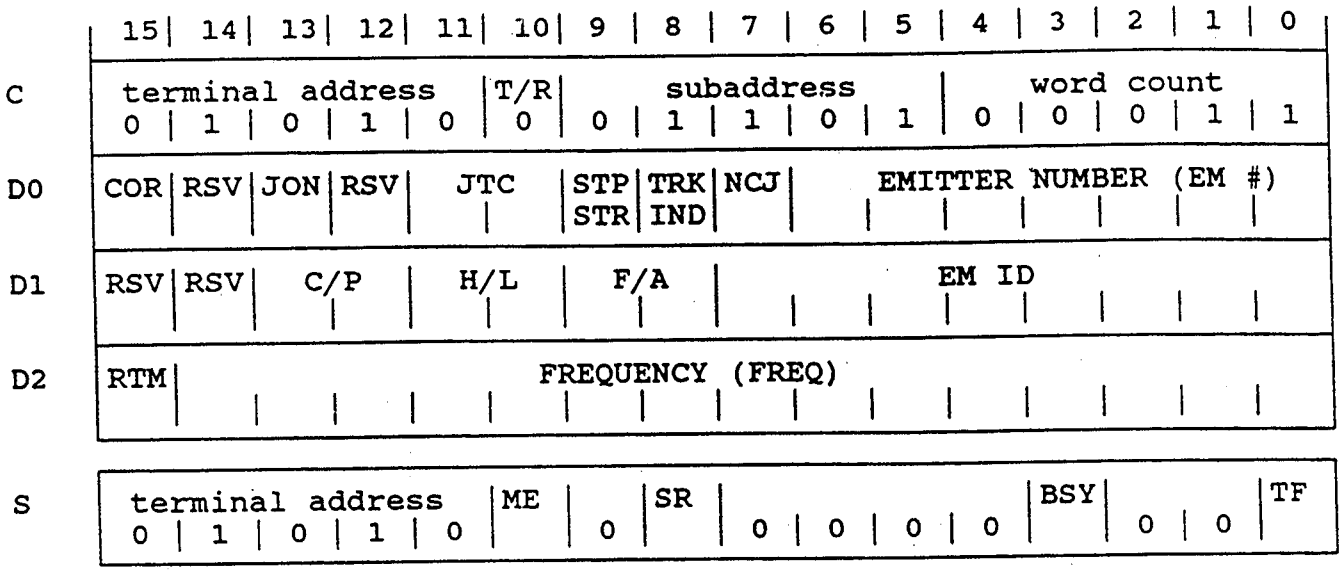
DATA WORD 2-4

BIT	FIELD	DESCRIPTION
15-0	RSV	ReSerVed (for AN/ALR-67B(V)2 use with other systems)

Remarks - Except for the definitions in data word 0 (D0) in this Self-Test Message, this ICD will serve to interface the AN/ALE-47 to both the AN/ALR-45F and the AN/ALR-67B(V)2. The AN/ALR-67B(V)2 form of D0 is specified in this ICD.

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JAMMING ASSIGNMENTS MESSAGE Sub-Address: 01101



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. Jammer Assignment Message = 01101.
4-0	W	Word count. The data word count shall be 3 words.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	COR	CORrelation bit. 1 = emitter is correlated with RWR (currently only AN/ALQ-126B will set the correlation bit)
14	RSV	ReSerVed (uncover tag not available)
13	JON	Jammer On/Off 1 = Jammer in transmit mode
12	RSV	ReSerVed (stop conflicting jammer technique)
11-10	JTC	Jammer Type Code 00 = AN/ALQ-126B 01 = AN/ALQ-162 10 = AN/ALQ-165 11 = Spare
9	STP STR	SToP/STaRt. 1 = Resources not assigned to this emitter or (if previously assigned) resources de-assigned from this emitter

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DATA WORD 0 (continued)

BIT	FIELD	DESCRIPTION
8	TRK IND	TRacK INDication AN/ALQ-126B 0 = Tracker A 1 = Tracker B AN/ALQ-162 0 (bit not used) AN/ALQ-165 0 = Untracked Signal 1 = Tracked Signal
7	NCJ	Non-Conflicting Jamming. 1 = non-transponder technique in use 0 = transponder technique in use
6-0	EM #	EMitter Number. Track list identification number. If correlated, then RWR track list identification is returned. If uncorrelated, then the Jammer returns the Jammer assigned number (Response Number). AN/ALQ-126B = 01 to 7Fh AN/ALQ-162 = 00 (not used) AN/ALQ-165 = 00 to 13h (presently uncorrelated only)

DATA WORD 1

BIT	FIELD	DESCRIPTION
15	RSV	ReSerVed (for HARM)
14	RSV	ReSerVed (used by AN/ALR-56M)
13-12	C/P	CW/Pulse Threat 00 = Undefined 01 = Pulse from the UDF 10 = CW from the UDF 11 = Both pulsed and CW from the UDF
11-10	H/L	High/Low Duty Cycle 00 = Low duty cycle response 01 = High duty cycle response 10 = Both high and low duty cycle 11 = Spare
9-8	F/A	Forward/Aft 00 = Invalid 01 = Aft response 10 = Forward response 11 = Forward and Aft response

Allowed states are:
AN/ALQ-126B = 00
AN/ALQ-162 = 00,01
AN/ALQ-165 = 00,01,10,11
(currently, 00 only)

Remark - These bits when received from AN/ALQ-126B are set to 00. The AN/ALR-67B(V)2 will convert the 00 to 11 before forwarding the message. In the A-6E (dual AN/ALQ-126B installation) the AN/ALR-67B(V)2 will set these bits depending on which unit sends the message. If from the forward unit (terminal address 04) the bits are set to 10. If from the aft unit (terminal address 05) the bits are set to 01.

7-0	EM ID	EMitter IDentification. Threat type code in RWR code format, regardless of whether or not the threat is correlated. For AN/ALQ-165, the Emitter ID is presently set to zero.

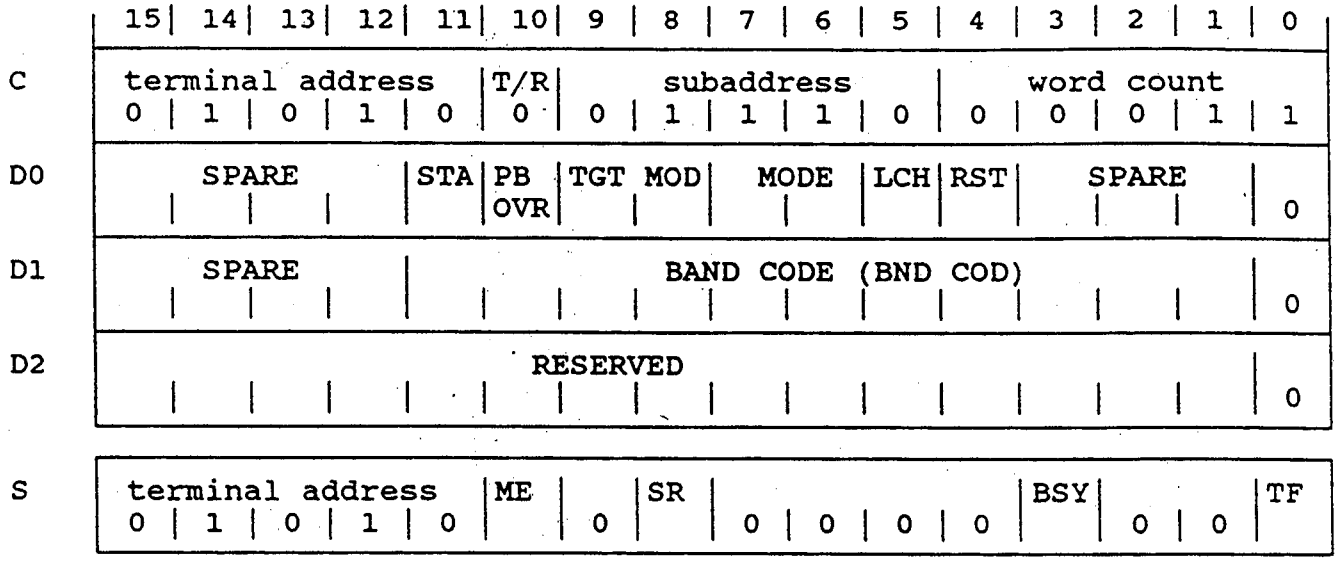
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DATA WORD 2

BIT	FIELD	DESCRIPTION
15	RTM	ReTransMit. This bit shall indicate why the Jamming Assignment Message is being sent and what processing is required by the AN/ALR-67B(V)2. RTM = 0. Indicates a normal Jamming Assignment Message that is sent because of a change in threat response required by the jammer. All receiving systems (except HARM) shall accept duplicate Jamming Assignment Messages. RTM = 1. Indicates that the Jamming Assignment Message is being sent to update a system (other than HARM) that has indicated a reset. A Jamming Assignment Message retransmitted as a result of a HARM reset shall have RTM = 0. When RTM = 1 the AN/ALR-67B(V)2 shall forward the Jamming Assignment Message to all applicable systems except HARM.
14-0	FREQ	See classified addendum for the meaning of this field. These bits indicate the frequency of jamming.

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HARM STATUS MESSAGE Sub-Address: 01110



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. HARM Status message = 01110.
4-0	W	Word count. 00011 = Three data words.

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DATA WORD 0

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11	STA	STAtion Select. 1 = HARM actively searching
10	PB OVR	Pull Back OVerRide. 1 = MSL will not automatically go to Self-Protect (SP) mode
9-8	TGT MOD	TarGeT MODe. Field indicates target type of threat environment. 00 = None 01 = A See classified addendum 10 = B 11 = C
7-6	MODE	MSL Mode. Field indicates the mode that the HARM is in. 00 = SP (Self Protection) 01 = TOO (Target of Opportunity) 10 = PB (Pull Back) 11 = BIT (Built In Test)
5	LCH	Launch of HARM. 1 = Missile has been fired by HARM Launch Computer
4	RST	HARM ReSeT Bit. 1 = reset
3-1	SPR	SPaRe
0	0	Zero

DATA WORD 1

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-1	BND COD	HARM Launch Sub-Bands. Description of which HARM sub-bands are active. (1 = HARM band active for search or launch.) [see classified addendum]
0	0	Zero

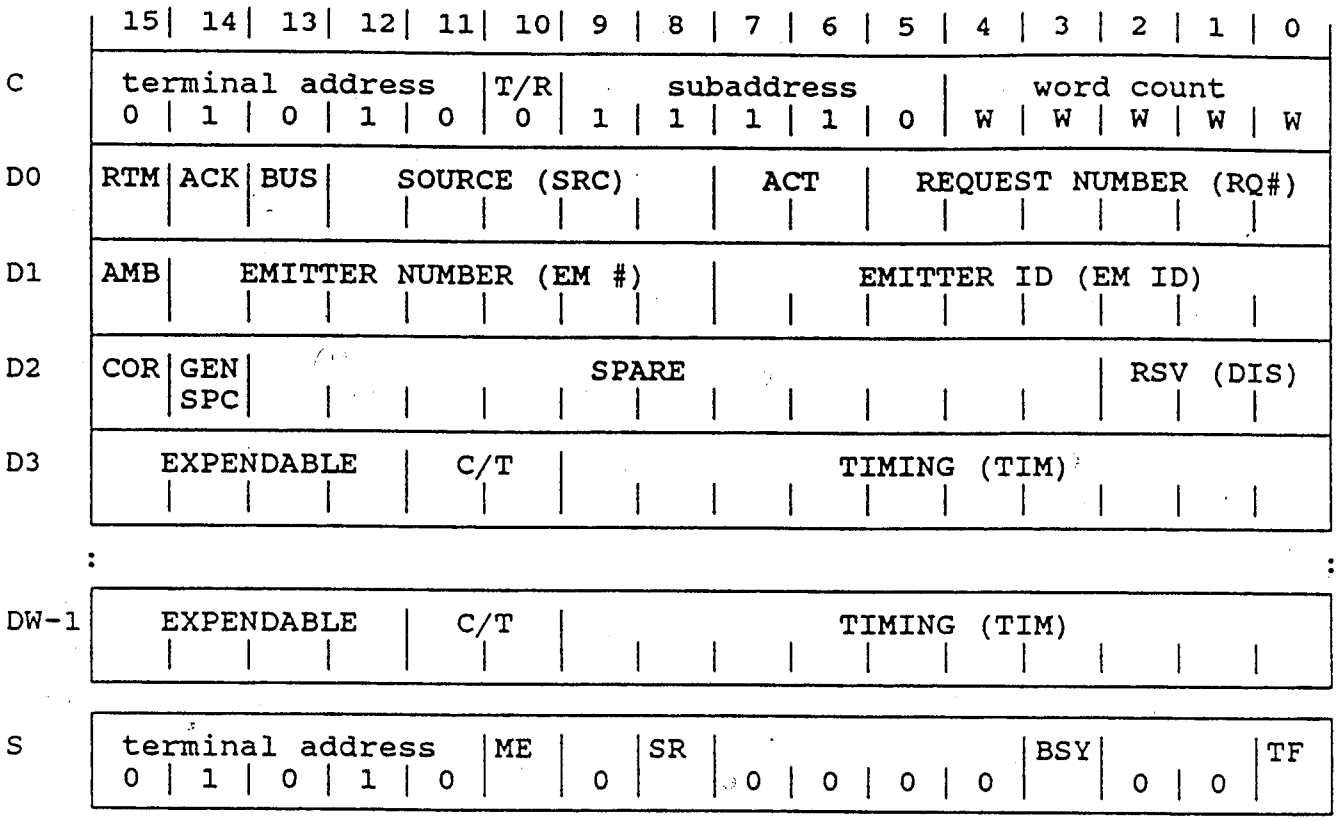
DATA WORD 2

BIT	FIELD	DESCRIPTION
15-1	RSV	ReSerVed (for HARM communication with a smart dispenser, not currently implemented)
0	0	Zero

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APPENDIX B

DISPENSE REQUEST MESSAGE Sub-Address: 11110



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 0 for Receive.
9-5	SA	SubAddress. Dispense Request Message = 11110.
4-0	W	Word count. Depending on the number of dispense requests the data word count shall be anywhere from 3 to 30 words.

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DATA WORD 0

BIT	FIELD	DESCRIPTION
15	RTM	ReTransMit. This bit shall be set to 1 to request that the Dispenser System to retransmit the Expendable Set-Up Message and the CMDS Status/Inventory Message. After a Source reset or power up sequence, if the presence of the Dispenser System is indicated in the current configuration, the Source remote terminal shall use this bit to instruct the Dispenser System to retransmit the Expendable Set-up Message. In this case the Word Count for the Dispense Request will be 1 and all other fields except BUS and SRC (bits 13 through 8 of Data Word 0) shall be set to 0. In all other cases the RTM bit shall be set to 0. (i.e., only at the requesting remote terminal power up or reset shall the retransmit request be made.) When a ReTransMit request is made of the AN/ALE-47 the ReSeT bit (bit 15 in Data Word 2) in the Expendable SET-Up Message response from AN/ALE-47 shall be set to zero to indicate to all other recipients of the Expendable Set-Up Message that no reset of the AN/ALE-47 has occurred. At Dispenser System power up/reset the Dispenser System is responsible for transmitting the Expendable Set-Up Message (with bit 15 of Data Word 2 = 1 to indicate an AN/ALE-47 reset) followed by the CMDS Status/Inventory Message. The retransmit request will be used by all applicable equipment except the AV Mux Bus Controller. At reset or power up of the AV Mux Bus Controller, the Dispenser System shall transmit the Expendable Set-Up Message (with bit 15 of Data Word 2 = 1 to indicate an AN/ALE-47 reset (to all other Mux Bus terminals) followed by the CMDS Status/Inventory Message, based on receipt of the first time flag in the Configuration Message from the Bus Controller. Upon receipt of a Dispense Request Message with RTM=1, the AN/ALE-47 should assume that all previous requests from the Source may be cancelled and that all previously received Jamming Assignment Messages from the Source are invalid.
14	ACK	ACKnowledge. The Acknowledge bit shall indicate receipt of a Frequency Inhibit request made by the AN/ALE-47 via the CMDS Status/Inventory Message. ACK = 1. Indicates that a Frequency Inhibit request has been received and processed. The action taken in response to the Frequency Inhibit request shall be as described in the ACTion field in bits 7-6. When ACK = 1 the Word Count for the Dispense Request Message shall be one. When ACK = 1 the BUS and SouRCe fields shall indicate the remote terminal that is acknowledging the AN/ALE-47 request, however, the RQ# shall contain

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the RQ# from the CMDS Status/Inventory Message that initiated the request.

ACK = 0. Indicates that the Dispense Request Message is being sent for some reason other than to acknowledge a Frequency Inhibit request from AN/ALE-47.

13 BUS — Mux BUS. This bit indicates which Mux Bus was used to originate the request to the AV Mux Bus Controller. This bit shall be set to the appropriate state by the requesting system.

0 = The source made the request over the AV Mux Bus

1 = The source made the request over the Avionics Mux Bus

12-8 SRC — SouRCe. The source is the terminal address of the Jammer making a dispense request. This field shall be set to the Jammer terminal address of the Jammer making the request. If the Expendable Request originates at the Bus Controller of either the AV or Avionics Mux Bus this field shall be set to 00000.

7-6 ACT — ACTion. The action field shall indicate what action is being requested of the AN/ALE-47 or what action has been taken by the Source in response to a Frequency Inhibit request from the AN/ALE-47.

When ACK = 0, indicating that a request is being made of the AN/ALE-47 by the Source, the ACT field shall describe the request being made as follows:

ACT = 00. No Dispense Sequence or Expendable Inhibit action is being requested of the AN/ALE-47. This condition is only valid when the Word Count of the Dispense Request Message equals one.

ACT = 01. Cancel previously requested Expendable Inhibit of specified type for a specified time.

ACT = 10. Execute the Dispense Sequence indicated in Data Words 3 to W-1.

ACT = 11. Enable a timed Expendable Inhibit for each expendable Type (or Category) as indicated by the contents of Data Words 3 to W-1.

Once a request has been made of the AN/ALE-47, the Source shall not make another request until the previous request has been acknowledged by the AN/ALE-47.

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When ACK = 1, indicating that an Frequency Inhibit request has been received via the CMDS Status/Inventory Message, the ACT field shall describe the action taken as follows:

ACT = 00. No action has or will be taken by the Source in response to the Frequency Inhibit request received in the CMDS Status/Inventory Message (Request denied).

ACT = 01. The Frequency Inhibit previously requested and honored has been canceled (either because of a CDMS Status/Inventory Message or some other reason internal to the Source) or the requested time-out is complete.

ACT = 10. The untimed Frequency Inhibit request has been honored and is currently in effect.

ACT = 11. The timed Frequency Inhibit request has been honored and is currently in effect.

When processing of a Frequency Inhibit request results in an interruption of jamming, the Dispense Request Message shall be transmitted followed by the appropriate Jamming Assignment Message (if any). When jamming is resumed, a Dispense Request Message shall be transmitted followed by the appropriate Jamming Assignment Message (if any). The Source shall generate a Dispense Request Message to acknowledge a Frequency Inhibit request from AN/ALE-47 within 100 msec of the receipt of the CMDS Status/Inventory Message that contained the Frequency Inhibit.

5-0 RQ# ReQuest Number. This field shall contain a value between 0 and 63 to indicate the sequential order in which each request was made by the Source. The next request after number 63 shall restart the count to indicate request number 0. The request number field shall indicate the request number of the requesting equipment when a request is being made (ACK = 0 and ACT = 01, 10, or 11) and shall indicate the AN/ALE-47 request number when a Frequency Inhibit request from AN/ALE-47 is being acknowledged (ACK = 1 and ACT = 00, 01, 10, or 11).

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DATA WORD 1

BIT	FIELD	DESCRIPTION
15	AMB	AMBiguity. AMB = 0. Indicates that the Emitter Identification is unambiguous. AMB = 1. Indicates that the Emitter Identification is ambiguous.
14-8	EM #	Emitter #. Track list identification number. If correlated, then RWR track list identification is returned. If uncorrelated, then the Jammer returns the Jammer assigned number (Response Number). AN/ALQ-126B = 01 to 7Fh AN/ALQ-162 = 00 (not used) AN/ALQ-165 = 00 to 13h (presently uncorrelated only)
7-0	EM ID	EMitter IDentification. This field, which has a range of 00 - FF hex, shall indicate the type of threat signal which is intended to be countered utilizing this Dispense Request. The assignment of field values shall be in accordance with the EID values used in the Mux Bus Controller Emitter Identification List. An EID of 0 shall indicate no identification.

DATA WORD 2

BIT	FIELD	DESCRIPTION
15	COR	CORrelation bit. 1 = correlation between Source and RWR emitter numbers
14	GEN SPC	GEN-X SPeCial handling. When set to 1, this field causes the RWR to precede the transmission of this message to the AN/ALE-47 with a Signal Message containing the parameters of the emitter specified by the Emitter Number and the Emitter ID fields of this message. The delayed transmission of this message and the generation of the Signal Message allows the source to request the dispense of a GEN-X expendable against a threat not selected by the ID Preselect Message from the AN/ALE-47.
13-3	SPR	SPaRe
2-0	RSV	ReSerVed for Discretes (DIS)

DATA WORDS 3 to W-1

BIT	FIELD	DESCRIPTION
15-12	EXP	EXPendable. The expendable field shall describe, the expendable to be dispensed. The meaning of the EXPendable field shall depend on the C/T indication in bits 11-10. When C/T = 00 = None, EXPendable shall = 0000 and shall indicate no dispense or inhibit of a Type or Category. When C/T=00 and EXP=0000 the AN/ALE-47 shall use the TIMe indicated to extend the interval between launches in a Dispense Sequence or to provide an initial delay

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(C/T=00 and EXP=0000 in Data Word 3). An initial delay may be used for a Dispense Sequence or an Expendable Inhibit request. When an Expendable Inhibit request is being made C/T=00 and EXP=0000 is the only valid in Data Word 3 (i.e. no time extensions other than an initial delay are permitted). Except in the case of Dispense Request Message with a Word Count of 1, the last Data Word in the Dispense Request Message shall not indicate C/T=00 and EXP=0000.

When C/T = 01 = Type, EXP shall have a range of 0-15 and the EXP number shall correspond to Expendable Type # 0-15 as defined in the Expendable Set-up Message.

When C/T = 10 = Category, EXP shall have a range of 0-3 and the EXP number shall correspond to the Expendable Category 0-3 as defined in the Expendable Set-Up Message.

11-10 C/T Category/Type

00 = Time Extension. When C/T = 00, the Dispenser System shall use the Time indicated to extend the time interval between expendable launches. This C/T field is used to extend the Time period beyond the maximum 10.23 second limit of the Time (TIM) field. The last Data Word in a Dispense Request Message shall not indicate C/T = 00.

01 = Type

10 = Category

11 = Invalid

BIT	FIELD	DESCRIPTION
9-0	TIM	TIMe. This field shall indicate the time that the Dispenser System is to wait, after launching the expendable called out in the EXP field of this Data Word, before launching the expendable called out in the EXP field of the next Data Word. The least significant bit shall indicate 10 msec. The TIMe value range is 0 to 1023 yielding time delays of 0 to 10.23 seconds in 10 msec. increments. When an Expendable Inhibit is being requested (ACK = 0 and ACT = 11) the TIMe field shall indicate the duration of the timed inhibit period. When a Dispense Sequence request is being made (ACK = 0 and ACT = 10) the last Data Word of the Dispense Request Message shall indicate a TIMe value of 0.

Remarks - The message sequence for a Jammer on the AV Mux Bus to request expendable dispensing, is for the Jammer to respond to the poll message on the AV Mux Bus with a request to transmit the Dispense Request Message. This transmit request is made by the Data Word in

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the Poll Message response. Data Word bit 10 in the Poll Message will be set to 1 to indicate a transmit request. Bits 9 through 5 in the Poll Message Data Word will be set to 11110 (1E hex) to indicate the subaddress of the Dispense Request Message. Bits 4 through 0 in the Poll Message Data Word will be set to indicate the number of words to be transmitted. The Word Count (W) indicated by WWWWW shall be one, or three plus the number of expendables or time extensions (EXP = 0000 and C/T = 00) in the dispense sequence. A single expendables request is limited to 27 sequences. (Data Words 0, 1, and 2 plus 27 sequences in Data Words 3 through 29). The Bus Controller for the EW Mux Bus will respond to the request in the Poll Message within 20 msec. by commanding the Jammer to transmit the Dispense Request Message. The Bus Controller on the EW Mux Bus will then send the Dispense Request Message to the Dispenser System within 5 msec. unless delayed for a Signal Message by the presence of a "1" in the GEN-X SPeCial handling field. Unless Data Word 3 indicates a time delay, the Dispenser System shall initiate the first dispense of an automatic dispense request in 75 $\pm$ 25 msec. The Dispenser System shall act on the remainder of the Data Words in this message within the time period(s) indicated in the message. The AN/ALE-47 will acknowledge a Dispense Request Message in less than 100 msec.

As a growth provision, it is also assumed that the AV Mux Bus Controller can receive requests over the Avionics Mux Bus to be forwarded to the Dispenser System over the AV Mux Bus. In this case the AV Mux Bus Controller acting as a Remote Terminal would receive the request from the Avionics Bus Controller in a format controlled by the aircraft interface control document and reformat the request as a Dispense Request Message to the Dispenser System in the format described herein.

The AN/ALE-47, when dispensing a GEN-X requested by a jammer, shall use the best information available at the time of generation of the program used to fulfill the Dispense Request message. If insufficient information is available to completely determine an optimized program for GEN-X, an appropriate default program will be used.

Field Interdependency - D2, bit 15 (correlation bit) set to 1 indicates that the emitter # (D1, bits 14-8) passed from the Source (D0, bits 12-8) is correlated with the RWR emitter #.

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APPENDIX B

POLL COMMAND AND RESPONSE MESSAGES Sub-Address: 00111

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress					word count				
	0	1	0	1	0	1	0	0	1	1	1	0	0	0	0	1
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	
D0	SPARE (SPR)					T/R	SUB ADDRESS (SA)					WORD COUNT (W)				
												W	W	W	W	W

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 1 for Transmit.
9-5	SA	SubAddress. Poll message = 00111.
4-0	W	Word count. 00001 = Transmit one data word.

STATUS WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47
8	SR	Service Request. This bit shall be set to one to indicate Data Word D0 is valid.
3	BSY	BuSY. If the AN/ALE-47 is currently too busy to process any more data it shall set the busy bit to one.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15-11	SPR	SPaRe
10	T/R	Transmit/Receive Bit. Set to zero for AN/ALR-67B(V)2 transmission, set to one for AN/ALE-47 transmission
9-5	SA	SubAddress. If the AN/ALE-47 set the service request bit to one it shall indicate in this field the subaddress of the message.
4-0	W	Word count. If the AN/ALE-47 set the service request bit to one it shall indicate in this field the word count of the message.

Remarks - The AN/ALR-67B(V)2 shall send the poll command word via the EW bus to test the status of the AN/ALE-47 to determine whether the AN/ALE-47 is busy or desires to send or request a message. The AN/ALE-47 shall respond with a poll response consisting of a status word and a single data word.

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APPENDIX B

The AN/ALR-67B(V)2 shall send the poll via the EW bus normally every 20 milliseconds, with normal maximum delays of 30 milliseconds.

If the busy bit is set in the status word, the AN/ALR-67B(V)2 shall inhibit sending data words to the AN/ALE-47 until the busy bit in the status word is cleared.

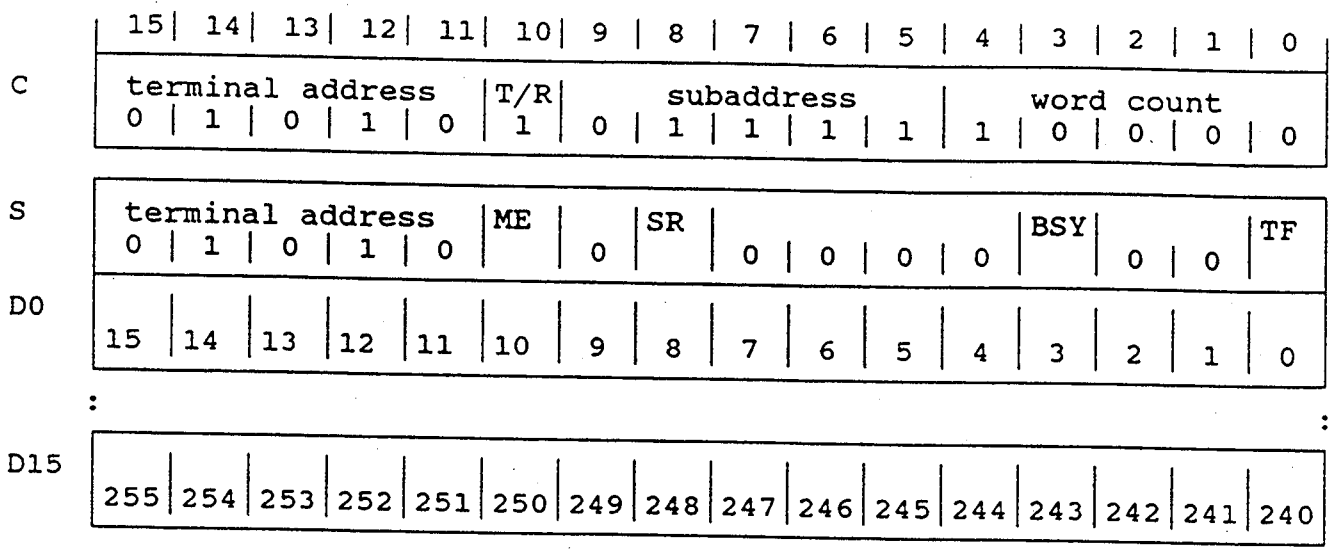
Field Interdependency - If the AN/ALE-47 has no messages to transmit or request reception of, the service request bit in the status word is cleared or BUSY is set. When the AN/ALR-67B(V)2 receives the status word with service request cleared or BUSY set, it shall ignore the associated data word. When the AN/ALE-47 has a message to send or request, it shall set the service request bit in the status word and shall indicate what message it wants to transmit or receive by using the fields in the data word.

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APPENDIX B

EMITTER ID PRESELECT MESSAGE

Sub-Address: 01111



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 1 for Transmit.
9-5	SA	SubAddress. ID Preselect message = 01111.
4-0	W	Word count. 10000 = Transmit sixteen data words.

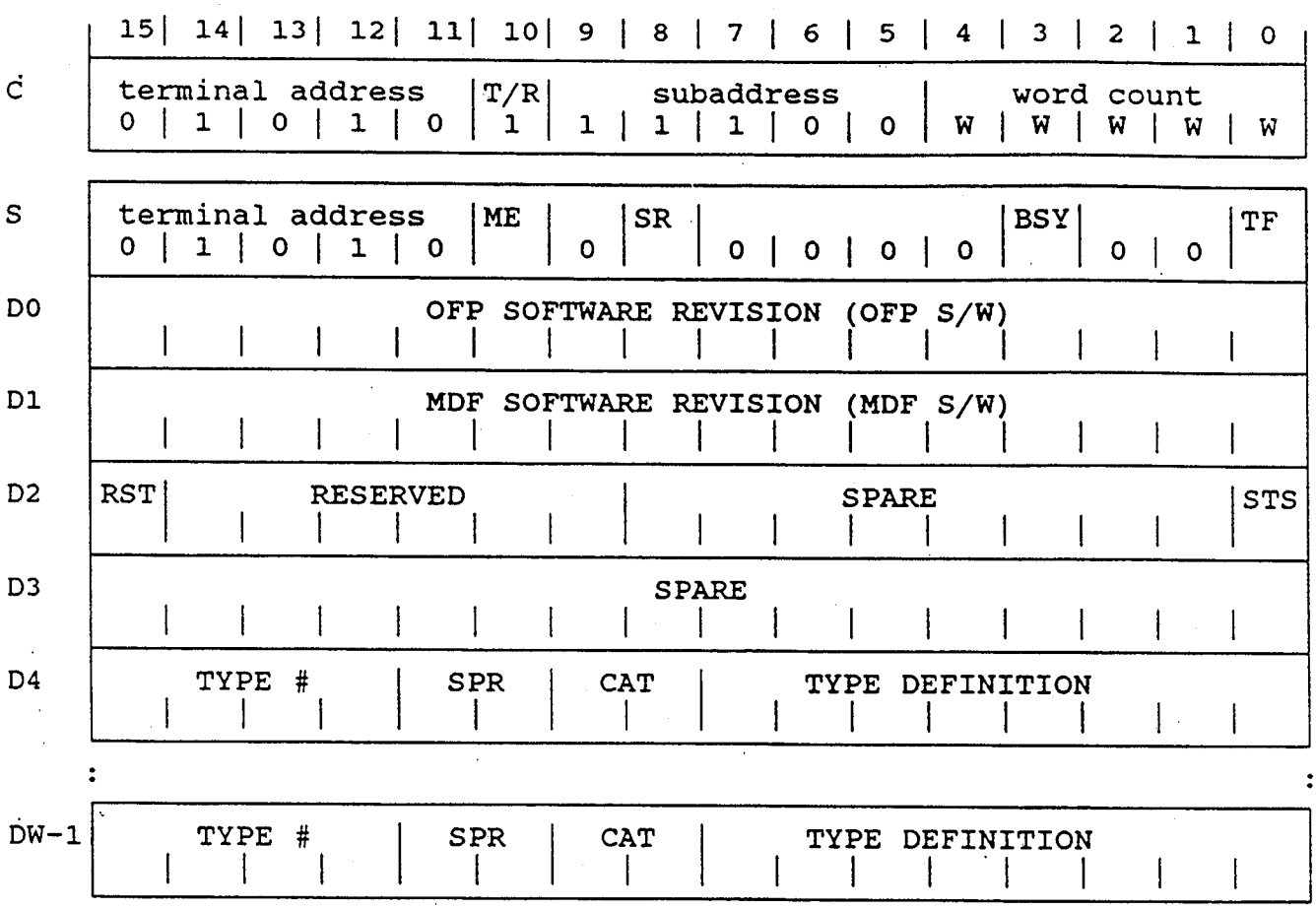
DATA WORDS 0 - 15

BIT	FIELD	DESCRIPTION
15-0		Each bit relates to a particular threat (See Classified Addendum) 1 = AN/ALR-67B(V)2 report threat 0 = AN/ALR-67B(V)2 should not report threat

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APPENDIX B

EXPENDABLE SET-UP MESSAGE Sub-Address: 11100



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 1 for Transmit.
9-5	SA	SubAddress. Expendable Set-Up = 11100.
4-0	W	Word count. WWWWW = # of data words. Maximum is 10100 (20 data words).

DATA WORDS 0 - 1

BIT	FIELD	DESCRIPTION
15-0		Same 16 bits used by MLV

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DATA WORD 2

BIT	FIELD	DESCRIPTION
15	RST	ReSet. This bit when set to 1 shall indicate that all systems receiving this message should assume that all previous Dispense Requests have been discarded and all previous information received from AN/ALE-47 is invalid. This bit shall be set to 1 when: the Dispenser System has just powered up or reset; or when the Dispenser System interface with the EW Mux Bus Controller has been reset because a Configuration Message with the first-time flag set was received by the Dispenser System. In all other cases (specifically in response to a Retransmit Request received in a Dispense Request Message) this bit shall be set to 0. Upon receipt of an Expendable Set-Up Message with RST = 1 each receiving system shall retransmit the applicable current threat information (Signal Message(s) from Bus Controller, Jamming Assignment Messages from other remote terminals, etc.)
14-9	RSV	ReServed (for Jammer discretes)
8-1	SPR	SPaRe = 0.
0	STS	STatuS. This bit shall indicate the status of AN/ALE-47 processing for Dispense Request Messages. STS = 0. Indicates that the AN/ALE-47 is capable of processing the Dispense Request Messages to produce a Dispense Sequence as requested and that the fields in Data Words 12 and 13 of the CMDS Status/Inventory Message may be interpreted by the recipient as described in this document. Also, discrete requests in Data Word 2 bits 2-0 for AN/ALQ-165 will be processed. STS = 1. Indicates that the fields in Data Words 12 and 13 of the CMDS Status/Inventory Message are not supported by AN/ALE-47 and/or the AN/ALE-47 may modify the Dispense Sequence in the Dispense Request Message. When STS = 1 the recipient shall not modify jamming techniques that depend on an AN/ALE-47 response to the Expendable Request Message for effectiveness. Also, Mux Bus definable discrete requests will not be available to AN/ALQ-165.

DATA WORD 3

BIT	FIELD	DESCRIPTION
15-0	SPR	SPaRe = 0

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DATA WORDS 4 - DW-1

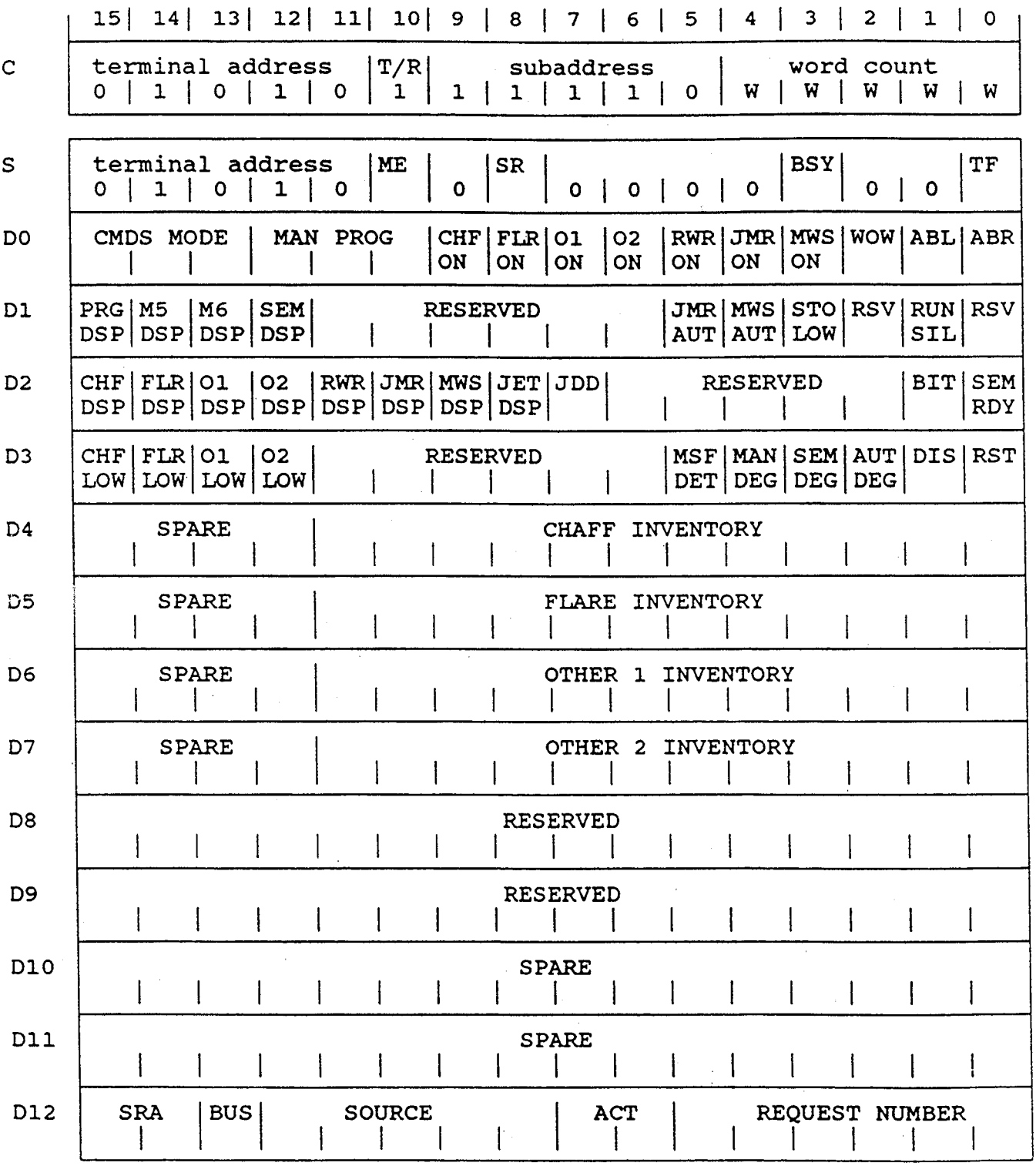
BIT	FIELD	DESCRIPTION
15-12	TYPE #	0-Fh corresponds to the 16 expendables the AN/ALE-47 will recognize
11-10	SPR	SPaRe

9-8	CAT	0-3 maps the type # to a category
		00 = Chaff
		01 = Flare
		10 = Other 1
		11 = Other 2

7-0	TYPE DEF	TYPE DEFinition. 00-FFh maps the "TYPE #" and "CAT" to one of 256 absolute expendable types (e.g. RR-180, MJU-22, GEN-X, etc.) 00h indicates no expendable with the given type # is carried.

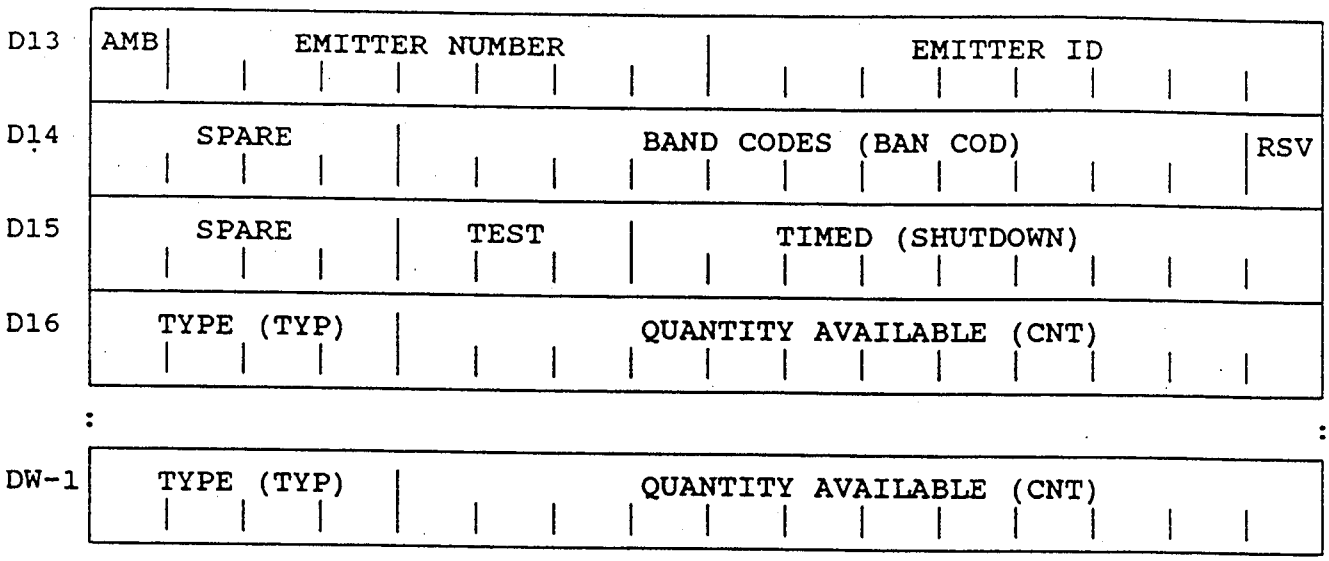
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CMDS STATUS/INVENTORY MESSAGE Sub-Address: 11110



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APPENDIX B



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47
10	T/R	Transmit Receive. Set to 1 for Transmit.
9-5	SA	SubAddress. Status/Expendable Inventory = 11110.
4-0	W	Word count. WWWWW = # of data words. Maximum is 11010 (26 data words).

STATUS WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Remote terminal address of the AN/ALE-47
8	SR	Service Request. This bit shall be set to one to indicate the AN/ALE-47 wishes to send a message to the AN/ALR-67B(V)2 via the transmit command.
3	BSY	BuSY. If the AN/ALE-47 is currently too busy to process any more data it shall set the busy bit to one.

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DATA WORD 0

BIT	FIELD	DESCRIPTION
15-13	CMDS MODE	CMDS MODE: Mode Switch Selection Code. Format: 3-bit code (one octal digit): 0 = Standby 1 = Manual 2 = Semi-Auto 3 = Auto 4-6 = Reserved 7 = Bypass
12-10	MAN PROG	Manual Program Selection Code 0 = Reserved 1 = Manual Program # 1 2 = Manual Program # 2 3 = Manual Program # 3 4 = Manual Program # 4 5 = Manual Program # 5 6 = Manual Program # 6 7 = Undefined
9	CHF ON	1 = Chaff dispensing enabled
8	FLR ON	1 = Flare dispensing enabled
7	O1 ON	1 = Other 1 dispensing enabled
6	O2 ON	1 = Other 2 dispensing enabled
5	RWR ON	1 = RWR dispensing enabled
4	JMR ON	1 = Jammer dispensing enabled
3	MWS ON	1 = MWS dispensing enabled
2	WOW	1 = Weight on Wheels asserted
1	ABL	1 = Left Afterburner on
0	ABR	1 = Right Afterburner on

DATA WORD 1

BIT	FIELD	DESCRIPTION
15	PRG DSP	1 = Dispense the selected manual program
14	M5 DSP	1 = Dispense Manual Program # 5
13	M6 DSP	1 = Dispense Manual Program # 6
12	SEM DSP	1 = Consent to Dispense (Semi-Auto Mode)
11-6	RSV	ReSerVed
5	JMR AUT	Jammer Dispense Requests AUTOmatic, without SEMI AUTO Initiate required in SEMI AUTO mode when = 1.
4	MWS AUT	MWS Dispense Requests AUTOmatic, without SEMI AUTO Initiate required in SEMI AUTO mode when = 1.
3	STO LOW	1 = STOres LOW would be exceeded by honoring this request
2	RSV	ReSerVed
1	RUN SIL	1 = Automatic dispense treated as in SEMI-AUTO mode (Run Silent)
0	RSV	ReSerVed (MC)

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DATA WORD 2

BIT	FIELD	DESCRIPTION
15	CHF DSP	1 = Chaff currently dispensing
14	FLR DSP	1 = Flare currently dispensing
13	O1 DSP	1 = Other 1 currently dispensing
12	O2 DSP	1 = Other 2 currently dispensing
11	RWR DSP	1 = RWR Initiated program currently dispensing
10	JMR DSP	1 = JMR Initiated program currently dispensing
9	MWS DSP	1 = MWS Initiated program currently dispensing
8	JET DSP	1 = JETTISON currently dispensing
7	JDD	Jammer Discrete initiated program currently Dispensing. JDD = 1. Indicates that a jammer discrete input initiated the program currently dispensing. JDD = 0. Not initiated by jammer discretes.
6-2	RSV	ReSerVed
1	BIT	1 = CMDS BIT in progress
0	SEM RDY	1 = Semi-Auto Program ready for dispense initiation

DATA WORD 3

BIT	FIELD	DESCRIPTION
15	CHF LOW	1 = Chaff Low Stores level reached or exceeded
14	FLR LOW	1 = Flare Low Stores level reached or exceeded
13	O1 LOW	1 = Other 1 Low Stores level reached or exceeded
12	O2 LOW	1 = Other 2 Low Stores level reached or exceeded
11-6	RSV	ReSerVed
5	MSF DET	1 = Misfire detected (since power-up)
4	MAN DEG	1 = Manual dispense Unavailable
3	SEM DEG	1 = Semi-Auto dispense Unavailable
2	AUT DEG	1 = Auto dispense Unavailable
1	DIS	DIScrete interface failure. The discrete bit shall indicate a failure of the discrete interface with AN/ALQ-165. DIS = 0. Indicates no failure. DIS = 1. Indicates failure.
0	RST	1 = DATA LOST/RESET. Track re-initialization required (asserted at CMDS PGMR power-up/reset)

DATA WORD 4

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-0	CHAFF	Chaff Inventory Quantity: Format: 12-bit unsigned binary integer LSB = 1 expendable; MSB = 2048 expendables

DATA WORD 5

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-0	FLARE	Flare Inventory Quantity: Format: 12-bit unsigned binary integer LSB = 1 expendable; MSB = 2048 expendables

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DATA WORD 6

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-0	OTHER 1	Other 1 Inventory Quantity: Format: 12-bit unsigned binary integer LSB = 1 expendable; MSB = 2048 expendables

DATA WORD 7

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-0	OTHER 2	Other 2 Inventory Quantity: Format: 12-bit unsigned binary integer LSB = 1 expendable; MSB = 2048 expendables

DATA WORDS 8 - 9

BIT	FIELD	DESCRIPTION
15-0	RSV	ReSerVed (Avionics Bus use)

DATA WORDS 10 - 11

BIT	FIELD	DESCRIPTION
15-0	SPR	SPaRe = 0

DATA WORD 12

BIT	FIELD	DESCRIPTION
15-14	SRA	Status/Request/Acknowledge. The Status/Request/Acknowledge (SRA) field shall describe the reason the CMDS Status/Inventory Message is being sent. The meaning of the SRA field shall be as follows: SRA = 00. Indicates that the CMDS Status/Inventory Message is being sent to indicate the current Status of the CMDS. In this case: The BUS field shall indicate 0 (EW Mux Bus); the SRC field shall indicate 11111 (1F hex = CMDS); the ACT field shall indicate 00 (0 hex); and the RQ# field shall indicate 000000 (00 hex). SRA = 01. Indicates that the CMDS Status/Inventory Message is being sent to acknowledge a request received via a Dispense Request Message. In this case: the BUS field shall indicate the BUS as indicated in the Dispense Request Message that is being acknowledged; the SRC field shall indicate the SRC field as indicated in the Dispense Request Message that is being acknowledged; and the RQ# field shall indicated the RQ# field as indicated in the Dispense Request Message that is being acknowledged. The CMDS reponse to the request being acknowledged shall be contained in the ACT field. SRA = 10. Indicates that the CMDS Status/Inventory Message is being sent to make a request of the recipient. In this case: the BUS field shall

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indicate the bus on which the recipient is communicating, 1 = Avionics Mux); the SRC field shall indicate the Terminal Address on the indicated bus to which the request is being directed; the RQ# field shall indicate the AN/ALE-47 Request Number. The request being made by the CMDS shall be as described in the ACT field.

SRA = 11. Indicates that a test of the jammer discretes is being requested. The test code (discrete code) to be generated shall be as described by the TeST code in bits 11-9 of Data Word 15. After receipt of the CMDS Status/Inventory Message the test code is to be generated by the recipient within 100 msec (may be as long as 200 msec relative to the time the AN/ALE-47 transmits the CMDS Status/Inventory Message) and shall remain active for a minimum of 40 msec and a maximum of 50 msec. Once a test of the discrete interface is requested by AN/ALE-47, the AN/ALE-47 shall inhibit normal processing of the discrete inputs for 250 msec.

13 BUS BUS. The BUS bit shall indicate which Mux Bus the intended recipient is located on, or the Mux Bus from which the request was received.

BUS = 1. Indicates the Avionics Mux Bus.

12-8 SRC SouRCe. The source field shall indicate the Terminal Address of the Source being acknowledged or the Terminal Address, on the intended bus, to which the CMDS is directing the request. When SRA=00, SRC shall equal 11111 (1F hex) to indicate the CMDS.

7-6 ACT ACTion. The action field shall indicate what action is being requested of the intended recipient or what action has been taken by the CMDS in response to a request to the AN/ALE-47. When SRA = 00 or 11, then ACT shall be 00 or shall be considered invalid.

When SRA = 01, indicating that a Dispense Request Message is being acknowledged the meaning of the ACT field shall be as follows:

ACT = 00. No action has or will be taken by AN/ALE-47 in response to the Dispense Request Message. When the request has been denied because the low stores limit would have been exceeded the STO LOW bit (bit 3) in Data Word 1 shall so indicate.

ACT = 01. The Expendable Inhibit or Dispense Sequence previously requested and honored has been canceled (either because of a Dispense Request Message or some other reason internal to the CMDS) or the requested time-out is complete.

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ACT = 10. The Dispense Sequence requested has been initiated.
ACT = 11. The timed Expendable Inhibit requested has been honored and is in effect.

When SRA = 10, indicating that a request is being made by the AN/ALE-47 the meaning of the ACT field shall be as follows:
ACT = 00. No action required by the Bus, Source recipient.
ACT = 01. Cancel the previously requested Frequency Inhibit indicated by the Band Code field in Data Word 14.
ACT = 10. Enable the untimed Frequency Inhibit indicated by the Band Code field in Data Word 14.
ACT = 11. Enable a timed Frequency Inhibit indicated by the Band Code field in Data Word 14 and the Timed (Shutdown) field in Data Word 15.

5-0 RQ# ReQuest Number (#). The request number field shall: indicate the AN/ALE-47 request number when the AN/ALE-47 is making a request (SRA = 10); indicate the request number of the requesting equipment when a request is being acknowledged (SRA = 01); and indicate 000000 in all other cases (SRA = 00 or 11).

DATA WORD 13

BIT	FIELD	DESCRIPTION
15	AMB	AMBiguity. AMB = 0. Indicates that the Emitter Identification is unambiguous. AMB = 1. Indicates that the Emitter Identification is ambiguous.
14-8	EM#	Emitter #. Track list identification number. If correlated, then RWR track list identification is returned. If uncorrelated, then the Jammer returns the Jammer assigned number (Response Number). AN/ALQ-126B = 01 TO 7F hex AN/ALQ-162 = 00 (not used) AN/ALQ-165 = to 13 hex (presently uncorrelated only)
7-0	EM ID	EMitter IDentification. This field, which has a range of 00 - FF hex, shall indicate the type of threat signal which is intended to be countered utilizing this Dispense Request. The assignment of field values shall be in accordance with the EID values used in the EW Mux Bus Controller Emitter Identification List. An EID of 0 shall indicate no identification.

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DATA WORD 14

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-1	BAN COD	BANd CODes for shutdown due to GEN-X launch (See classified addendum)
0	RSV	ReSerVed (HARM)

DATA WORD 15

BIT	FIELD	DESCRIPTION
15-12	SPR	SPaRe
11-9	TST	TeST. The test field shall indicate a test code to be generated by the recipient on the Expendables Output discrete lines. Bit 9 shall indicate the state of the least significant bit. The test code will only be requested by AN/ALE-47 when the Mission Data File indicates use of the discrete lines and the Configuration Message from the Bus Controller indicates that the system that the test applies to is present. A bit value of "0" in a bit of this field indicates the corresponding discrete line is to be grounded (i.e., asserted); a "1" means a line is de-asserted for the test.
8-0	TIMED	TIMED shutdown timing word: Format: 9-bit unsigned binary integer LSB = 100 msec Range: 0 - 51.1 sec

DATA WORDS 16 - DW-1

BIT	FIELD	DESCRIPTION
15-12	TYPE	Expendable Type code, as described by the Expendable Set-up Message.
11-0	CNT	CouNT. Format: 12-bit unsigned binary integer LSB = 1 available; MSB = 2048 available

Spares always set to zero. Typically, at most one of the Dispense Initiate flags is asserted. If more than one is asserted simultaneously, the CMDS will execute the appropriate response according to its normal threat processing prioritization methods. These flags, however, will show the CMDS perception of the current states of these control inputs.

The RST (data lost) flag is asserted after power-up of the CMDS PGMR. Whenever it is asserted, the CMDS Mode field of this message, and the actual CMDS mode, is STANDBY. Upon recovery from power-up, the CMDS de-asserts RST and changes to the current user-selected mode.

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APPENDIX B

The degraded mode indicators indicate the un-availability of various dispense modes (or MANUAL, in case MAN DEG is not asserted while SEM DEG and/or AUT DEG is asserted). When asserted, these indicate the operator should operate the CMDS in BYPASS mode.

The RUN SILENT flag (AUTOMATIC DISPENSE DISABLE) causes AUTO mode RWR, JMR, and MWS programs/inputs to be treated as if the mode were SEMI-AUTO. If the MDF flags indicate that JMR and MWS programs are treated as AUTO mode during SEMI-AUTO mode as reflected by the JMR AUT and MWS AUT fields of the CMDS STATUS RESPONSE message, RUN SILENT has no effect on programs initiated by those sources.

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APPENDIX B

ADDENDUM B

DESIGN SPECIFICATION DIFFERENCES BETWEEN LITTON AND TRACOR

The differences described in this addendum are design differences which may cause incompatibilities between the AN/ALE-47 and the AN/ALR-67B(V)2.

B.1 MESSAGE UPDATE CRITERIA FOR AOA CHANGES

The ICD is written to require a new Signal Message from the AN/ALR-67B(V)2 whenever the Angle of Attack changes by ± 10 degrees. (This is documented in text on pages A-9 and A-14 in reference to the angle in Data Word 2 of Signal Message on pages A-11 and A-12.) Litton has suggested that a new message be sent when change greater than ± 30 degrees is made. However Tracor requires new messages upon ± 10 degrees to fully execute the algorithm.

B.2 INCLUSION OF SIGNAL UPDATE MESSAGE

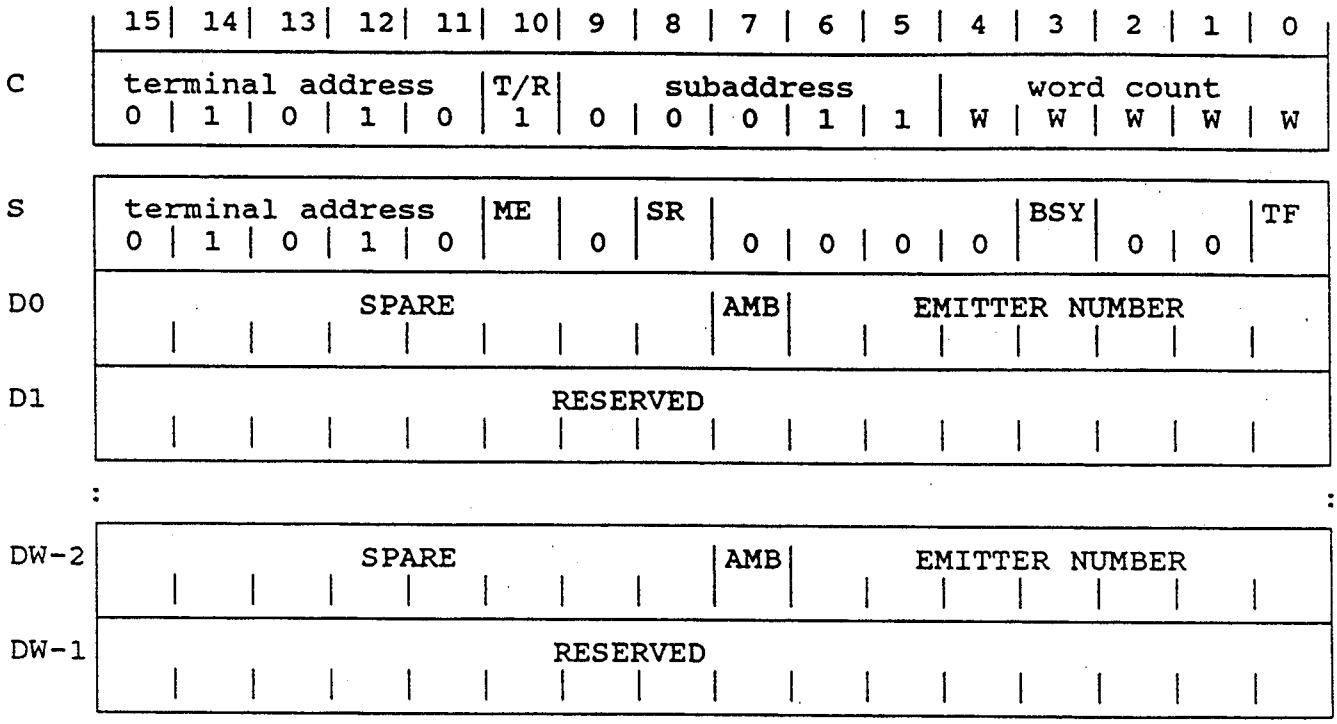
Removed by joint agreement.

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Check the source to verify that this is the current version before use.

B.3 SELECTED SIGNAL REQUEST MESSAGE

Litton desires to document the format of this message now so they can be assured that it is compatible with formats of messages in other systems. Tracor's position is that this message is not required in the AN/ALE-47 baseline.

SELECTED SIGNAL REQUEST MESSAGE Sub-Address: 00011



COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 1 for Transmit.
9-5	SA	SubAddress. Selected Signal Request Message = 00011.
4-0	W	Word count. (Variable from 2 to 30)

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DATA WORD 0 (All Even Data Words)

BIT	FIELD	DESCRIPTION
15-8	SPR	Spare
7	AMB	AMBiguous. If this bit is set the given emitter number was associated with a list of ambiguous with emitter identification codes.
6-0	EM #	EMitter Number. This is the track list reference number assigned by the AN/ALR-67B(V)2 to the signal being deleted.

DATA WORD 1 (All Odd Data Words)
Reserved

Indications of from 1 to 15 signals requested for data refresh, one per 2-word record, with count determined by the word count field (i.e., half the word count is the number of signals selected). The Emitter Number field is the correlation number used by the RWR (and thus the CMDS) for all threat (emitter/signal) related messages, and is maintained across any dis-ambiguation of signals. The data requested will be returned via Signal Messages.

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Check the source to verify that this is the current version before use.

B.4 JAMMER DECOYS MESSAGE

Litton desires to document the format of this message now so they can be assured that it is compatible with formats of messages in other systems. Tracor's position is that this message is not required in the AN/ALE-47 baseline.

JAMMER DECOYS MESSAGE Sub-Address: 00110

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress						word count			
	0	1	0	1	0	1	0	0	1	1	0	0	0	1	0	0
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	
D0	COR	RSV	JON	RSV	JTC			STP STR	TRK IND	NCJ	EMITTER NUMBER (EM #)					
D1	RSV	RSV	C/P		H/L		F/A		EM ID							
D2	RSV	FREQUENCY (FREQ)														
D3	INH	C/P		SPR	BAND CODE (BND COD)											ZRO 0

COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 1 for Transmit.
9-5	SA	SubAddress. Jammer Assignment Message = 00110.
4-0	W	Word count. The data word count shall be 4 words.

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Check the source to verify that this is the current version before use.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	COR	CORrelation bit. 1 = emitter is correlated with RWR
14	RSV	ReSerVed
13	JON	Jammer On/Off. 1 = Jammer in transmit mode
12	RSV	ReSerVed
11-10	JTC	Jammer Type Code 00 = AN/ALQ-126B 01 = AN/ALQ-162 10 = AN/ALQ-165 11 = AN/ALE-47 Controlled
9	STP STR	SToP/STaRt. 1 = Resources not assigned to this emitter
8	TRK IND	TRacK INDication AN/ALQ-126B 0 = Tracker A 1 = Tracker B AN/ALQ-162 0 (bit not used) AN/ALQ-165 0 = Untracked Signal 1 = Tracked Signal AN/ALE-47 0 (bit not used)
7	NCJ	Non Conflicting Jamming 1 = non-transponder technique in use 0 = transponder technique in use
6-0	EM #	EMitter Number. Track list identification number. If correlated, then RWR track list identification is returned. If uncorrelated, then the Jammer (or AN/ALE-47) returns the Jammer assigned number (Response Number).

DATA WORD 1

BIT	FIELD	DESCRIPTION
15-14	RSV	ReSerVed
13-12	C/P	CW Pulse Threat 00 = CW threat 01 = Pulse threat 10 = Both CW and Pulse threats 11 = Spare
11-10	H/L	High/Low Duty Cycle 00 = Low duty cycle response 01 = High duty cycle response 10 = Both high and low duty cycle 11 = Spare (invalid)
9-8	F/A	Forward/Aft 00 = Invalid 01 = Aft response 10 = Forward response 11 = Forward and Aft response
7-0	EM ID	EMitter IDentification. Threat type code in RWR code format, regardless of whether or not the threat is correlated. For AN/ALQ-165, the Emitter ID is presently set to zero.

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DATA WORD 2

BIT	FIELD	DESCRIPTION
15	RSV	ReSerVed
14-0	FREQ	See classified addendum for the meaning of this field. These bits indicate the frequency of jamming.

DATA WORD 3

BIT	FIELD	DESCRIPTION
15	INH	1 = Inhibit jamming in specified BND COD (i.e. bits 11-1)
14-13	C/P	CW Pulse Threat 00 = CW threat 01 = Pulse threat 10 = Both CW and Pulse threats 11 = Spare
12	SPR	SPaRe
11-1	BND COD	See classified addendum
0	ZRO	Always = 0

Remarks - Upon receipt of this message, the AN/ALR-67B(V)2 shall forward this message to all jammers on the AV Bus.

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B.5 BIT DISPLAY MESSAGE

Litton desires to document the format of this message now so they can be assured that it is compatible with formats of messages in other systems. Tracor's position is that this message is not required in the AN/ALE-47 baseline.

Character codes, enhancement codes, and location coordinate explanations will be supplied by Litton prior to implementation. The detailed meaning of the pages displayed with the BIT Display Message will be supplied by Tracor when the AN/ALE-47 implements the message.

BIT DISPLAY MESSAGE

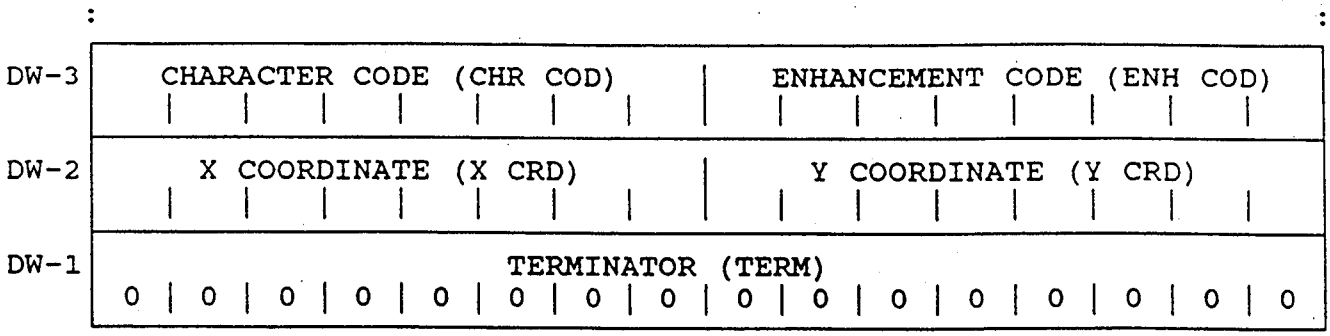
Sub-Address: 01010

	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
C	terminal address					T/R	subaddress					word count				
	0	1	0	1	0	1	0	1	0	1	0	W	W	W	W	W
S	terminal address					ME		SR					BSY			TF
	0	1	0	1	0		0		0	0	0	0		0	0	
D0	MAN BIT	PAGE				SYSTEM (SYS)										
						0	0	0	0	0	0	0	0	0	1	0	0
D1																
	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
D2																
	1	1	1	1	0	1	0	1	0	1	1	0	0	1	0	1
D3																
	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0
D4																
	0	0	0	0	0	1	0	1	0	1	1	0	0	1	0	1
D5	CHARACTER CODE (CHR COD)							ENHANCEMENT CODE (ENH COD)								
D6	X COORDINATE (X CRD)							Y COORDINATE (Y CRD)								

:

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COMMAND WORD

BIT	FIELD	DESCRIPTION
15-11	TA	Terminal Address. Remote terminal address of the AN/ALE-47.
10	T/R	Transmit Receive bit. Set to 1 for Transmit.
9-5	SA	SubAddress. BIT Display = 01010.
4-0	W	Word count. The word count will vary from 6 to 32 depending on the number of characters to display.

DATA WORD 0

BIT	FIELD	DESCRIPTION
15	MANBIT	MANual BIT. This bit is set to one to indicate this information is for the manual bit display.
14-12	PAGE	000 = Page 0 of the bit display. This page shall have 47 displayed at the top and shall also list any critical failures of the AN/ALE-47. 001 = Page 1 of the bit display. This page shall display a 47 at the top and the amount of each type of expendable that is available. 010 = Page 2 of the bit display. This page shall include a 47 at the top and a 3 row by 4 column hex character matrix indicating what the specific failures may be. This page shall be used for maintenance of the system. 011-111 = Not used.
11-0	SYS	SYStem. This field shall be used to tell the AN/ALR-67B(V)2 which system the bit display data belongs to. The field shall be bit mapped with only one of the 12 bits set to one. The AN/ALE-47 shall set bit 2 to one.

DATA WORD 1

BIT	NUM	DESCRIPTION
15-8	CHR COD	CHaRacter CODe. Words one and two of all three pages of the bit display message shall be reserved for the four characters that will be displayed at the top of each of the pages.
7-0	ENH COD	ENHancement CODe. The enhancement code field shall not be used for this character.

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DATA WORD 2

BIT	NUM	DESCRIPTION
15-8	X CRD	X CooRDinate. The X coordinate of the 4 shall be at the top of the display.
7-0	Y CRD	Y CooRDinate. The Y coordinate of the 4 shall be slightly offset to the left of the y axis.

DATA WORD 3

BIT	NUM	DESCRIPTION
15-8	CHR COD	CHaracteR CODe. Words three and four of all three pages of the bit display message shall be reserved for the seven that will be displayed at the top of each of the pages.
7-0	ENH COD	ENHancement CODe. Enhancement code field shall not be used for this character.

DATA WORD 4

BIT	NUM	DESCRIPTION
15-8	X CRD	X CooRDinate. The X coordinate of the 7 shall be at the top of the display.
7-0	Y CRD	Y CooRDinate. The Y coordinate of the 7 shall be slightly offset to the right of the y axis.

ODD NUMBERED DATA WORDS 5 TO W-3

BIT	NUM	DESCRIPTION
15-8	CHR COD	CHaRacter CODe. The odd data words 5 to W-3 specify character codes to be displayed on the AN/ALE-47 bit display pages. The information that is displayed is dependent of the page number and the AN/ALE-47 status. See the PAGE field description in word 0 for more information.
7-0	ENH COD	ENHancement CODe. An enhancement code can be specified to be displayed at the same location as the above characters.

EVEN NUMBERED DATA WORDS 6 TO W-2

BIT	NUM	DESCRIPTION
15-8	X CRD	X CooRDinate. These bits in the even data words 6 to W-2 specify the X coordinate on the AN/ALR-67B(V)2 display for the character in the previous word.
7-0	Y CRD	Y CooRDinate. These bits in the even data words 6 to W-2 specify the Y coordinate on the AN/ALR-67B(V)2 display for the character in the previous word.

DATA WORD W-1

BIT	NUM	DESCRIPTION
15-0	TERM	TERMinator. The last character and x-y coordinate data word pair of each bit display message shall be terminated by an all 0 data word.

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Remarks - This message will be sent by the AN/ALE-47 whenever the interface with the AN/ALR-67B(V)2 is initialized or reinitialized and whenever a change in the status of the AN/ALE-47 built in test (BIT) is detected. The AN/ALE-47 shall respond to the AN/ALR-67B(V)2 poll command with a transmit request. The AN/ALR-67B(V)2 will then command the AN/ALE-47 to transmit the message with a transmit command message.

When the interface with the AN/ALR-67B(V)2 is initialized or reinitialized this message shall be sent three times, once for each bit display page. After that the message will be sent only for bit display pages that are effected by a change in AN/ALE-47 BIT status.

The word count (W) indicated by WWWWW will vary depending on how many characters are to be displayed. Each character to be displayed by the AN/ALR-67B(V)2 requires 2 data words. The first word describes the character to be displayed and the second word tells where on the AN/ALR-67B(V)2 display the character is to be located. Character codes and how to locate them on the AN/ALR-67B(V)2 display will be added to this document prior to implementation. The interpretation of the data on the individual data pages will also be added to this document.

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APPENDIX B

ADDENDUM C

CHANGE REQUEST PROPOSAL SHEETS

Changes to this document will be made using the Change Request Proposal Sheets contained in this Addendum. Forward proposed changes to:

Commander, Naval Air Warfare Center
Weapons Division - Point Mugu
Electronic Warfare System Integration Code P1343 (4043)
Point Mugu, CA 93042-5001

PH (805) 989-1343/1344
AV 351-1343/1344

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CHANGE PROPOSAL

CHANGE
REQUEST
NUMBER _____

SHEET ___ OF ___

THIS CHANGE NOTICE, APPLICABLE TO DWG _____ REV ____
UPON APPROVAL WILL BE SHOWN IN REVISION _____

CHANGE REQUEST TITLE:

TECHNICAL APPROVAL					
NAVAIR					
	APPROVAL	DATE		APPROVAL	DATE
PMA 253			TRACOR		
			LITTON		

TITLE
AN/ALE-47 // AN/ALR-67(V)2
AVIONICS
INTERFACE CONTROL
DOCUMENT

SIZE	CODE IDENT. NO.	DRAWING NO.	REV.
A	07421	000047067	

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APPENDIX B

SAMPLE III. SYSTEM-TO-SYSTEM ICD

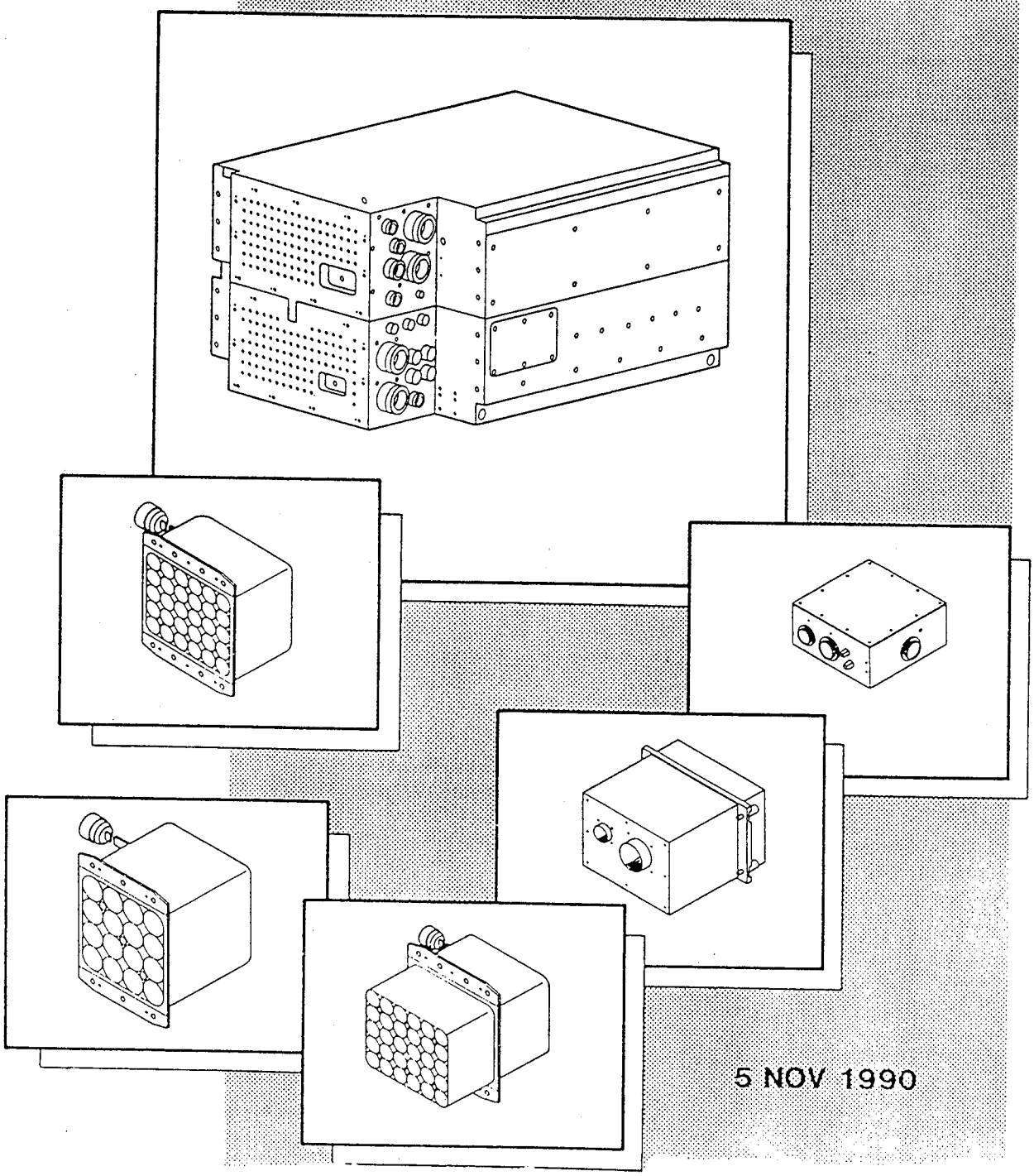
(INDIRECT INTERFACE)

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PACIFIC MISSILE TEST CENTER

AN/ALE-47 // AN/ALQ-126B INTERFACE CONTROL DOCUMENT #000047126



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APPENDIX B

TABLE iA

INTERFACE DESIGN DATA REQUIRED (IDDR) TABLE					
NUMBER	IDDR TITLE	SHEET NUMBER	DUE DATE	SUPPLIER	ACCEPTANCE
2	SUPPLY BASELINE SOFTWARE CONFIGURATION	7.1		TRACOR/SANDERS	

TABLE iB

COMPLETED IDDR's					
NUMBER	IDDR TITLE	SHEET NUMBER	DUE DATE	SUPPLIER	ACCEPTANCE
1	ALE-47 // ALQ-126B INTERFACE COMMAND, DATA, AND STATUS WORD MESSAGES	5.1		TRACOR/SANDERS	

INTERFACE CONTROL DOCUMENT (ICD)

TECHNICAL APPROVAL					
NAVAIR			AN/ALE-47 // AN/ALQ-126B		
APPROVAL		DATE	APPROVAL		DATE
PMA 253			TRACOR	Jim Sander	5/11/90
			SANDERS	John M Sylvain	10/4/90

INTERSTATE ELECTRONICS CORPORATION
A FIGGIE INTERNATIONAL COMPANY
1001 E. BALL ROAD P.O. BOX 3117 ANAHEIM CA. 92803 (714) 758-0500

TITLE
AN/ALE-47 // AN/ALQ-126B
INTERFACE CONTROL
DOCUMENT

SIZE	CAGE CODE	DRAWING NO.	REV
A	07421	000047126	-

SCALE NONE | REV IN-FILE DATE | SHEET 1

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EFFECTIVE SHEETS

SHEET NUMBERS	CHANGE IN EFFECT	DATE ISSUED
i - vii	Original Issue	7 September 1990
1.1	Original Issue	7 September 1990
3.1	Original Issue	7 September 1990
5.1	Original Issue	7 September 1990
7.1	CR 001	5 November 1990
ADDENDUM A	Original Issue	7 September 1990
ADDENDUM B	Original Issue	7 September 1990
ADDENDUM C	Original Issue	7 September 1990

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CHANGE REQUEST (CR) RECORD

CR #	DATE	TITLE/DESCRIPTION	SHEETS AFFECTED	COGNIZANT CONTRACTOR
001	05/11/90	Clarification of system reprogramability	7.1	IEC / VSE

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TABLE OF CONTENTS

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1.1	Scope	1.1
1.2	Description	1.1
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"B-Size" Documentation		3.1 - 7.1
AVMUX Command, Data, and Status Message Format	Addendum A	(1 pg)
Design Specification Contractor Differences	Addendum B	(1 pg)
Change Request Proposal Sheets	Addendum C	(3 pgs)

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SUB-CONTENTS FOR "B-SIZE" DOCUMENTATION

TITLE	SHEET
Not Applicable and Intentionally Deleted	2.1
AN/ALE-47 // AN/ALQ-126B Interface Block Diagram	3.1
Not Applicable and Intentionally Deleted	4.1
AV Mux Bus Interface Requirements	5.1
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System Software Configuration	7.1

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LIST OF ABBREVIATIONS/ACRONYMS

ASPJ	AIRBORNE SELF-PROTECTION JAMMER
AV	AVIONICS
CMDS	COUNTERMEASURES DISPENSING SYSTEM
DECM	DECEPTIVE ELECTRONIC COUNTERMEASURES
ICD	INTERFACE CONTROL DOCUMENT
IDDR	INTERFACE DESIGN DATA REQUIRED
MD	MISSILE DETECTOR
MLV	MEMORY LOADER VERIFIER
MUX	MULTIPLEXER
PMTC	PACIFIC MISSILE TEST CENTER
RWR	RADAR WARNING RECEIVER
SDL	SEQUENCER DATA LINK
SW	SWITCH
SYNC	SYNCHRONOUS
T/R	TRANSMIT/RECEIVE
WOW	WEIGHT ON WHEELS

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APPLICABLE DOCUMENTS

The following documents of the exact issue indicated form a part of this ICD to the extent specified herein.

GOVERNMENT DOCUMENTS

ICD 000047067	AN/ALE-47 // AN/ALR-67(V)2 Interface Control Document

McDONNELL DOUGLAS DOCUMENTS

MDC A3818, Rev B	F/A-18 Avionics Multiplex Design Specification

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APPENDIX B

1.0 INTRODUCTION

1.1 Scope

This document is an Interface Control Document (ICD) between the AN/ALE-47 Countermeasures Dispensing System (CMDS) and the AN/ALQ 126B Deceptive Electronic Countermeasures (DECM) Jammer. This document establishes the requirements for data interfaces.

For information only, additional interface requirements will be contained in the following documents:

000047067 (AN/ALE-47 // AN/ALR-67(V)2 RWR)
000067126 (AN/ALR-67(V)2 // AN/ALQ-126B DECM Jammer)

1.2 Description

The AN/ALE-47 CMDS is a threat-adaptive dispenser system designed to interface with an aircraft's AV suite, determine the nature of incoming threats, and automatically dispense appropriate expendables. The CMDS is compatible with the same payload types as the AN/ALE-39 with greater integration and automation for the pilot. With its data fusion capability from other AV suite sensors and aircraft avionics, threat adaptive dispensing will provide the optimum and most efficient countermeasures deployment, thus freeing the pilot to concentrate on maneuvering techniques and mission goals.

The AN/ALQ-126B DECM Jammer is a multi-mode, power-managed, reprogrammable deceptive electronic countermeasures system with digital instantaneous frequency measurement.

2.0 NOTATION

2.1 Symbols



Indicates interface 1. Interface numbers located within triangles throughout this document are referenced to interfaces on sheet 3.1.



Indicates additional data shown on sheet 5.1, zone 8H. In the event that related information is spread out over more than one sheet, only the first sheet will be indicated in this manner.

2.2 Table Nomenclature

Table iA indicates the first table on sheet i.

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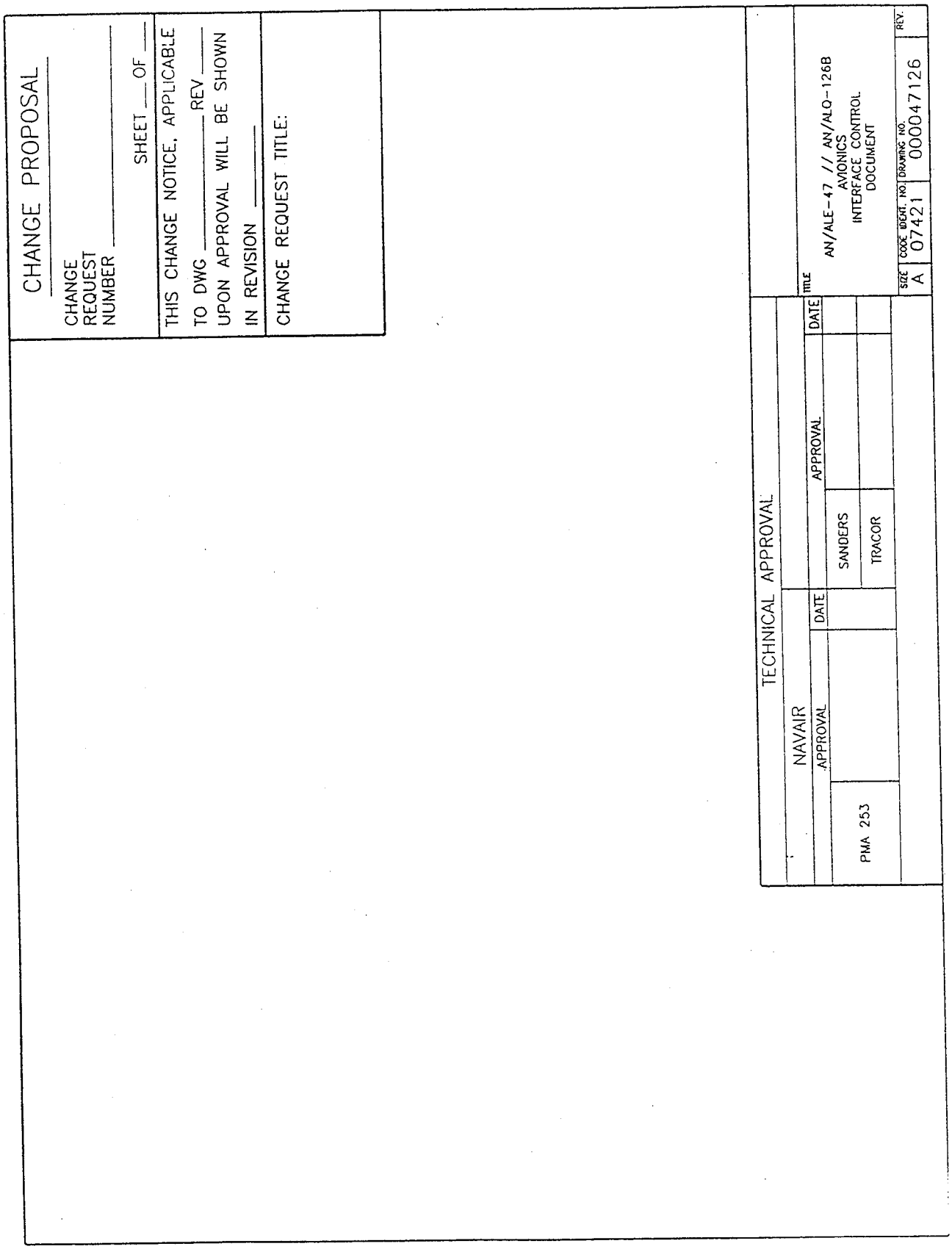
APPENDIX B

ADDENDUM A

For the AV MUX command, data, and status messages format for the AN/ALE-47 CMDS and the AN/ALQ-126B, refer to the current AN/ALE-47 CMDS and AN/ALR-67(V)2 ICD # 000047067.

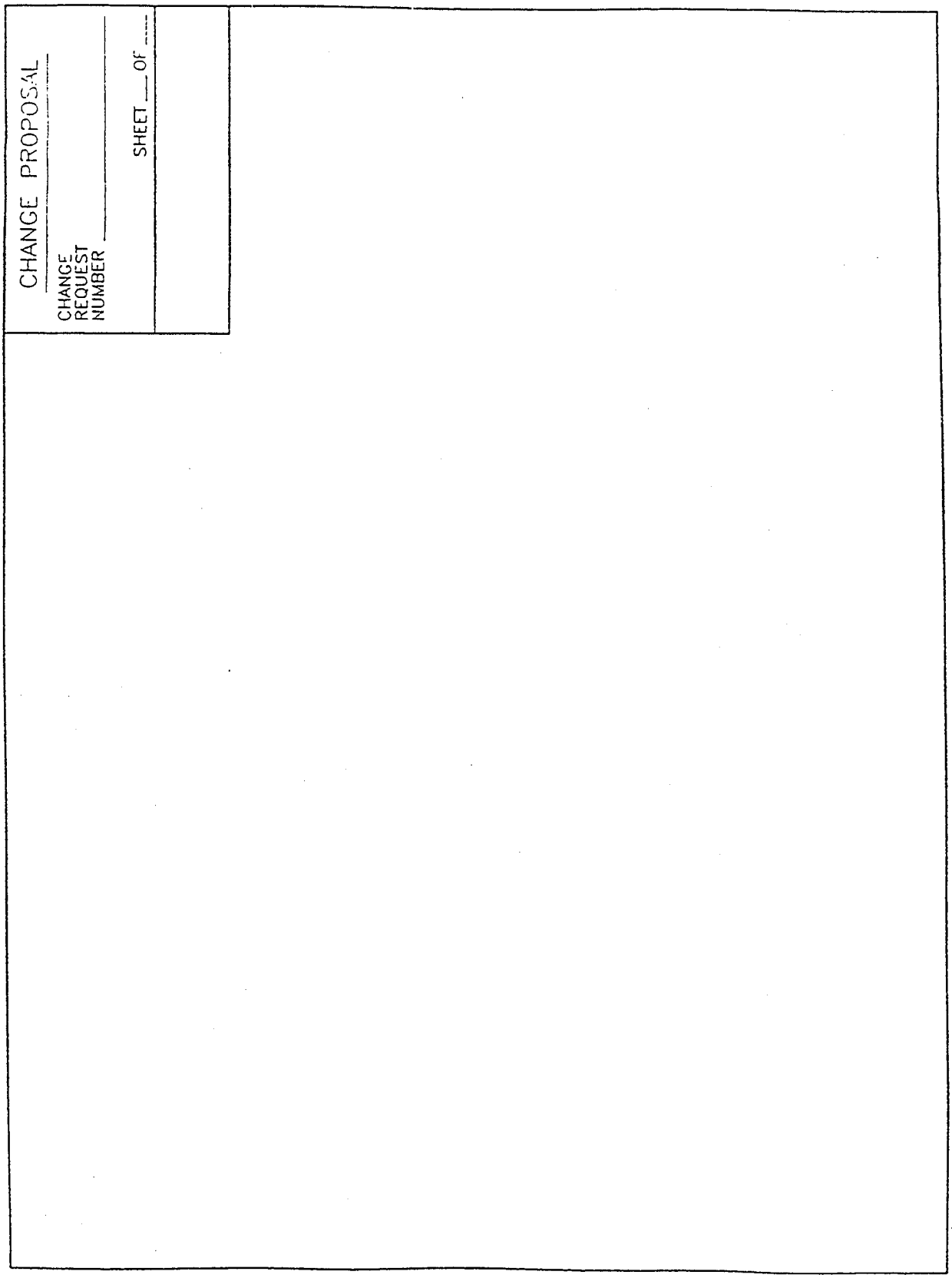
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ADDENDUM B

Design Specification Contractor Differences

Not Applicable

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ADDENDUM C

CHANGE REQUEST PROPOSAL SHEETS

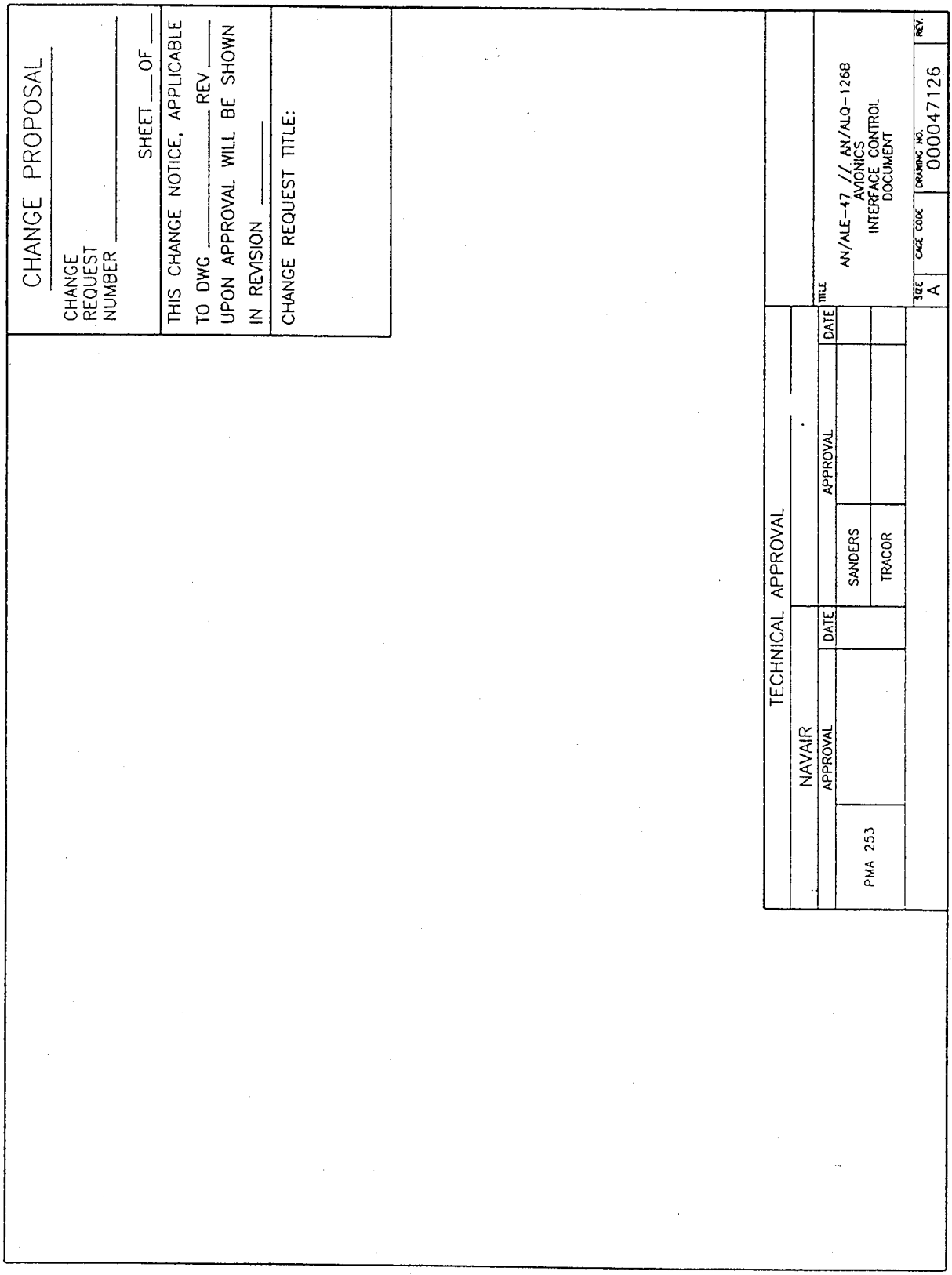
Changes to this document will be made using the Change Request Proposal Sheets contained in this Addendum. Forward proposed changes to:

Commander, Naval Air Warfare Center
Weapons Division - Point Mugu
Electronic Warfare System Integration Code P1343 (4043)
Point Mugu, CA 93042-5001

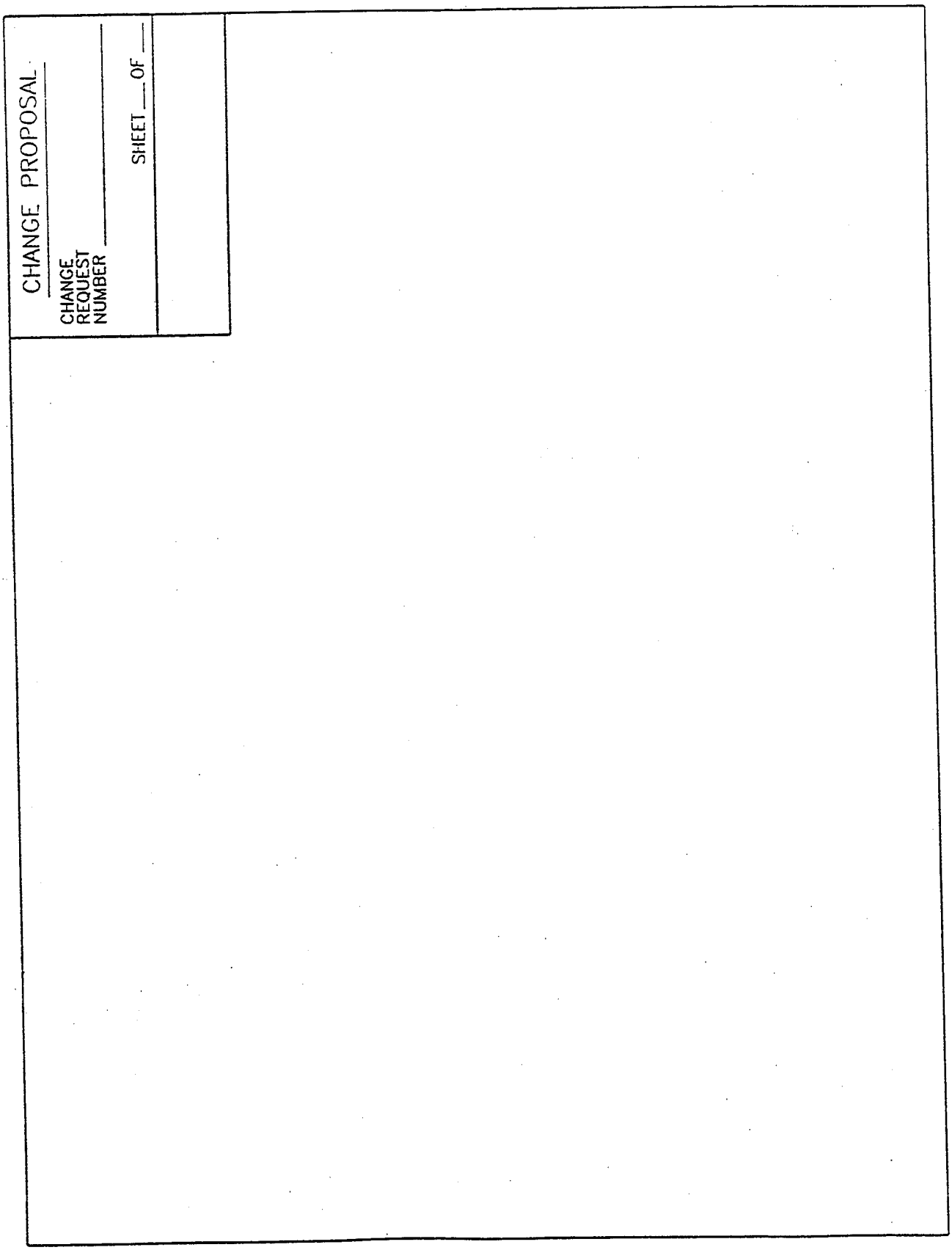
PH (805) 989-1343/1344
AV 351-1343/1344

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SAMPLE IV. PLATFORM-TO-SYSTEM ICDS WITH RF INTERFACES

(INDIRECT INTERFACE)

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Department of the Navy
Naval Air Warfare Center
Aircraft Division (Code SR3)
Lakehurst, NJ 08733-5100

Commanding Officer
Naval Air Warfare Center Aircraft Division Lakehurst
Systems Requirements Department (Code SR3)
Lakehurst, NJ 08733-5100

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-STD-2222(AS)	920812

3. DOCUMENT TITLE
INTERFACE CONTROL DOCUMENTATION FOR AVIONICS SYSTEMS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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